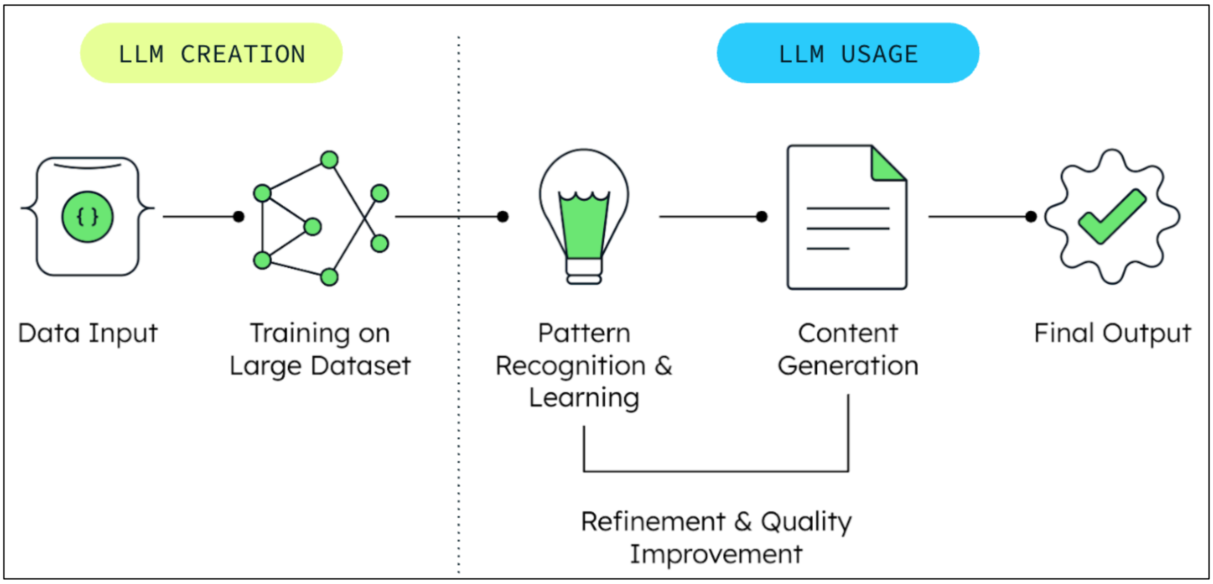
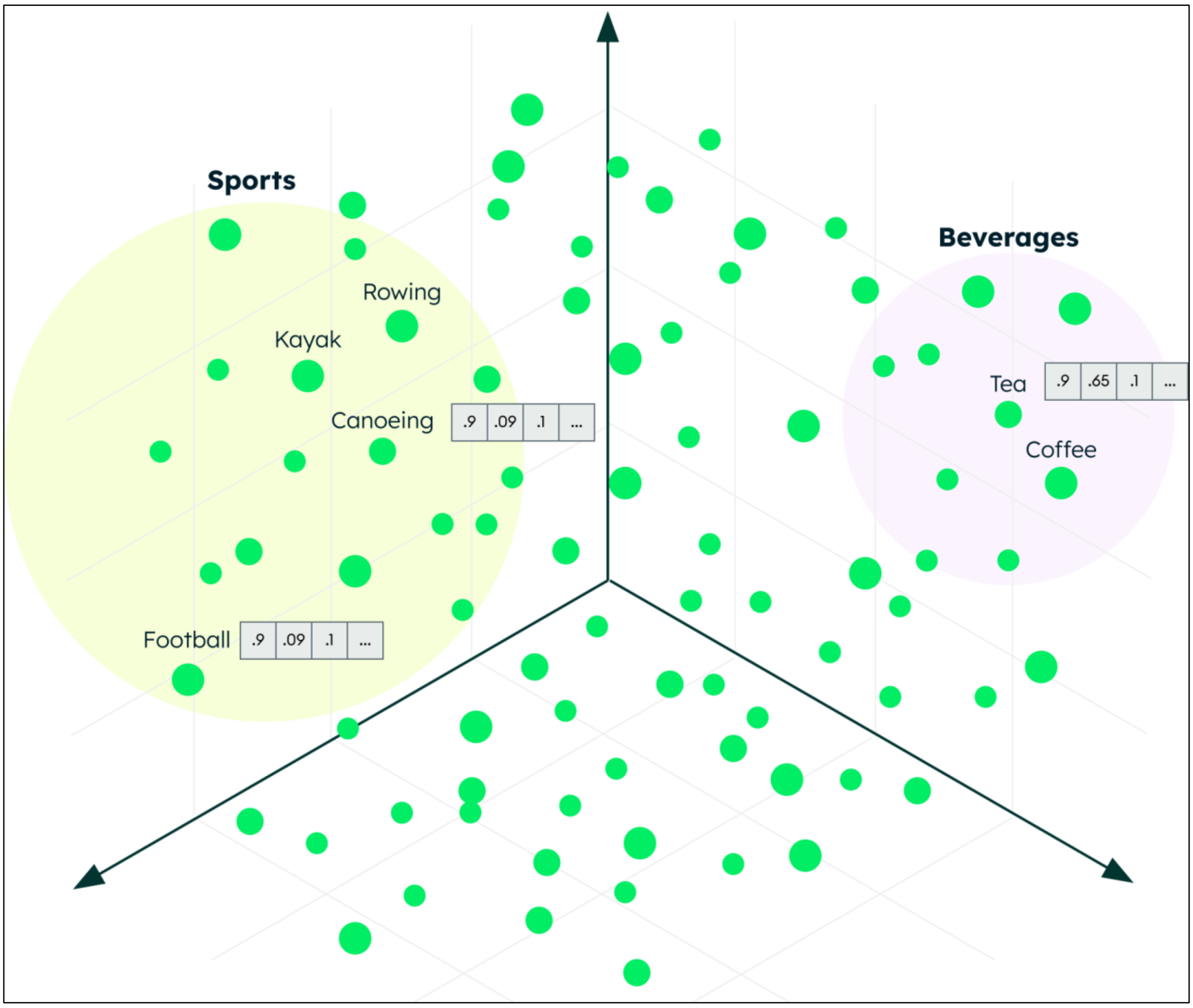
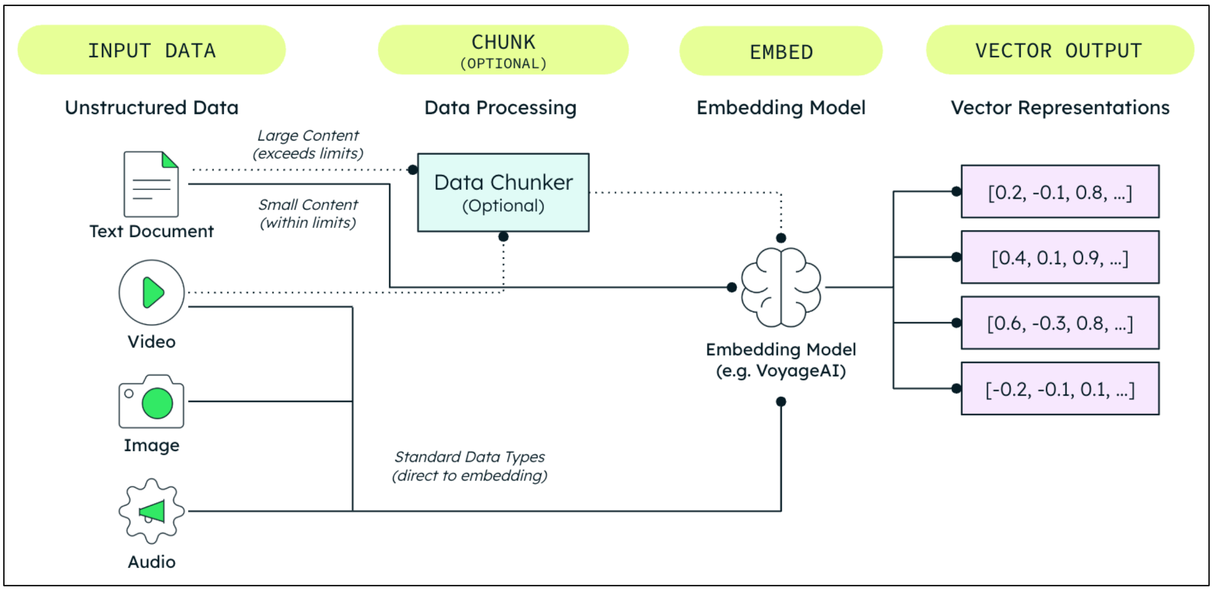
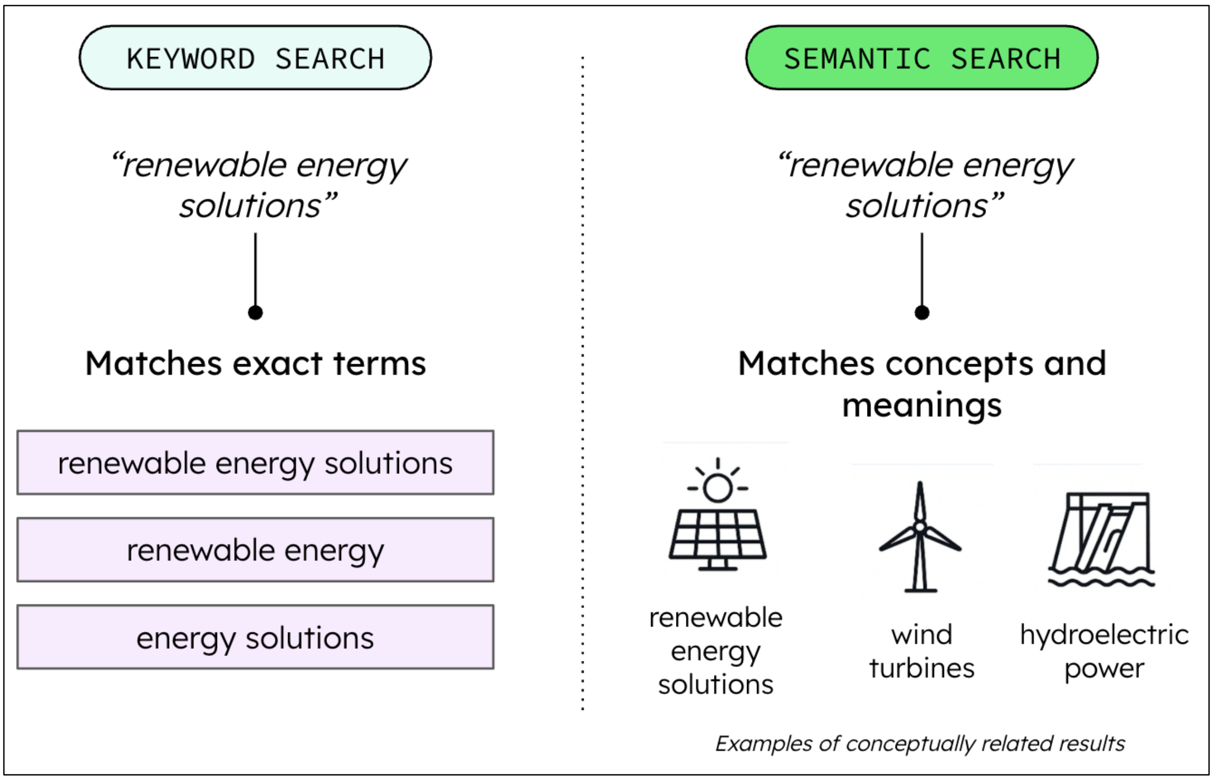
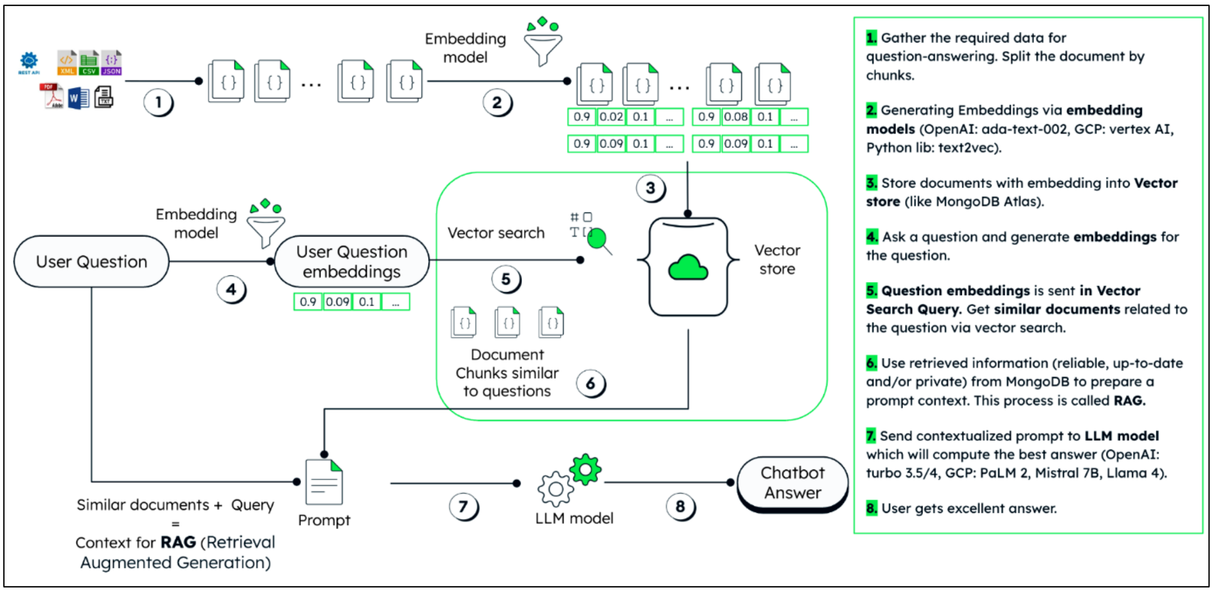
# Chapter 2: What Sets GenAI, RAG, and Agentic AI Apart

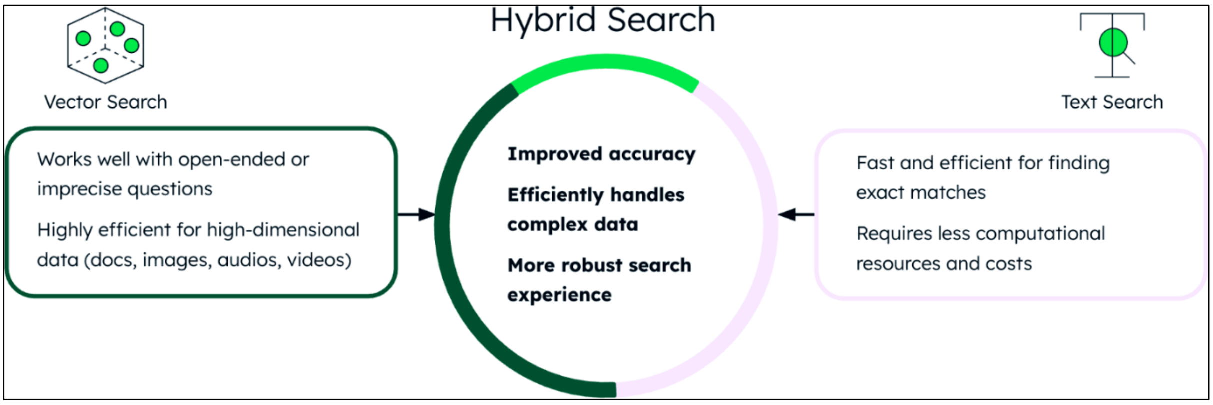


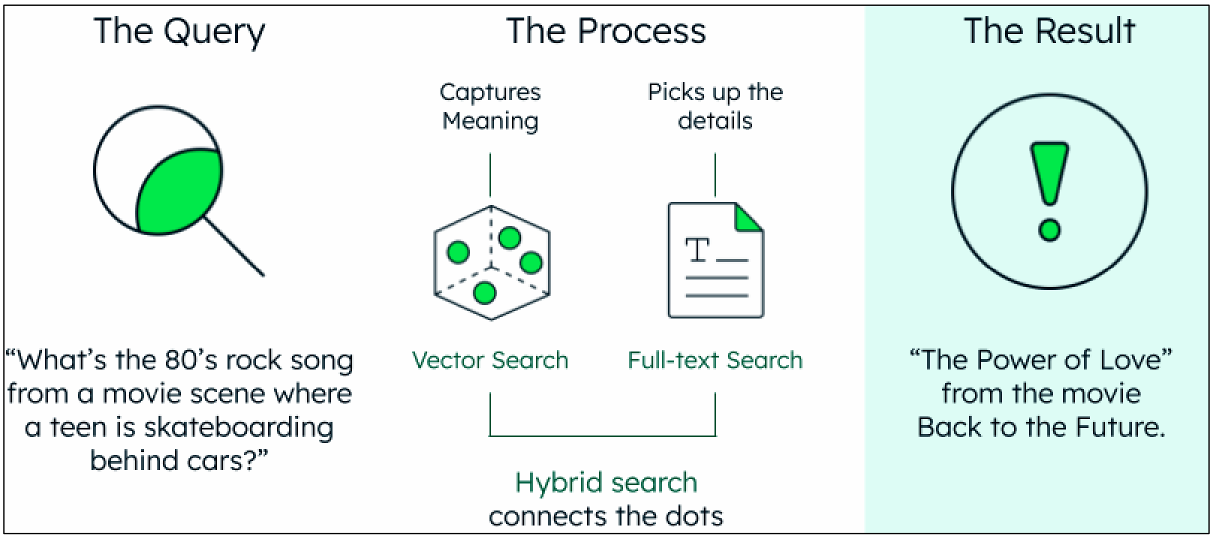



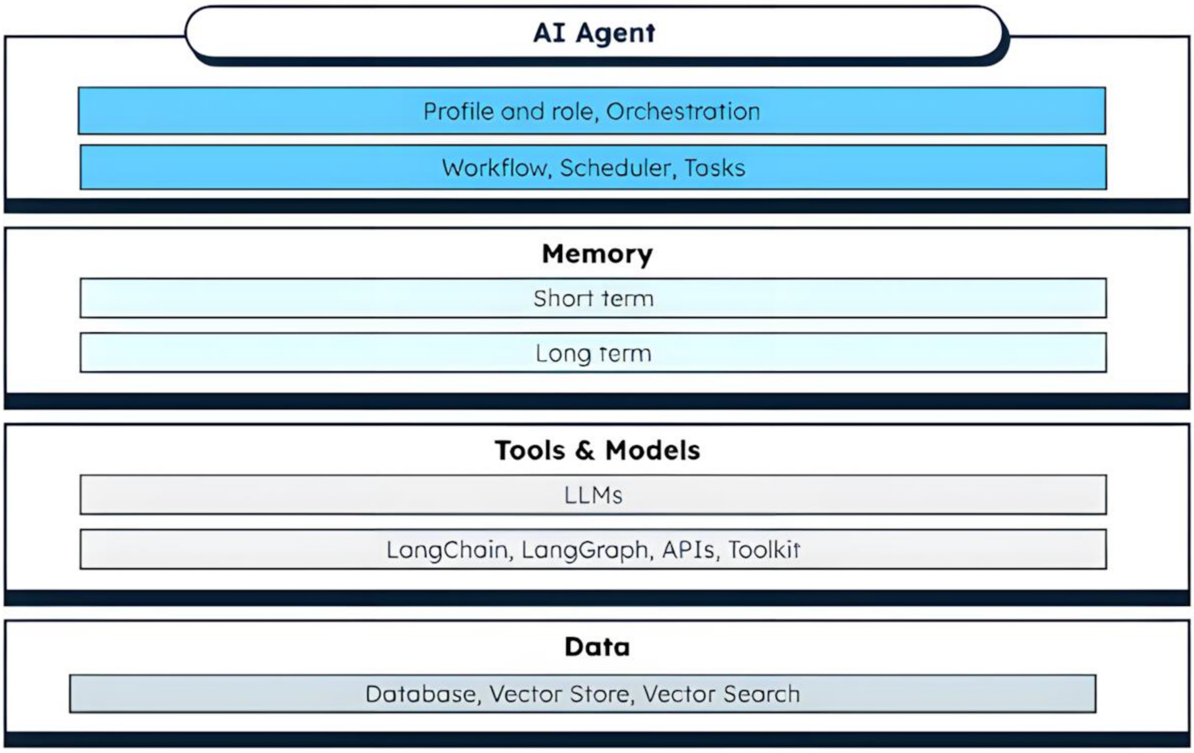

INPUT DATA
CHUNK (OPTIONAL)
EMBED
VECTOR OUTPUT
Unstructured Data
Data Processing
Embedding Model
Vector Representations
Large Content (exceeds limits)
Small Content (within limits)
Text Document
Data Chunker (Optional)
Video
Image
Audio
Embedding Model (e.g. VoyageAI)
Standard Data Types (direct to embedding)
[0.2, -0.1, 0.8, ...]
[0.4, 0.1, 0.9, ...]
[0.6, -0.3, 0.8, ...]
[-0.2, -0.1, 0.1, ...]

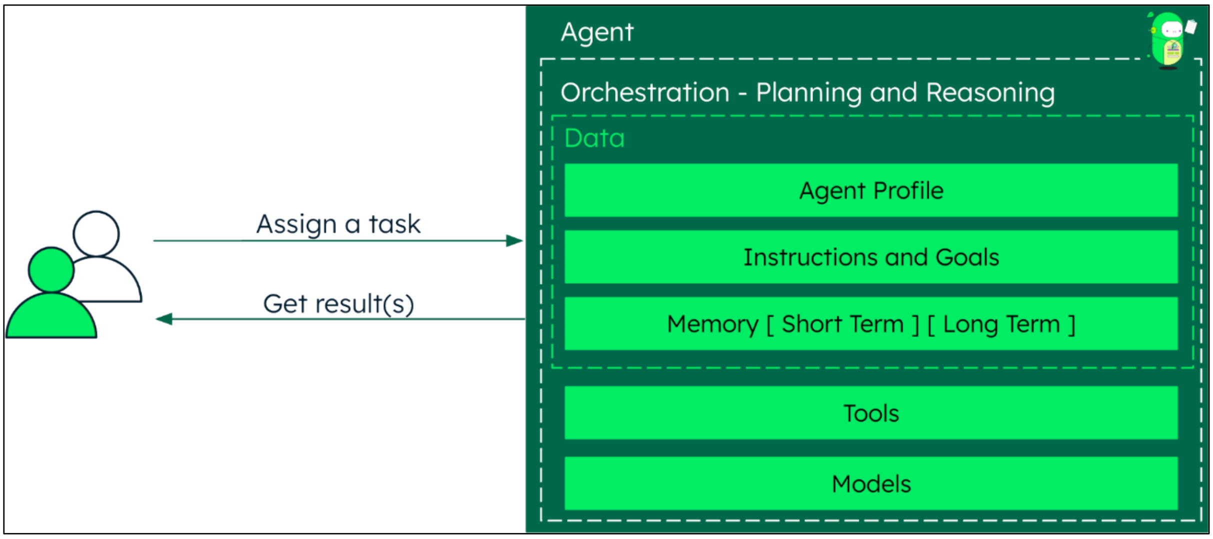

KEYWORD SEARCH
"renewable energy solutions"
Matches exact terms
renewable energy solutions
renewable energy
energy solutions
SEMANTIC SEARCH
"renewable energy solutions"
Matches concepts and meanings
renewable energy solutions
wind turbines
hydroelectric power
Examples of conceptually related results

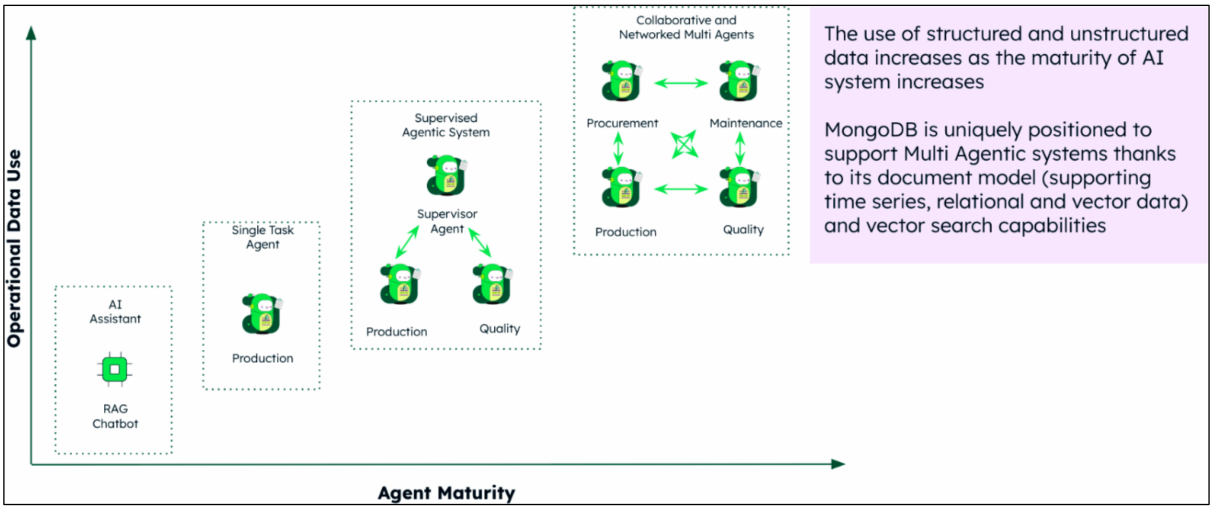

Embedding model
1
2
0.9 0.02 0.1 ...
0.9 0.08 0.1 ...
0.9 0.09 0.1 ...
0.9 0.09 0.1 ...
3
Embedding model
User Question
4
User Question embeddings
0.9 0.09 0.1 ...
Vector search
5
Vector store
Document Chunks similar to questions
6
Similar documents + Query
=
Context for **RAG** (Retrieval Augmented Generation)
Prompt
7
LLM model
8
Chatbot Answer
**1.** Gather the required data for question-answering. Split the document by chunks.
**2.** Generating Embeddings via **embedding models** (OpenAI: ada-text-002, GCP: vertex AI, Python lib: text2vec).
**3.** Store documents with embedding into **Vector store** (like MongoDB Atlas).
**4.** Ask a question and generate **embeddings** for the question.
**5. Question embeddings** is sent **in Vector Search Query.** Get **similar documents** related to the question via vector search.
**6.** Use retrieved information (reliable, up-to-date and/or private) from MongoDB to prepare a prompt context. This process is called **RAG.**
**7.** Send contextualized prompt to **LLM model** which will compute the best answer (OpenAI: turbo 3.5/4, GCP: PaLM 2, Mistral 7B, Llama 4).
**8.** User gets excellent answer.


Hybrid Search
Vector Search
Works well with open-ended or imprecise questions
Highly efficient for high-dimensional data (docs, images, audios, videos)
**Improved accuracy**
**Efficiently handles complex data**
**More robust search experience**
Text Search
Fast and efficient for finding exact matches
Requires less computational resources and costs

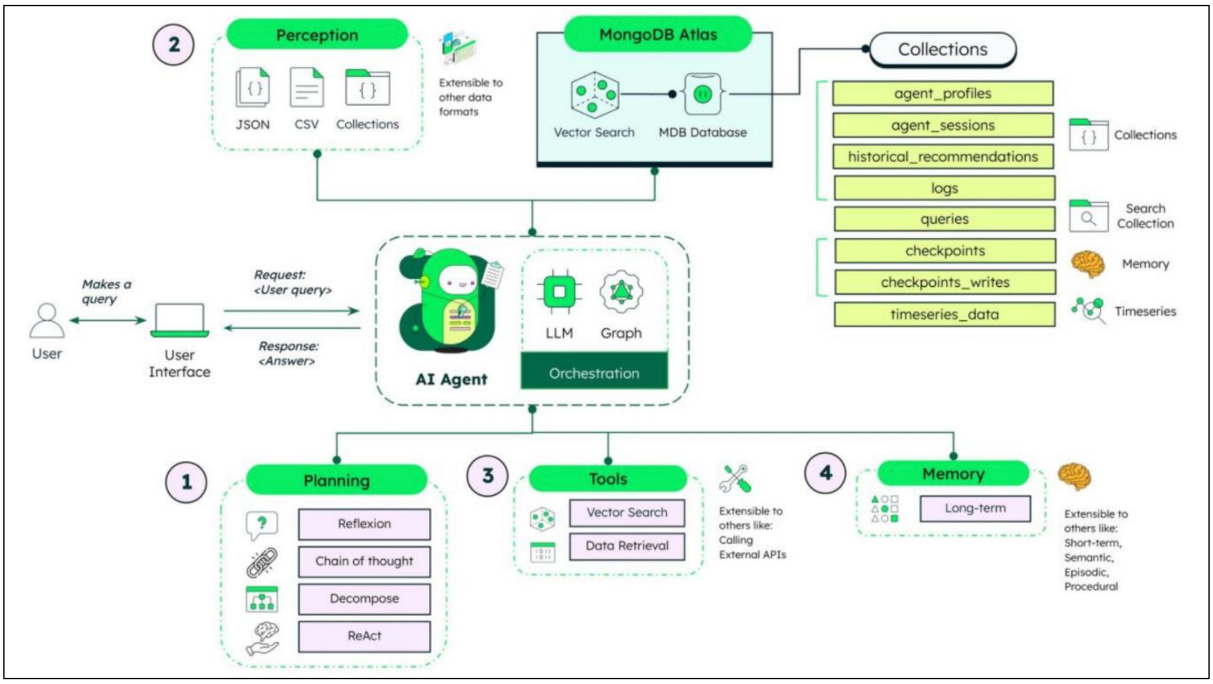

The Query
"What's the 80's rock song from a movie scene where a teen is skateboarding behind cars?"
The Process
Captures Meaning
Picks up the details
Vector Search
Full-text Search
Hybrid search connects the dots
The Result
"The Power of Love" from the movie Back to the Future.


AI Agent
Profile and role, Orchestration
Workflow, Scheduler, Tasks
Memory
Short term
Long term
Tools & Models
LLMs
LongChain, LangGraph, APIs, Toolkit
Data
Database, Vector Store, Vector Search


Assign a task
Get result(s)
Agent
Orchestration - Planning and Reasoning
Data
Agent Profile
Instructions and Goals
Memory [ Short Term ] [ Long Term ]
Tools
Models

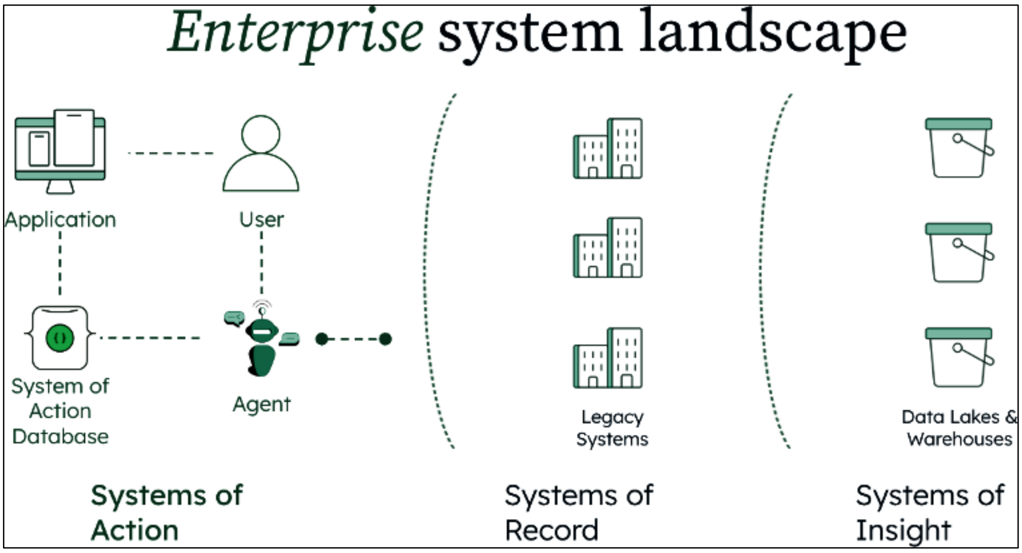


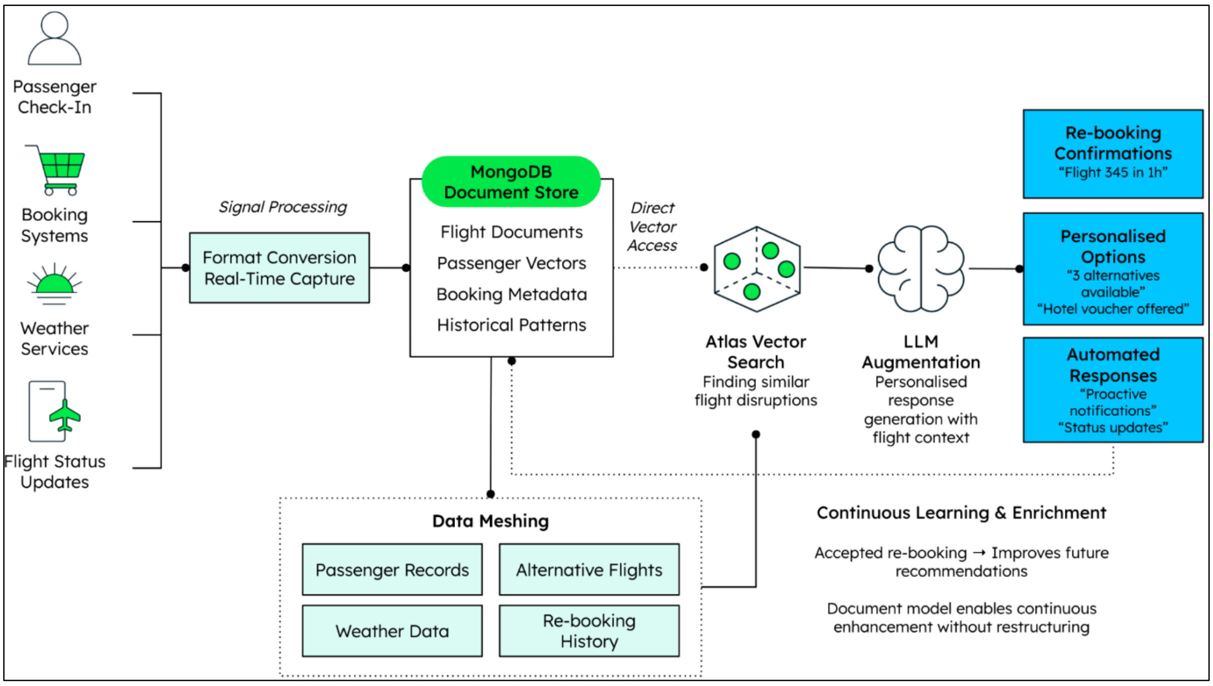
Operational Data Use
Agent Maturity
AI Assistant
RAG Chatbot
Single Task Agent
Production
Supervised Agentic System
Supervisor Agent
Production
Quality
Collaborative and Networked Multi Agents
Procurement
Maintenance
Production
Quality
The use of structured and unstructured data increases as the maturity of AI system increases
MongoDB is uniquely positioned to support Multi Agentic systems thanks to its document model (supporting time series, relational and vector data) and vector search capabilities

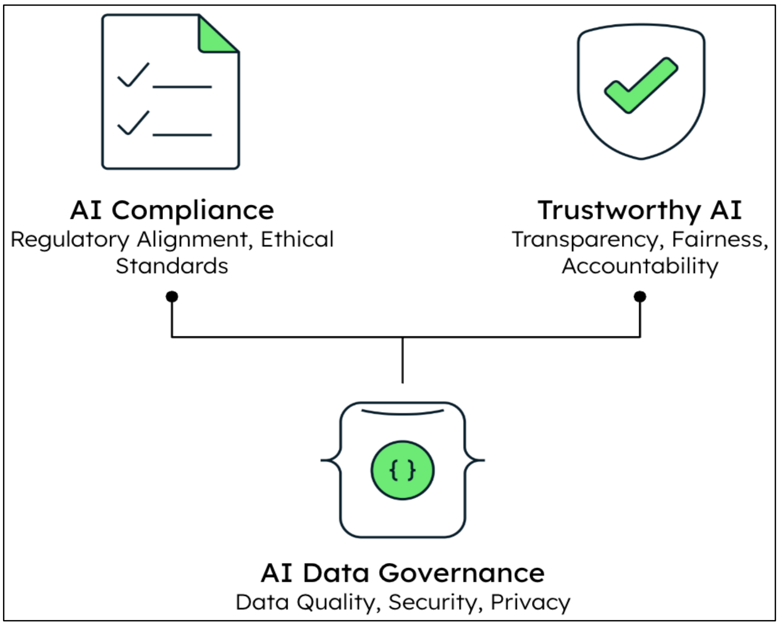
2
Perception
JSON
CSV
Collections
Extensible to other data formats
MongoDB Atlas
Vector Search
MDB Database
Collections
agent_profiles
agent_sessions
historical_recommendations
logs
queries
checkpoints
checkpoints_writes
timeseries_data
Collections
Search Collection
Memory
Timeseries
User
Makes a query
User Interface
Request: <User query>
Response: <Answer>
AI Agent
LLM
Graph
Orchestration
1
Planning
Reflexion
Chain of thought
Decompose
ReAct
3
Tools
Vector Search
Data Retrieval
Extensible to others like: Calling External APIs
4
Memory
Long-term
Extensible to others like: Short-term, Semantic, Episodic, Procedural

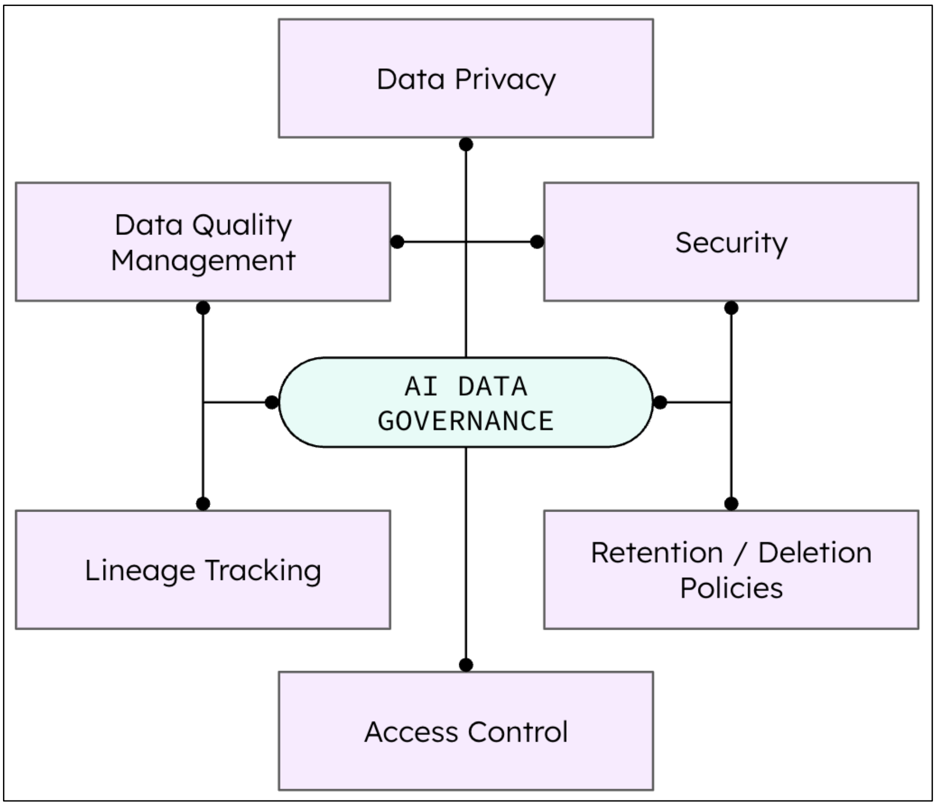
# Chapter 3: The System of Action



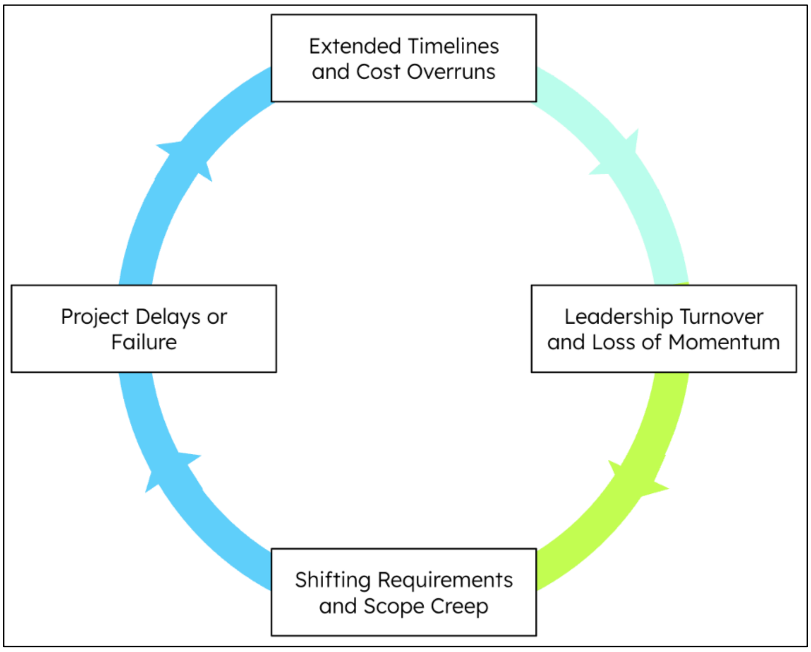
# Chapter 4: Trustworthy AI, Compliance, and Data Governance

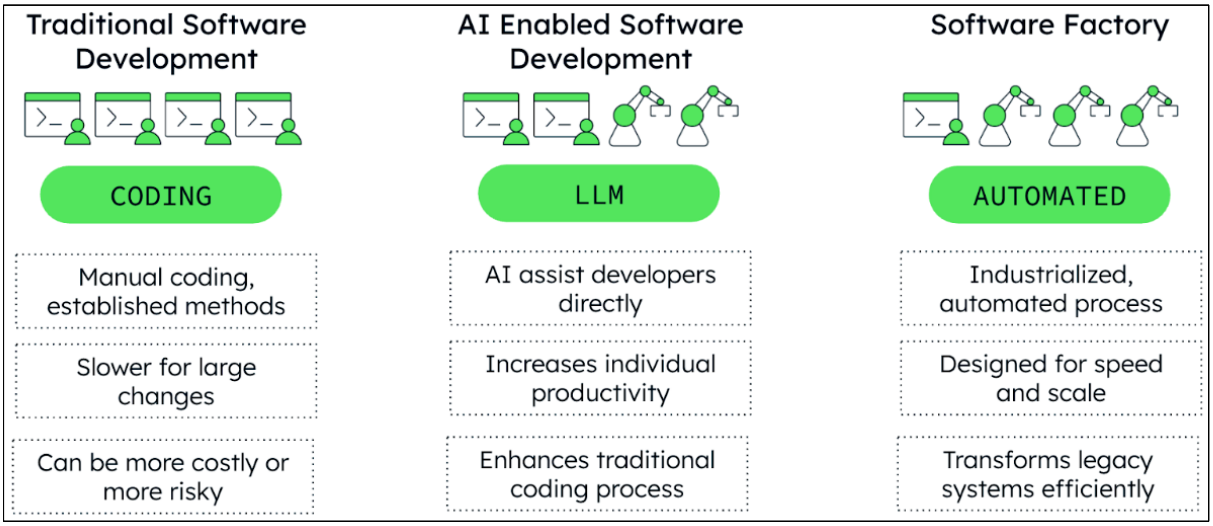


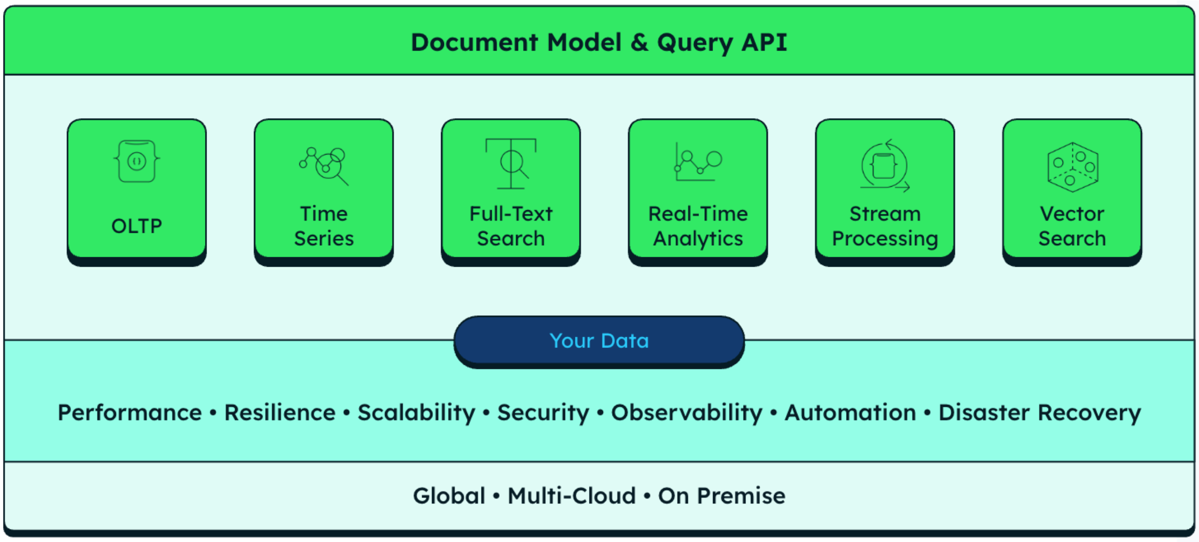



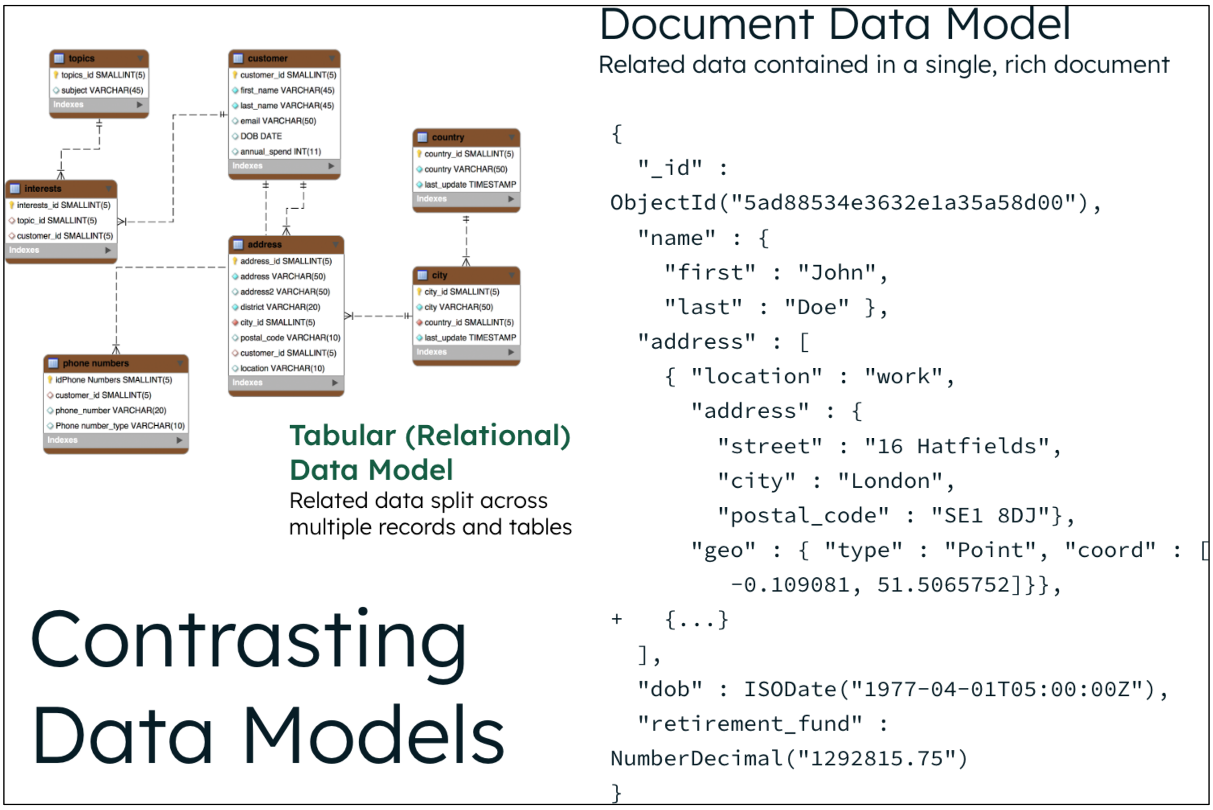
| Technique | Strengths | Weaknesses | Typical Use Cases |
| --- | --- | --- | --- |
| **LIME**<br>(Local Interpretable Model-agnostic Explanations) | Model-agnostic; fast local explanations; highlights key features per prediction | Valid only locally; unstable for complex models; approximate, not exact | Quick checks, debugging individual predictions, fast instance-level insights |
| **SHAP**<br>(Shapley Additive exPlanations) | Grounded in game theory; consistent feature attribution; supports local and global explanations | Computationally intensive; slow on large datasets or complex models | Regulatory reporting, fairness audits, high-stakes decisions needing detailed explanations |
| **Attention Mechanisms** | Visualizes model focus; naturally integrated in deep learning models (NLP and vision); intuitive for sequence data | Not a guaranteed explanation (shows focus, not causality); limited scope beyond attention layers | NLP (translation, summarization), vision tasks, any case where knowing model focus aids understanding |
| **Intrinsically Interpretable Models** | Transparent by design; decision logic traceable; fast and lightweight | Limited accuracy; misses complex patterns; weak on unstructured data | Finance, healthcare, or domains needing critical interpretability with simple models (e.g., decision trees, linear regression) |

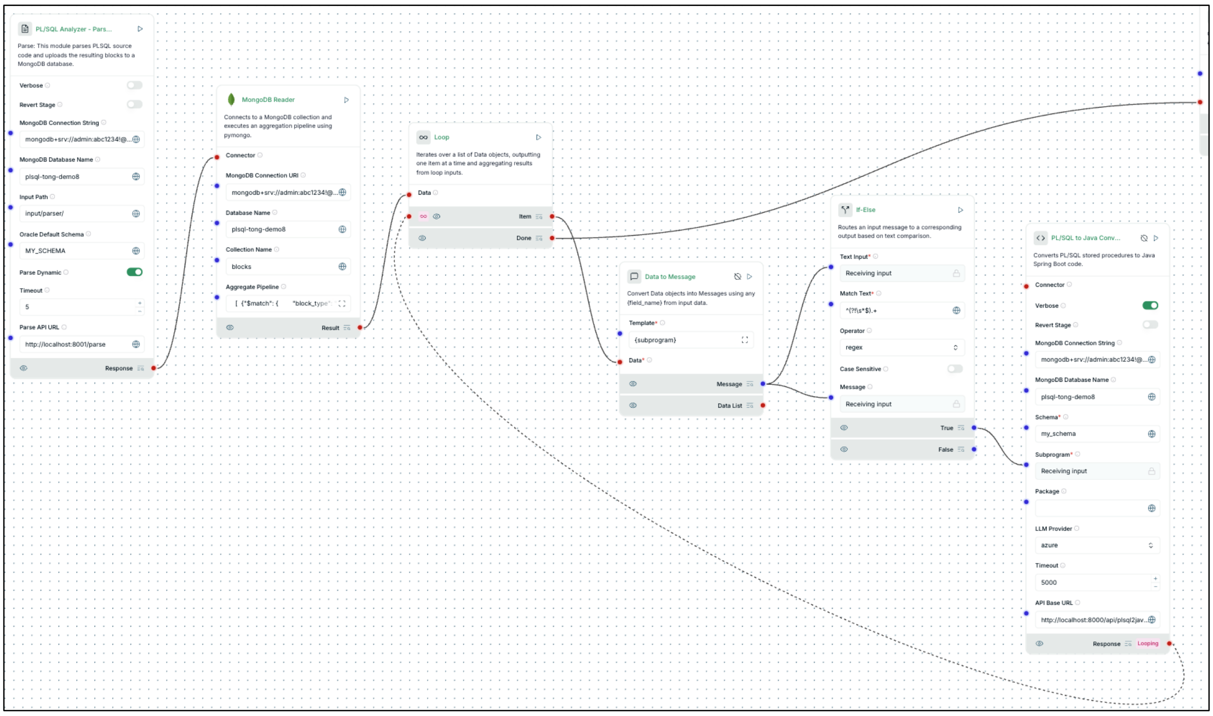
# Chapter 5: Modernization Using AI

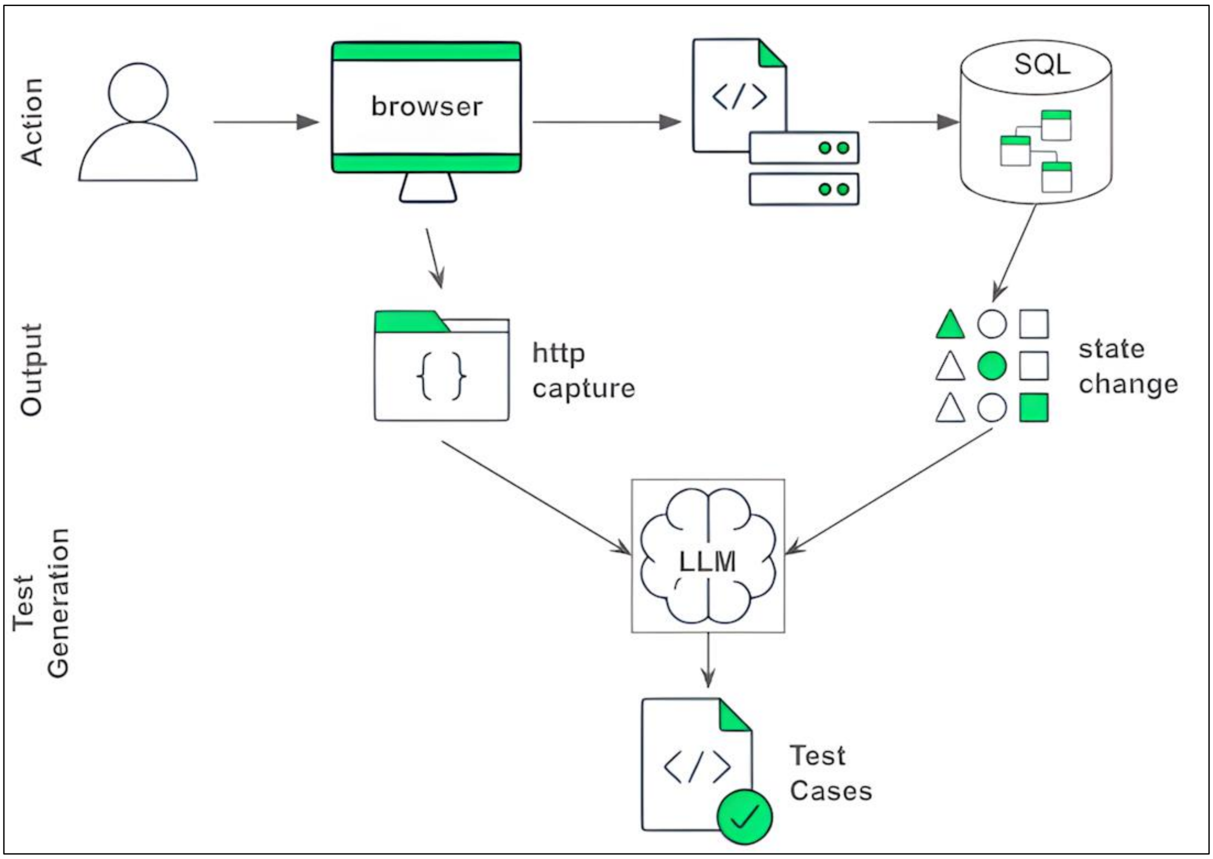


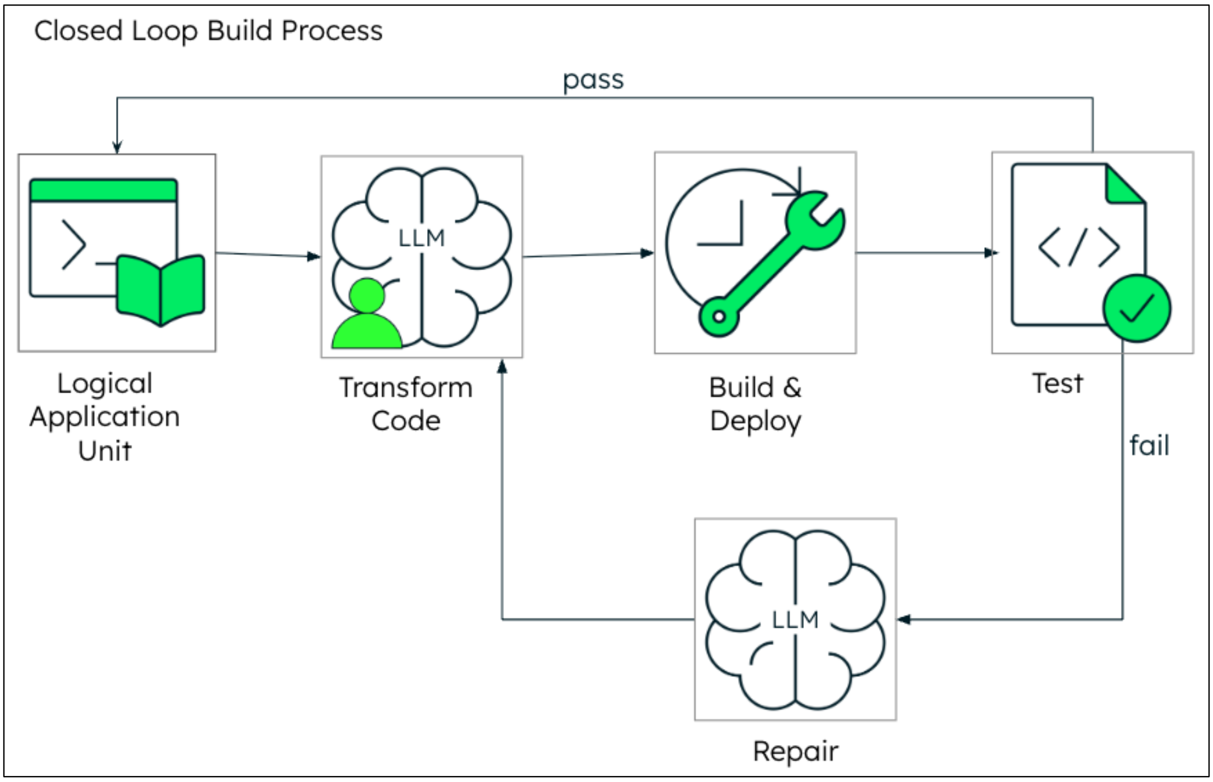



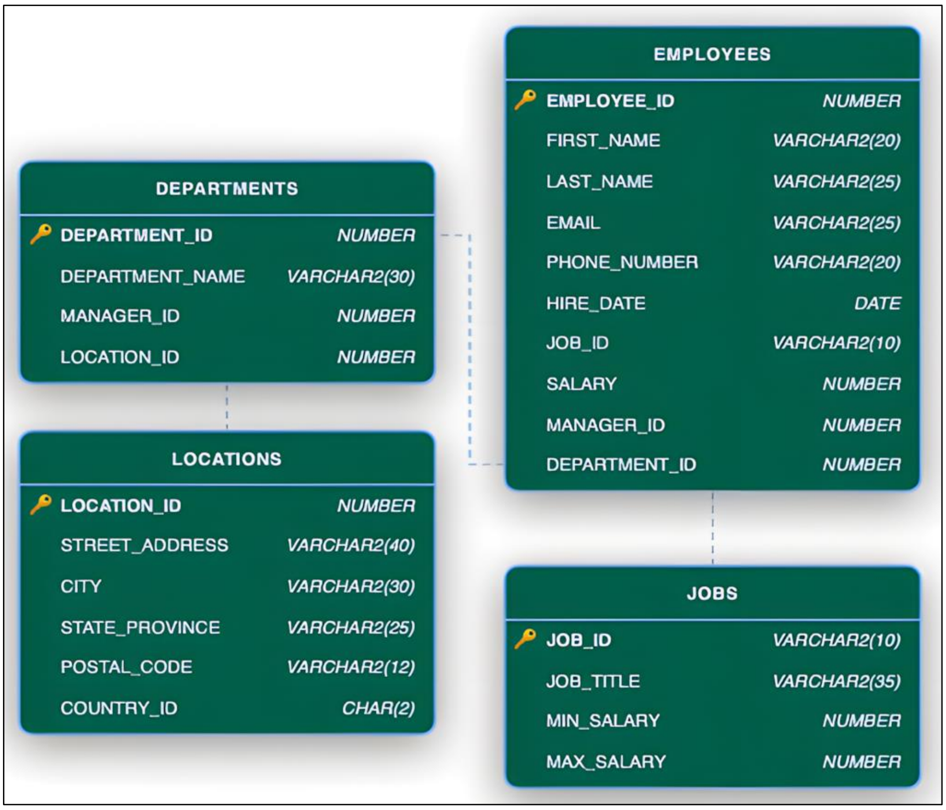



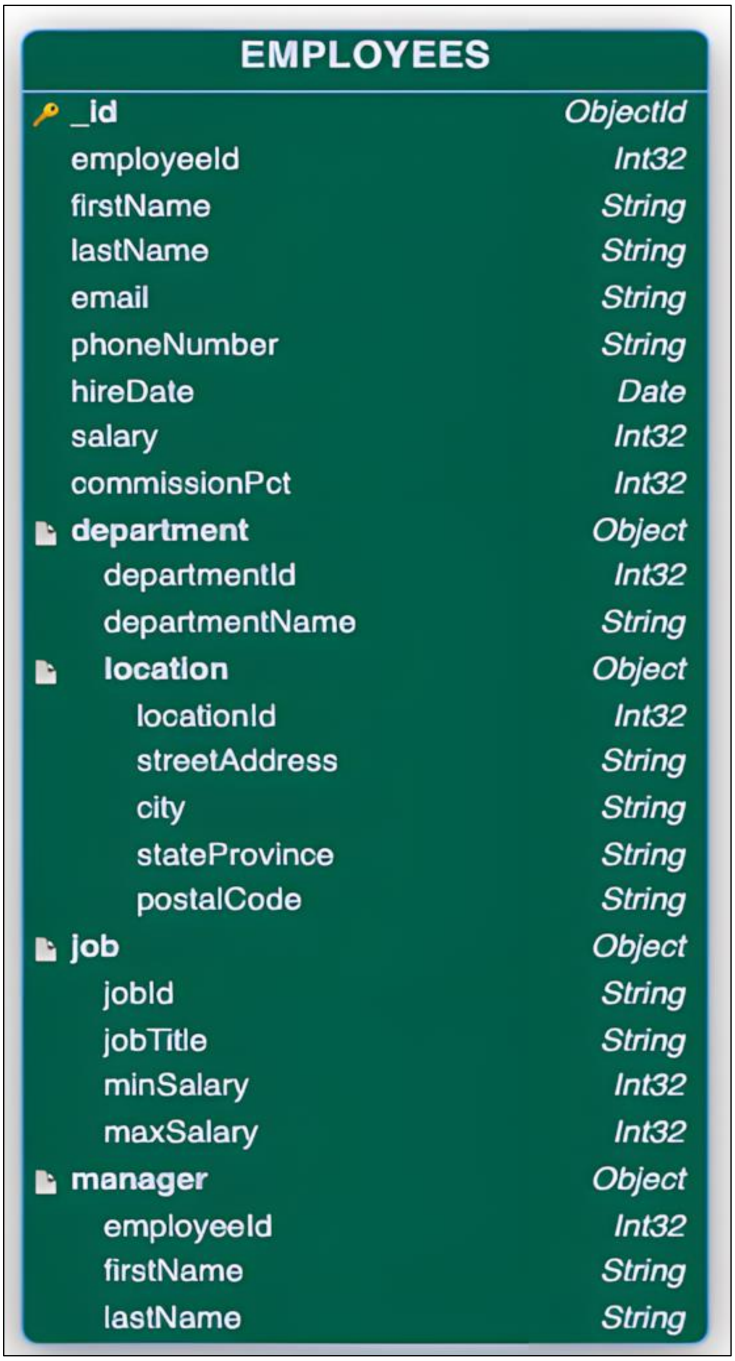

Document Data Model
Related data contained in a single, rich document
{
"_id" :
ObjectId("5ad88534e3632e1a35a58d00"),
"name" : {
"first" : "John",
"last" : "Doe" },
"address" : [
{ "location" : "work",
"address" : {
"street" : "16 Hatfields",
"city" : "London",
"postal_code" : "SE1 8DJ"},
"geo" : { "type" : "Point", "coord" : [
-0.109081, 51.5065752]}},
+ {...}
],
"dob" : ISODate("1977-04-01T05:00:00Z"),
"retirement_fund" :
NumberDecimal("1292815.75")
}
topics
customer
country
interests
address
city
phone numbers
Tabular (Relational) Data Model
Related data split across multiple records and tables
Contrasting Data Models



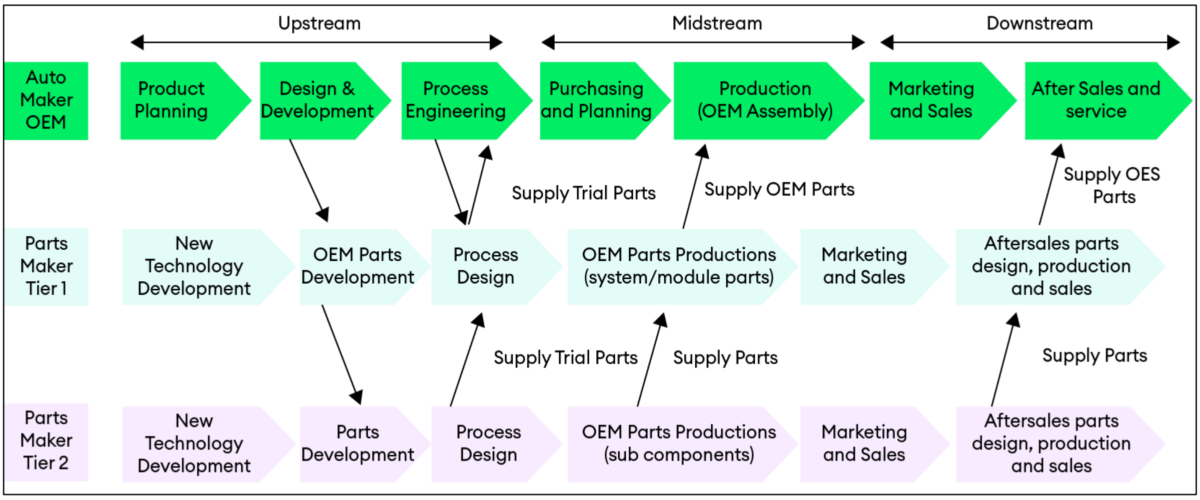

Action
Output
Test Generation
browser
SQL
http capture
state change
LLM
Test Cases

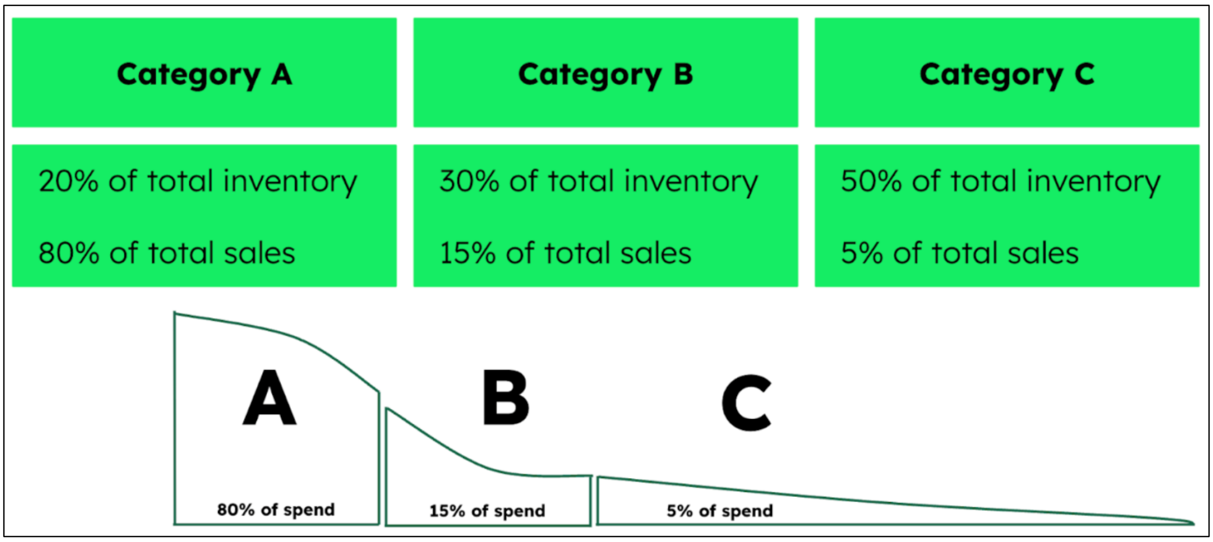

Closed Loop Build Process
pass
LLM
Logical Application Unit
Transform Code
Build & Deploy
Test
fail
LLM
Repair

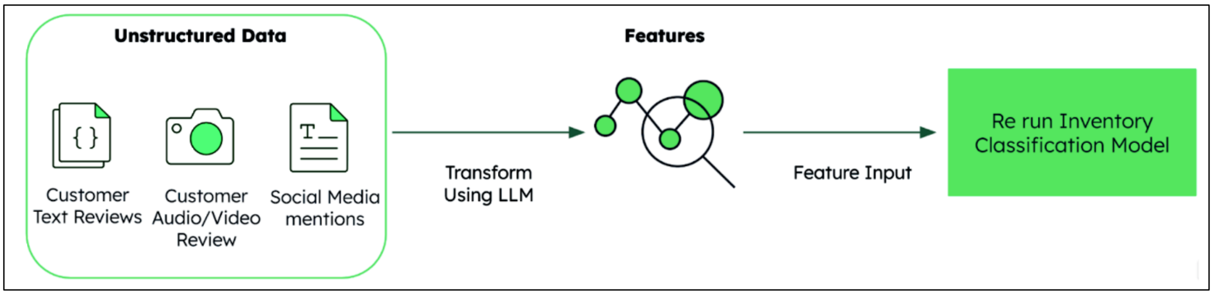

EMPLOYEES
EMPLOYEE_ID NUMBER
FIRST_NAME VARCHAR2(20)
LAST_NAME VARCHAR2(25)
EMAIL VARCHAR2(25)
PHONE_NUMBER VARCHAR2(20)
HIRE_DATE DATE
JOB_ID VARCHAR2(10)
SALARY NUMBER
MANAGER_ID NUMBER
DEPARTMENT_ID NUMBER
DEPARTMENTS
DEPARTMENT_ID NUMBER
DEPARTMENT_NAME VARCHAR2(30)
MANAGER_ID NUMBER
LOCATION_ID NUMBER
LOCATIONS
LOCATION_ID NUMBER
STREET_ADDRESS VARCHAR2(40)
CITY VARCHAR2(30)
STATE_PROVINCE VARCHAR2(25)
POSTAL_CODE VARCHAR2(12)
COUNTRY_ID CHAR(2)
JOBS
JOB_ID VARCHAR2(10)
JOB_TITLE VARCHAR2(35)
MIN_SALARY NUMBER
MAX_SALARY NUMBER

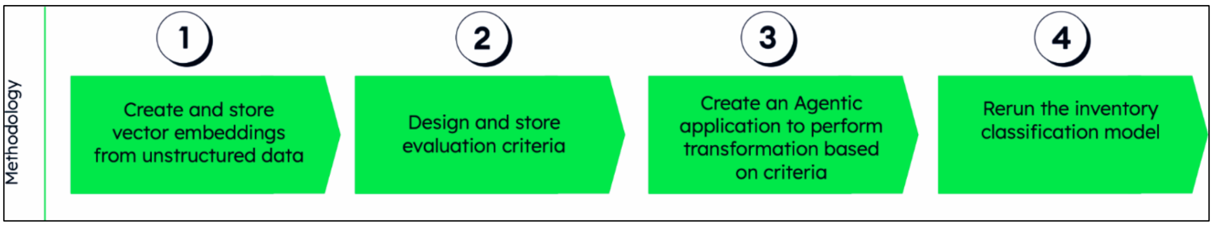

EMPLOYEES
_id ObjectId
employeeId Int32
firstName String
lastName String
email String
phoneNumber String
hireDate Date
salary Int32
commissionPct Int32
department Object
departmentId Int32
departmentName String
location Object
locationId Int32
streetAddress String
city String
stateProvince String
postalCode String
job Object
jobId String
jobTitle String
minSalary Int32
maxSalary Int32
manager Object
employeeId Int32
firstName String
lastName String

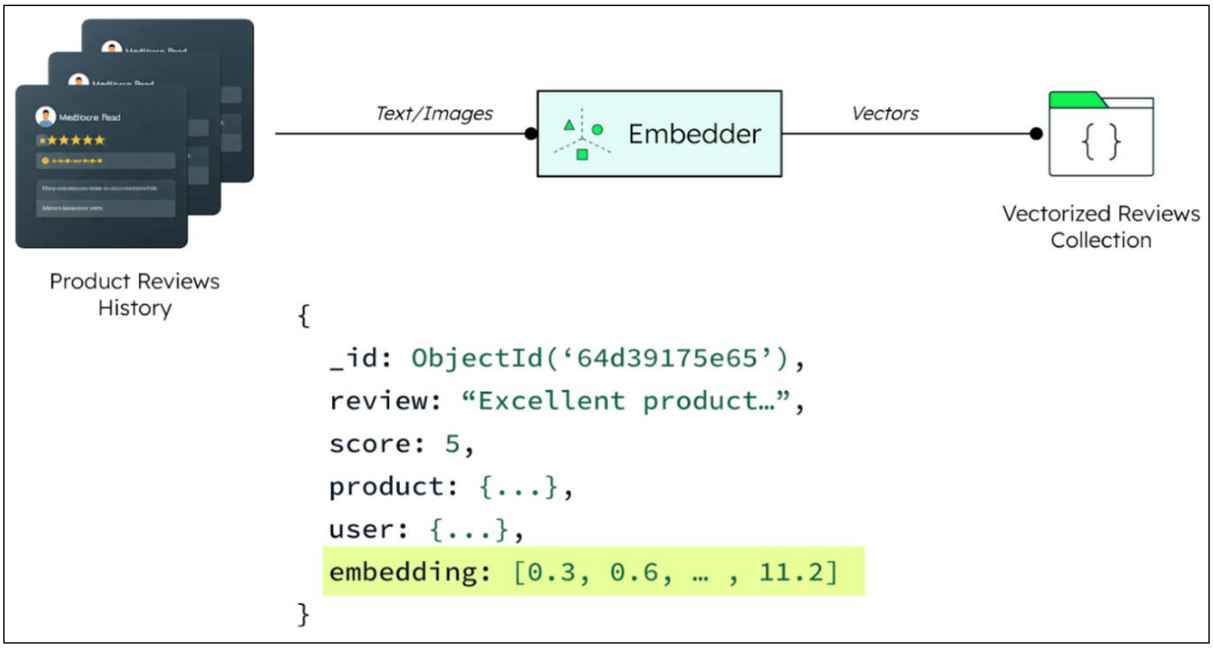
# Chapter 6: Practical Applications of Agentic and GenAI in Manufacturing – Part 1

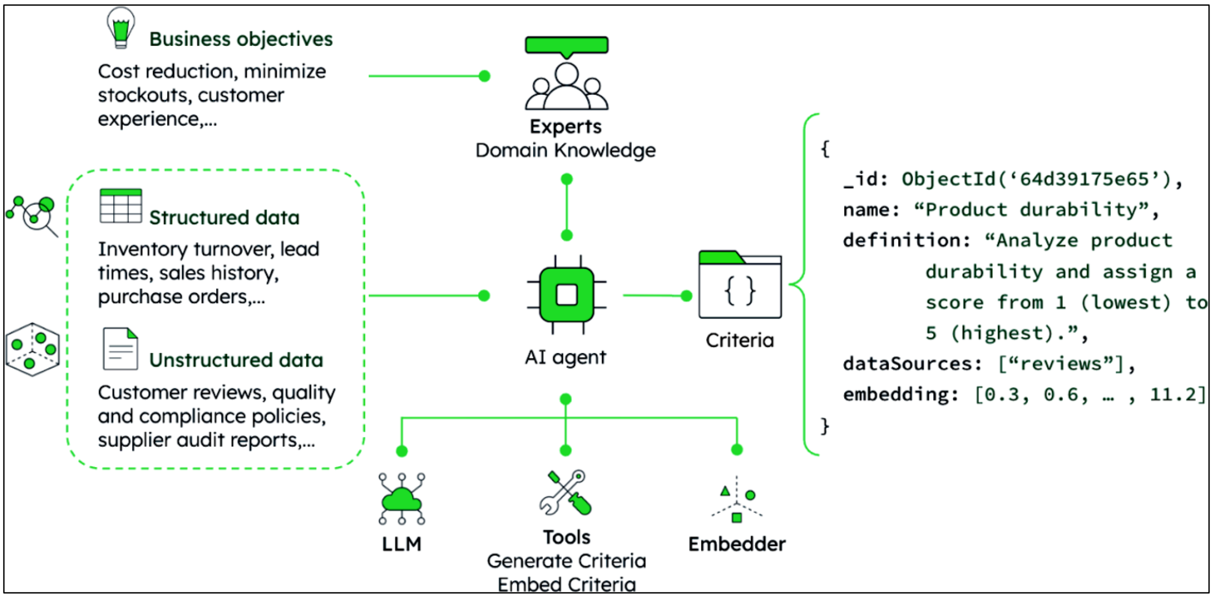


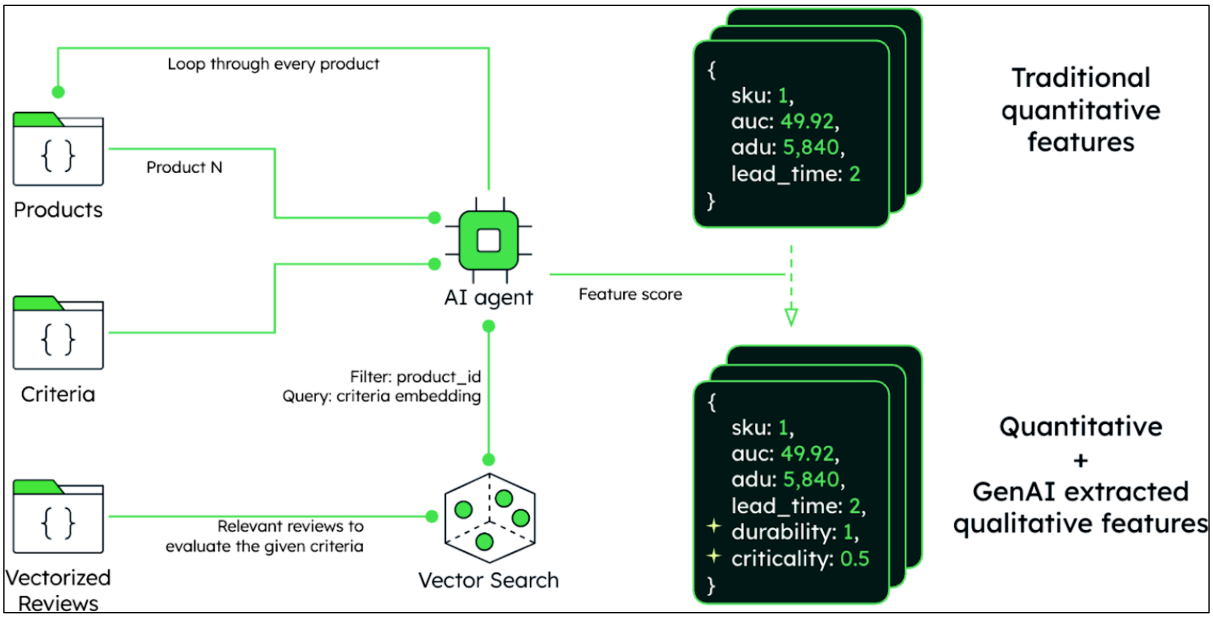



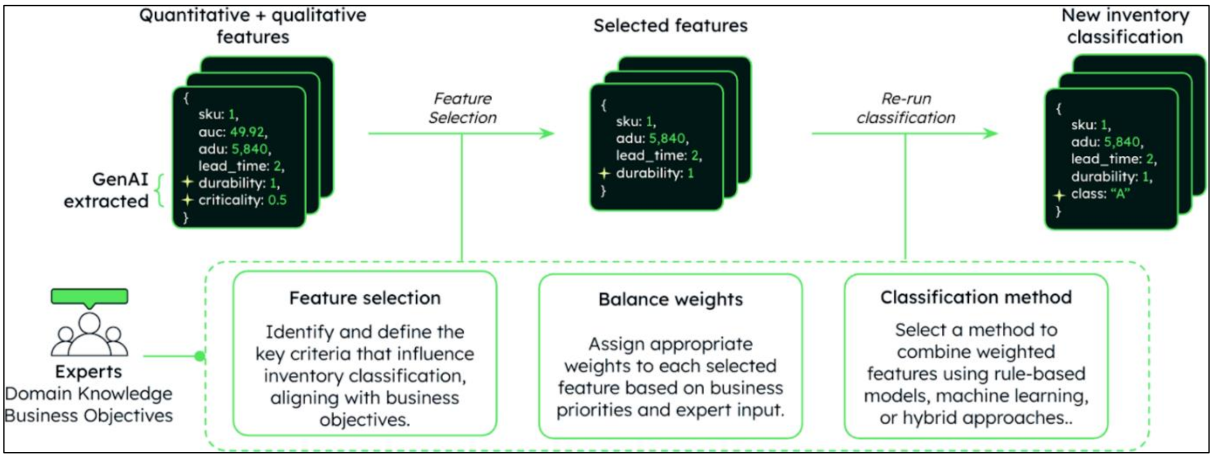




Product Reviews History
Text/Images
Embedder
Vectors
{ }
Vectorized Reviews Collection
{
_id: ObjectId('64d39175e65'),
review: "Excellent product…",
score: 5,
product: {...},
user: {...},
embedding: [0.3, 0.6, … , 11.2]
}

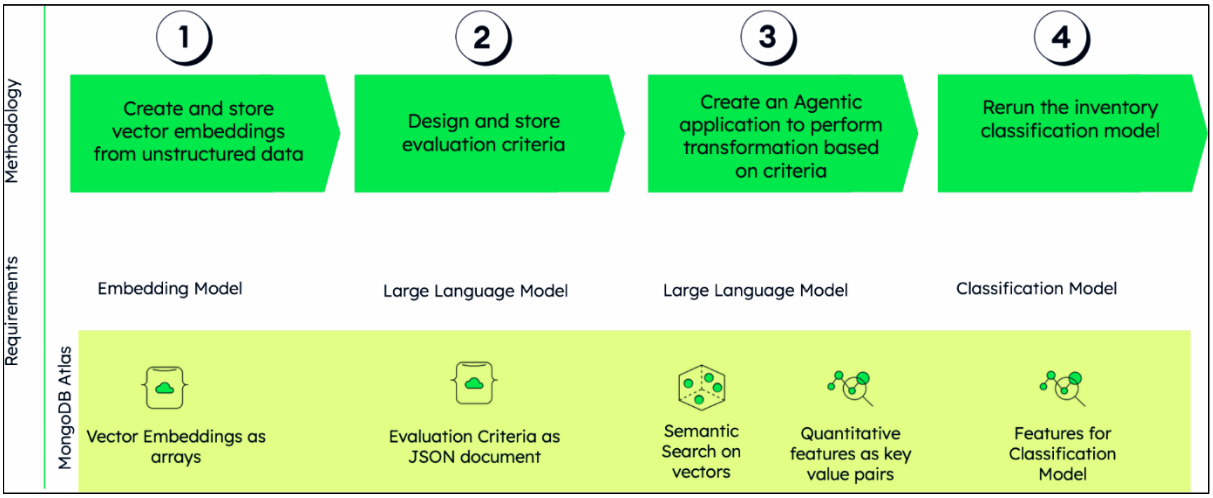

Business objectives
Cost reduction, minimize stockouts, customer experience,...
Experts
Domain Knowledge
Structured data
Inventory turnover, lead times, sales history, purchase orders,...
Unstructured data
Customer reviews, quality and compliance policies, supplier audit reports,...
AI agent
{ }
Criteria
{
_id: ObjectId('64d39175e65'),
name: "Product durability",
definition: "Analyze product durability and assign a score from 1 (lowest) to 5 (highest).",
dataSources: ["reviews"],
embedding: [0.3, 0.6, … , 11.2]
}
LLM
Tools
Generate Criteria
Embed Criteria
Embedder



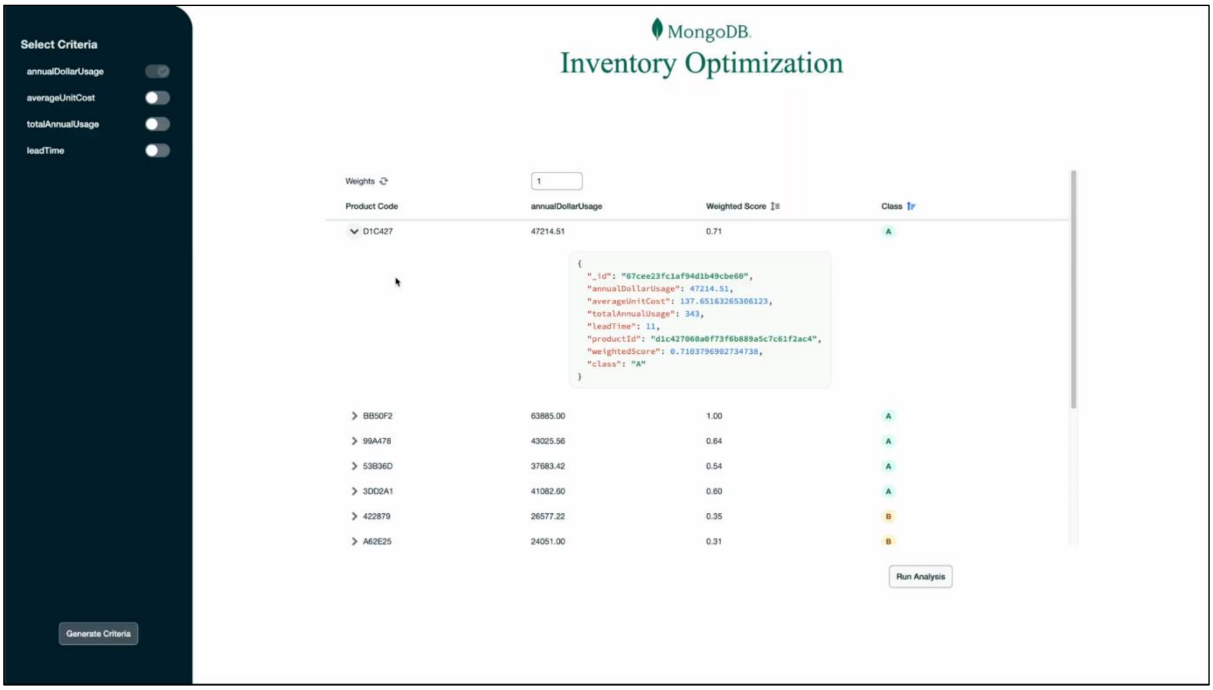
Loop through every product
Product N
Products
Criteria
Vectorized Reviews
AI agent
Feature score
Filter: product_id
Query: criteria embedding
Relevant reviews to evaluate the given criteria
Vector Search
{ sku: 1, auc: 49.92, adu: 5,840, lead_time: 2 }
Traditional quantitative features
{ sku: 1, auc: 49.92, adu: 5,840, lead_time: 2, + durability: 1, + criticality: 0.5 }
Quantitative + GenAI extracted qualitative features

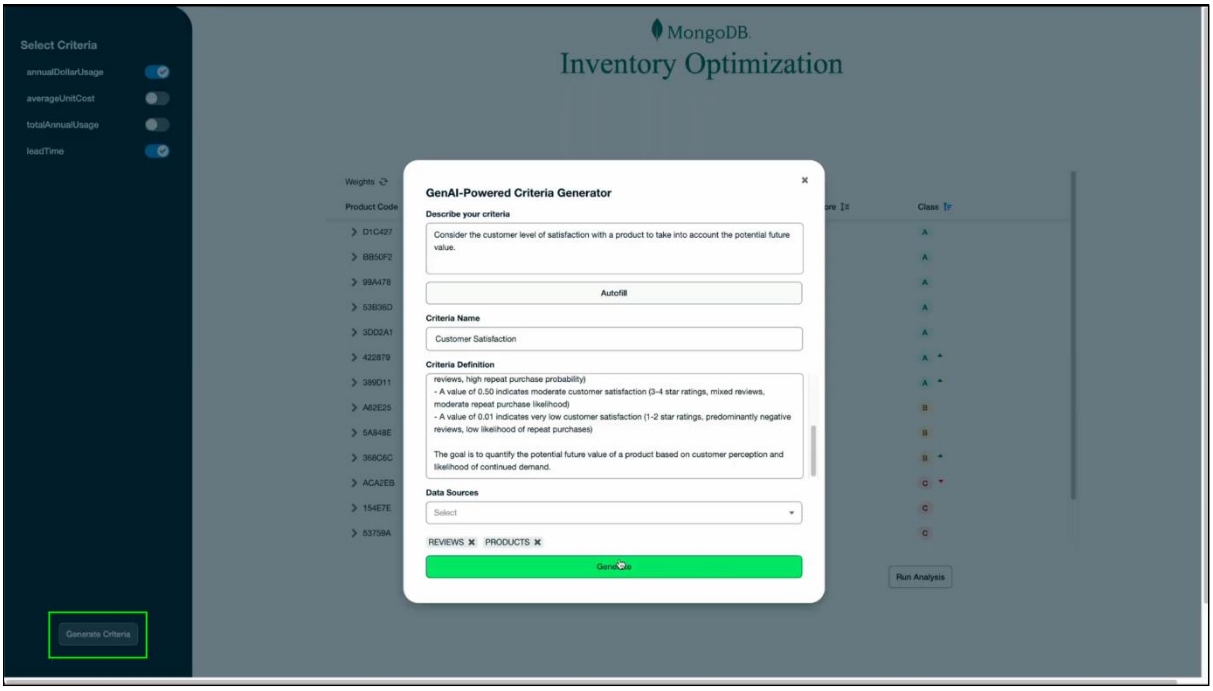



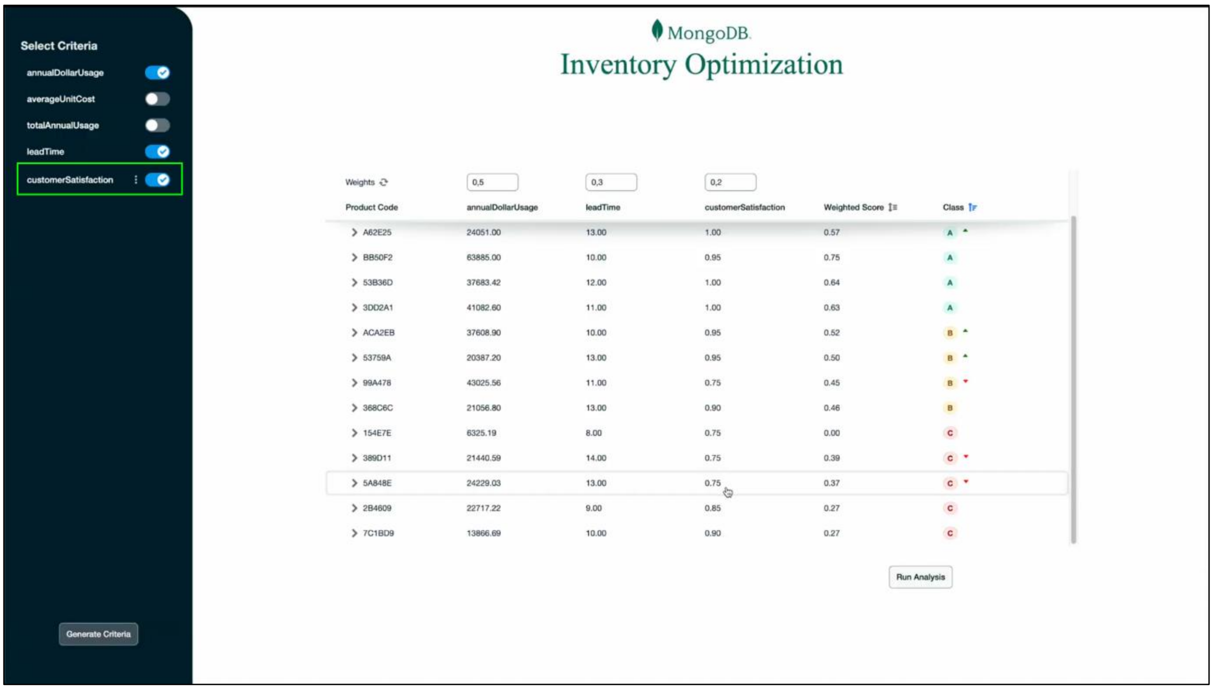
Quantitative + qualitative features
{ sku: 1, auc: 49.92, adu: 5,840, lead_time: 2, + durability: 1, + criticality: 0.5 }
GenAI extracted
Feature Selection
Selected features
{ sku: 1, adu: 5,840, lead_time: 2, + durability: 1 }
Re-run classification
New inventory classification
{ sku: 1, adu: 5,840, lead_time: 2, durability: 1, + class: "A" }
Experts
Domain Knowledge
Business Objectives
Feature selection
Identify and define the key criteria that influence inventory classification, aligning with business objectives.
Balance weights
Assign appropriate weights to each selected feature based on business priorities and expert input.
Classification method
Select a method to combine weighted features using rule-based models, machine learning, or hybrid approaches..

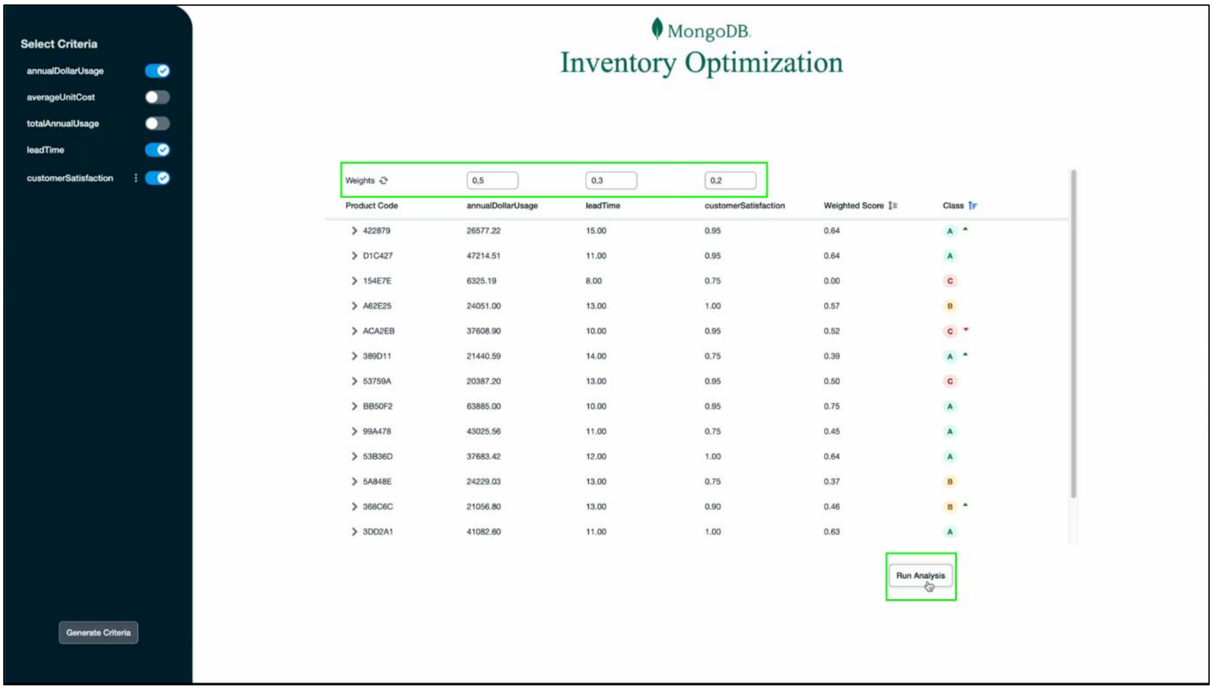



Methodology
Requirements
MongoDB Atlas
1
Create and store vector embeddings from unstructured data
2
Design and store evaluation criteria
3
Create an Agentic application to perform transformation based on criteria
4
Rerun the inventory classification model
Embedding Model
Large Language Model
Large Language Model
Classification Model
Vector Embeddings as arrays
Evaluation Criteria as JSON document
Semantic Search on vectors
Quantitative features as key value pairs
Features for Classification Model

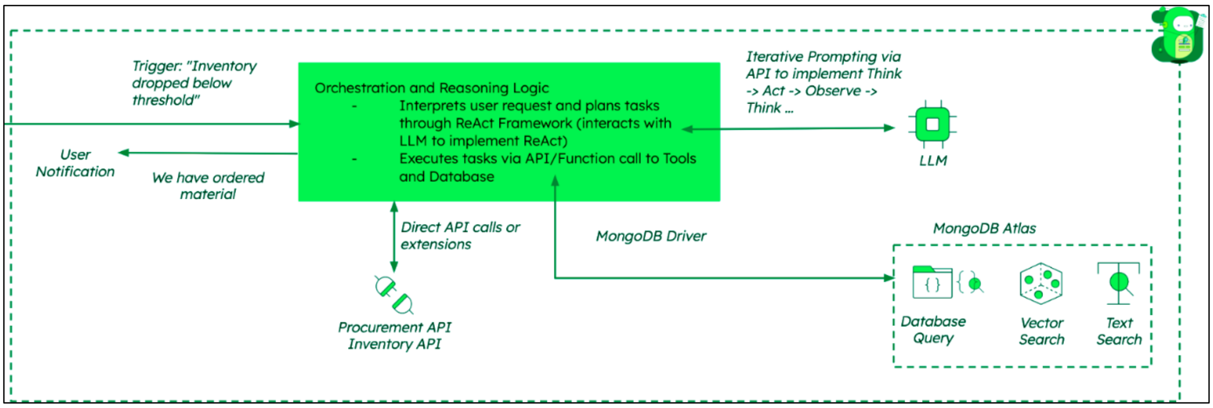

Select Criteria
annualDollarUsage
averageUnitCost
totalAnnualUsage
leadTime
Generate Criteria
MongoDB.
Inventory Optimization
Weights
1
Product Code
annualDollarUsage
Weighted Score
Class
D1C427 47214.51 0.71 A
{
"_id": "67cee23fc1af94d1b49cbe60",
"annualDollarUsage": 47214.51,
"averageUnitCost": 137.65163265306123,
"totalAnnualUsage": 343,
"leadTime": 11,
"productId": "d1c427060a0f73f6b889a5c7c61f2ac4",
"weightedScore": 0.7103796902734738,
"class": "A"
}
BB50F2 63885.00 1.00 A
99A478 43025.56 0.64 A
53B36D 37683.42 0.54 A
3DD2A1 41082.60 0.60 A
422879 26577.22 0.35 B
A62E25 24051.00 0.31 B
Run Analysis

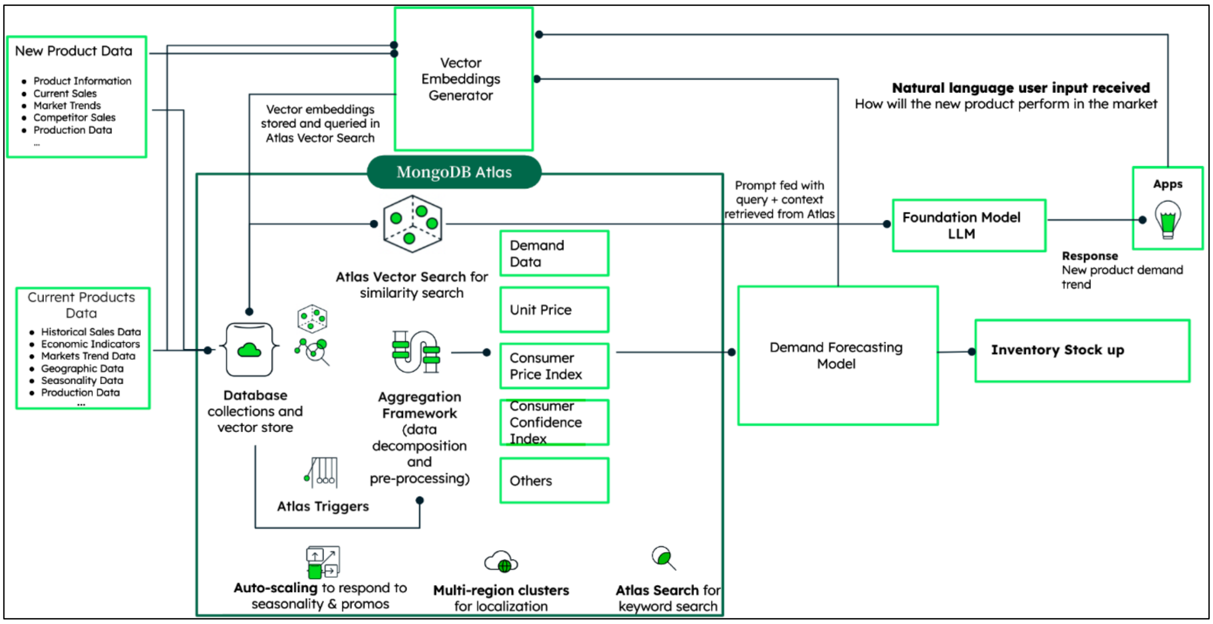

Select Criteria
annualDollarUsage
averageUnitCost
totalAnnualUsage
leadTime
Generate Criteria
MongoDB.
Inventory Optimization
GenAI-Powered Criteria Generator
Describe your criteria
Consider the customer level of satisfaction with a product to take into account the potential future value.
Autofill
Criteria Name
Customer Satisfaction
Criteria Definition
reviews, high repeat purchase probability)
- A value of 0.50 indicates moderate customer satisfaction (3-4 star ratings, mixed reviews, moderate repeat purchase likelihood)
- A value of 0.01 indicates very low customer satisfaction (1-2 star ratings, predominantly negative reviews, low likelihood of repeat purchases)
The goal is to quantify the potential future value of a product based on customer perception and likelihood of continued demand.
Data Sources
Select
REVIEWS
PRODUCTS
Generate
Run Analysis


Select Criteria
annualDollarUsage
averageUnitCost
totalAnnualUsage
leadTime
customerSatisfaction
Generate Criteria
MongoDB.
Inventory Optimization
Weights 0,5 0,3 0,2
Product Code annualDollarUsage leadTime customerSatisfaction Weighted Score Class
A62E25 24051.00 13.00 1.00 0.57 A
BB50F2 63885.00 10.00 0.95 0.75 A
53B36D 37683.42 12.00 1.00 0.64 A
3DD2A1 41082.60 11.00 1.00 0.63 A
ACA2EB 37608.90 10.00 0.95 0.52 B
53759A 20387.20 13.00 0.95 0.50 B
99A478 43025.56 11.00 0.75 0.45 B
368C6C 21056.80 13.00 0.90 0.46 B
154E7E 6325.19 8.00 0.75 0.00 C
389D11 21440.59 14.00 0.75 0.39 C
5A848E 24229.03 13.00 0.75 0.37 C
2B4609 22717.22 9.00 0.85 0.27 C
7C1BD9 13866.69 10.00 0.90 0.27 C
Run Analysis

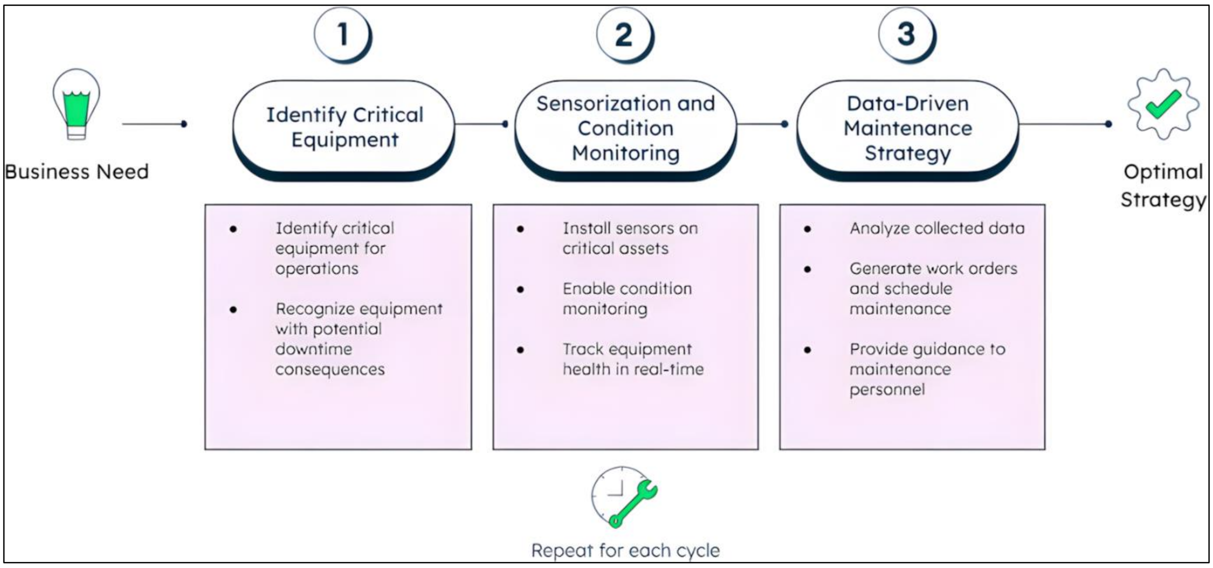

Select Criteria
annualDollarUsage
averageUnitCost
totalAnnualUsage
leadTime
customerSatisfaction
Generate Criteria
MongoDB.
Inventory Optimization
Weights 0,5 0,3 0,2
Product Code annualDollarUsage leadTime customerSatisfaction Weighted Score Class
422879 26577.22 15.00 0.95 0.64 A
D1C427 47214.51 11.00 0.95 0.64 A
154E7E 6325.19 8.00 0.75 0.00 C
A62E25 24051.00 13.00 1.00 0.57 B
ACA2EB 37608.90 10.00 0.95 0.52 C
389D11 21440.59 14.00 0.75 0.39 A
53759A 20387.20 13.00 0.95 0.50 C
BB50F2 63885.00 10.00 0.95 0.75 A
99A478 43025.56 11.00 0.75 0.45 A
53B36D 37683.42 12.00 1.00 0.64 A
5A848E 24229.03 13.00 0.75 0.37 B
368C6C 21056.80 13.00 0.90 0.46 B
3DD2A1 41082.60 11.00 1.00 0.63 A
Run Analysis

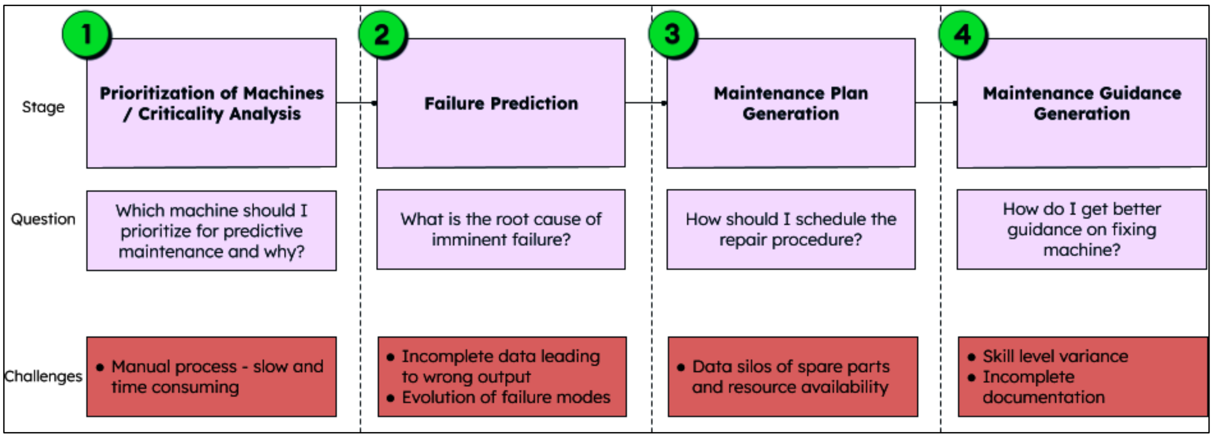

Orchestration - Thought -> Action -> Observation -> Thought ...
Input Triggers
Inventory Drops Below Threshold
New Production Order is Created
Supplier Shipment is Delayed
Price Volatility is Detected
Thought: "Check if we have enough raw materials for the next {user_setting} weeks of production"
Action: Query SAP ERP / Material Inventory DB for real-time stock levels.
Observation: "API X stock is low, requires reorder within 7 days to maintain production"
Thought: "What is the supplier lead time and historical performance for API X?"
Action: Query Supplier Database for historical order lead times and reliability..
Observation: "The usual supplier takes 15 days, but last quarter, they had a 20% delay rate."
Thought: "find an alternative supplier who can deliver within 10 days"
Action: Use Atlas Hybrid Search to find the best matching supplier based on lead time, price, reliability and past performance.
Observation: "Supplier B can deliver in 9 days at a slightly higher cost (+3%)"
Thought: "To avoid production delays, we will split the order: 70% from Supplier A, 30% from Supplier B"
Action: Call Procurement API to generate POs.
Data
Agent Profile
A Raw Material AI Procurement Agent
Instructions and Goals
Ensure just-in-time (JIT) raw material availability while minimizing costs and storage overhead
Memory
Short Term: Live stock levels for raw materials, recent orders, supplier lead times, shipment info
Long Term: production consumption patterns, supplier performance, quality issues, cost
Tools
[Atlas Database Search and Query] [Procurement API] [Atlas Search] [Inventory API]
Models
[Language Model]

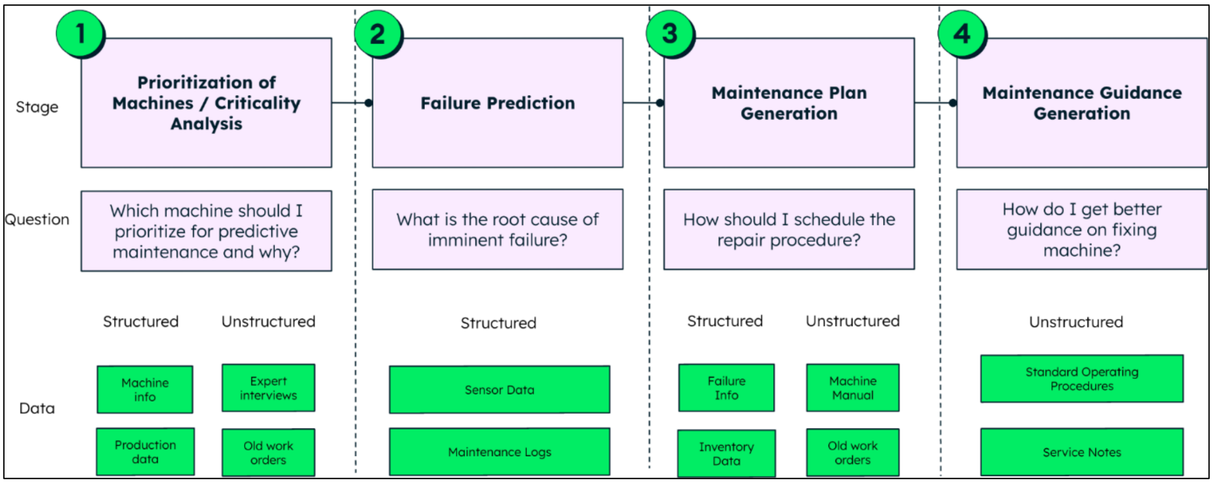

Trigger: "Inventory dropped below threshold"
User Notification
We have ordered material
Orchestration and Reasoning Logic
- Interprets user request and plans tasks through ReAct Framework (interacts with LLM to implement ReAct)
- Executes tasks via API/Function call to Tools and Database
Iterative Prompting via API to implement Think -> Act -> Observe -> Think ...
LLM
Direct API calls or extensions
MongoDB Driver
MongoDB Atlas
Procurement API
Inventory API
Database Query
Vector Search
Text Search

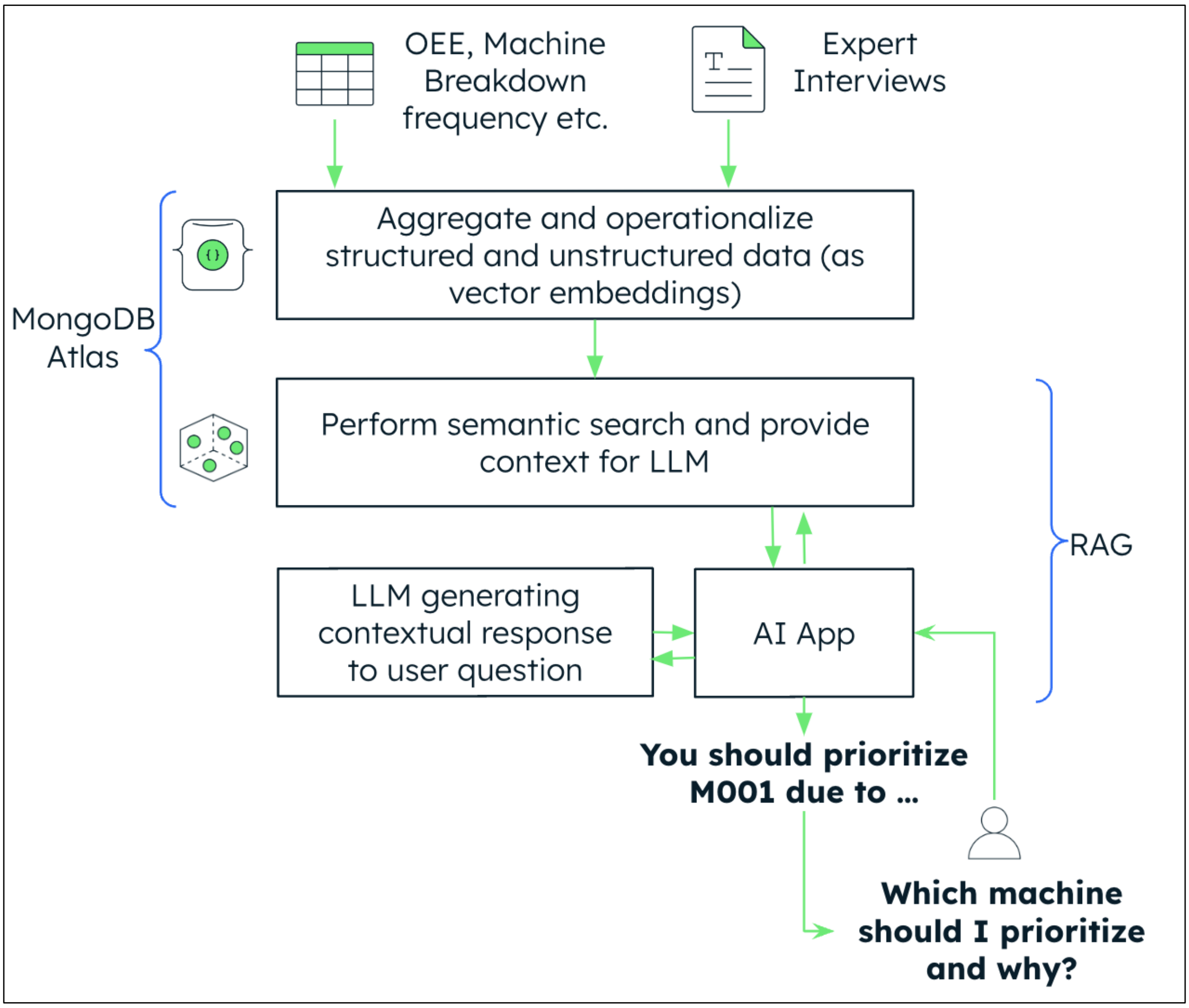

New Product Data
Product Information
Current Sales
Market Trends
Competitor Sales
Production Data
...
Vector Embeddings Generator
Vector embeddings stored and queried in Atlas Vector Search
Natural language user input received
How will the new product perform in the market
MongoDB Atlas
Prompt fed with query + context retrieved from Atlas
Foundation Model LLM
Apps
Response
New product demand trend
Atlas Vector Search for similarity search
Demand Data
Unit Price
Consumer Price Index
Consumer Confidence Index
Others
Current Products Data
Historical Sales Data
Economic Indicators
Markets Trend Data
Geographic Data
Seasonality Data
Production Data
...
Database collections and vector store
Aggregation Framework (data decomposition and pre-processing)
Atlas Triggers
Demand Forecasting Model
Inventory Stock up
Auto-scaling to respond to seasonality & promos
Multi-region clusters for localization
Atlas Search for keyword search

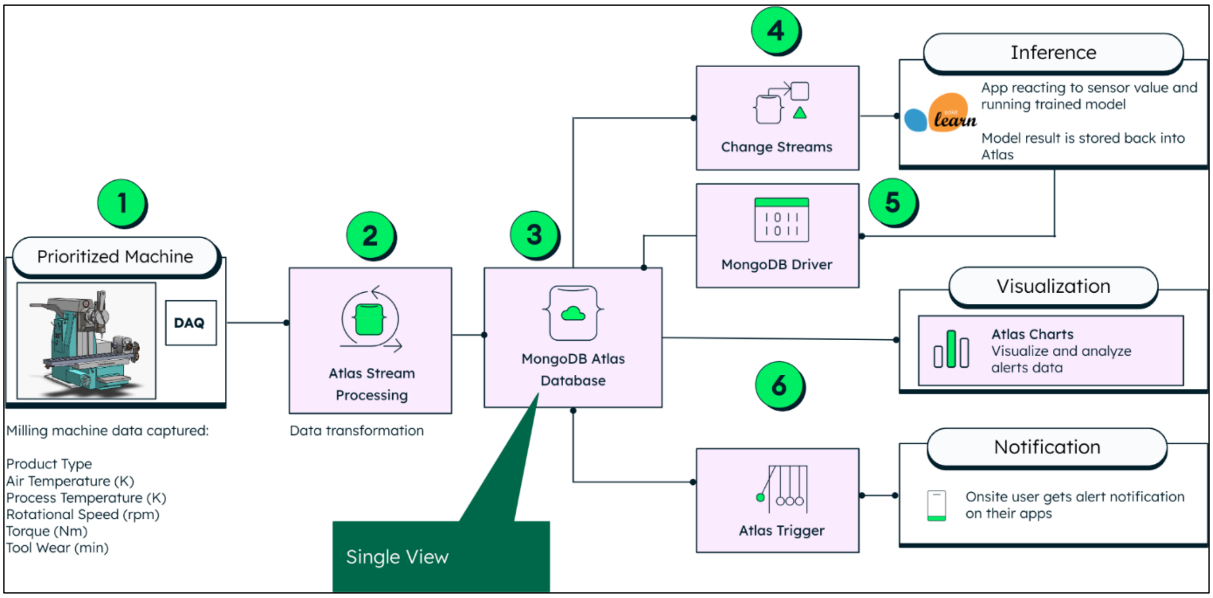
# Chapter 7: Practical Applications of Agentic and GenAI in Manufacturing – Part 2

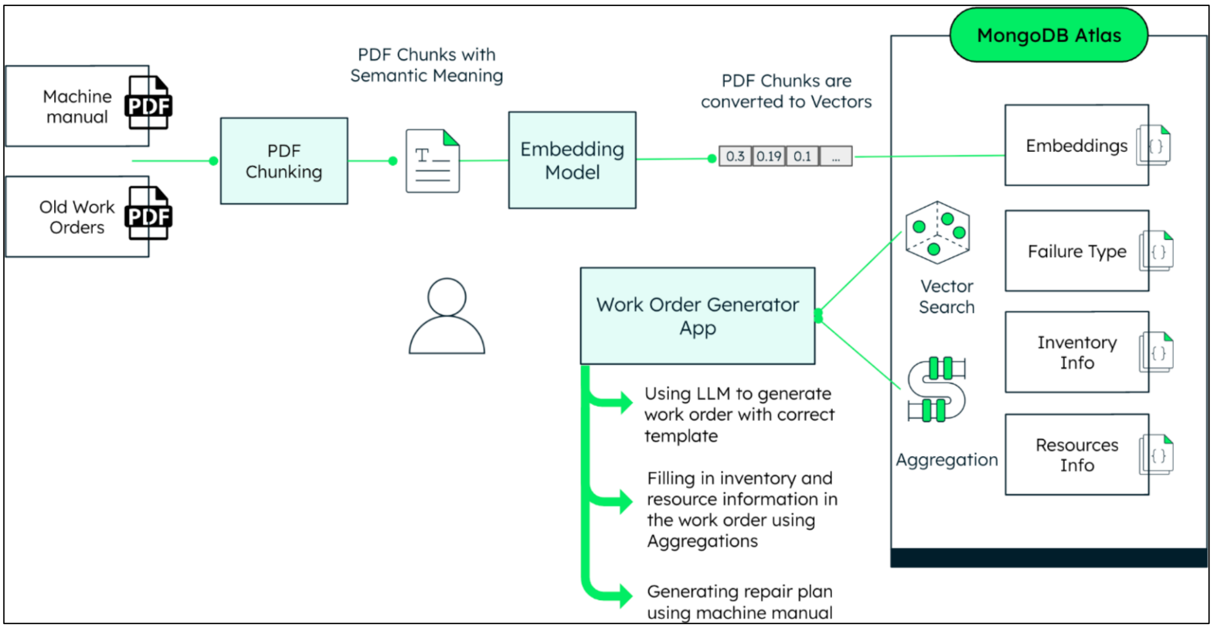


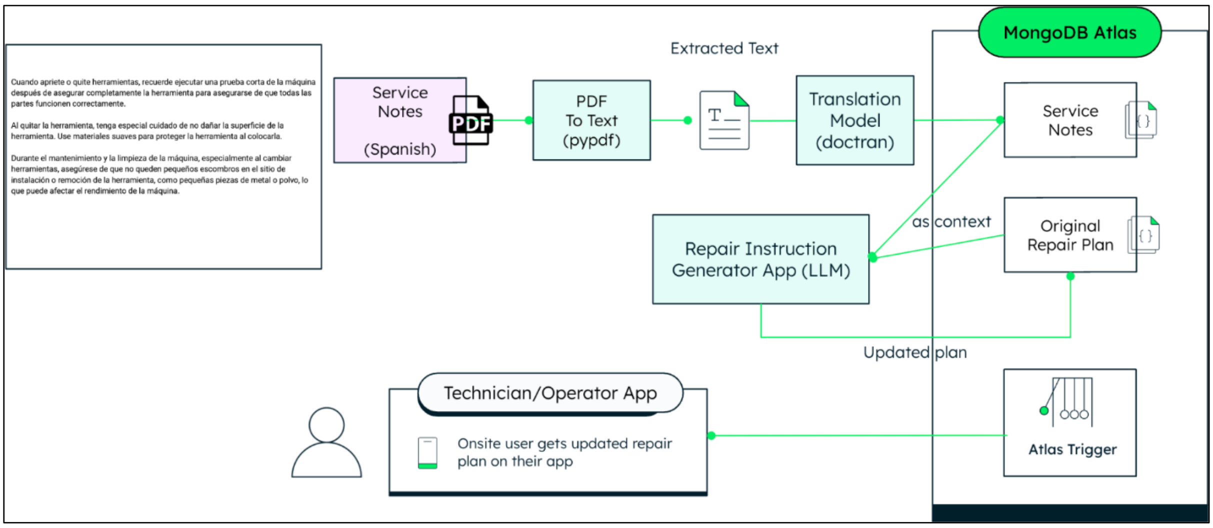




OEE, Machine Breakdown frequency etc.
Expert Interviews
MongoDB Atlas
Aggregate and operationalize structured and unstructured data (as vector embeddings)
Perform semantic search and provide context for LLM
RAG
LLM generating contextual response to user question
AI App
You should prioritize M001 due to ...
Which machine should I prioritize and why?

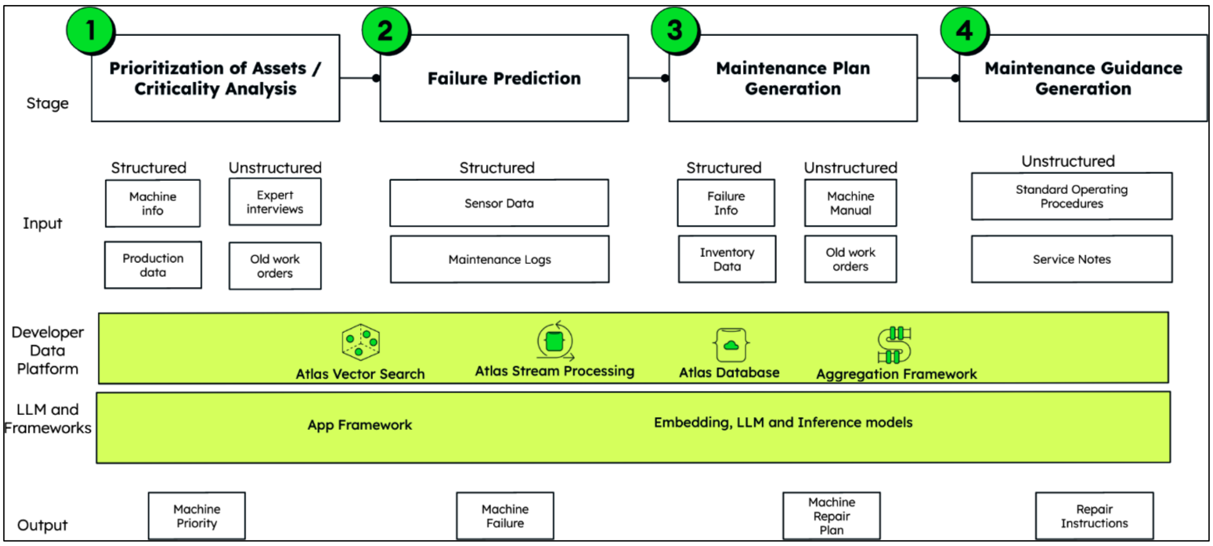

1
Prioritized Machine
DAQ
Milling machine data captured:
Product Type
Air Temperature (K)
Process Temperature (K)
Rotational Speed (rpm)
Torque (Nm)
Tool Wear (min)
2
Atlas Stream Processing
Data transformation
3
MongoDB Atlas Database
Single View
4
Change Streams
5
MongoDB Driver
6
Atlas Trigger
Inference
App reacting to sensor value and running trained model
Model result is stored back into Atlas
Visualization
Atlas Charts
Visualize and analyze alerts data
Notification
Onsite user gets alert notification on their apps



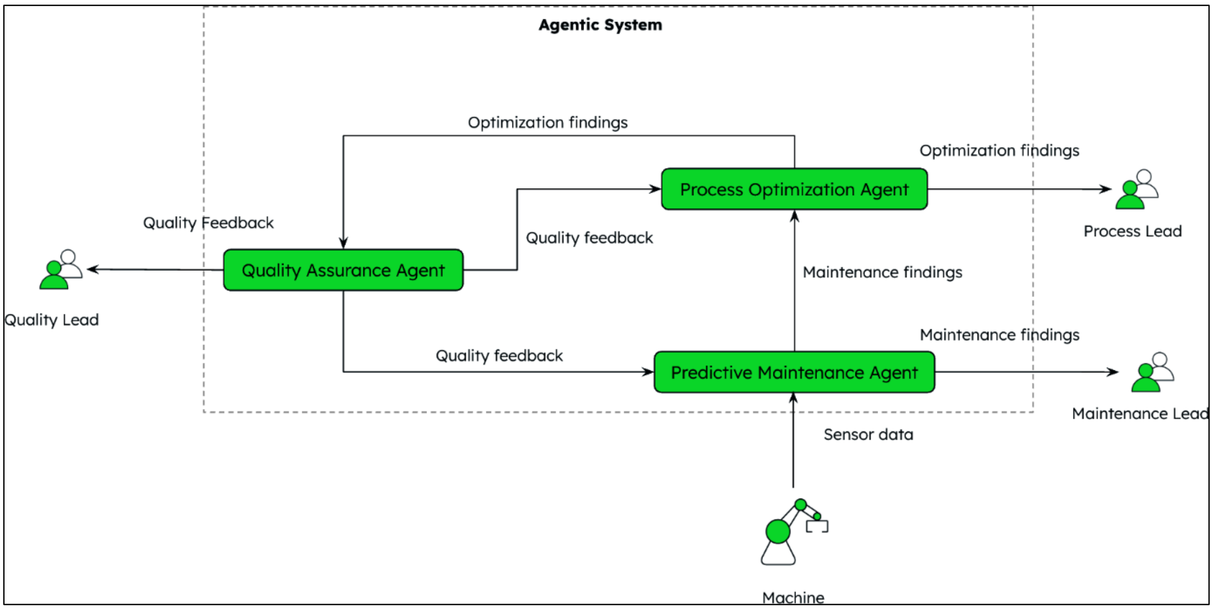
MongoDB Atlas
Machine manual
PDF
Old Work Orders
PDF
PDF Chunks with Semantic Meaning
PDF Chunking
Embedding Model
PDF Chunks are converted to Vectors
0.3
0.19
0.1
...
Embeddings
Failure Type
Inventory Info
Resources Info
Vector Search
Aggregation
Work Order Generator App
Using LLM to generate work order with correct template
Filling in inventory and resource information in the work order using Aggregations
Generating repair plan using machine manual

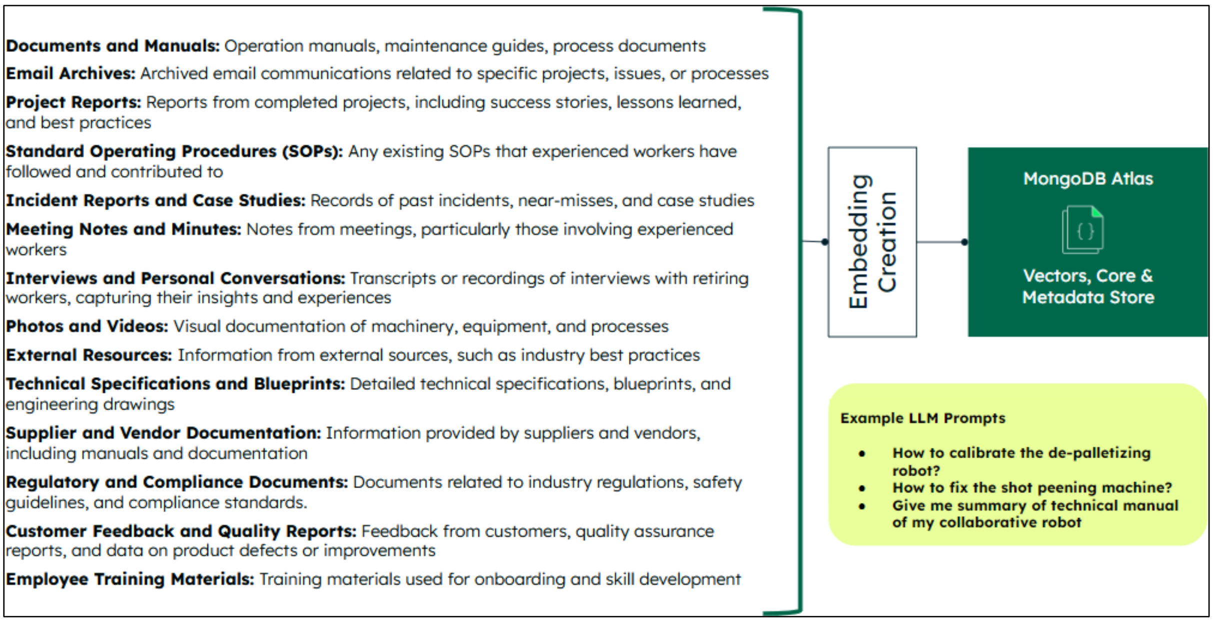



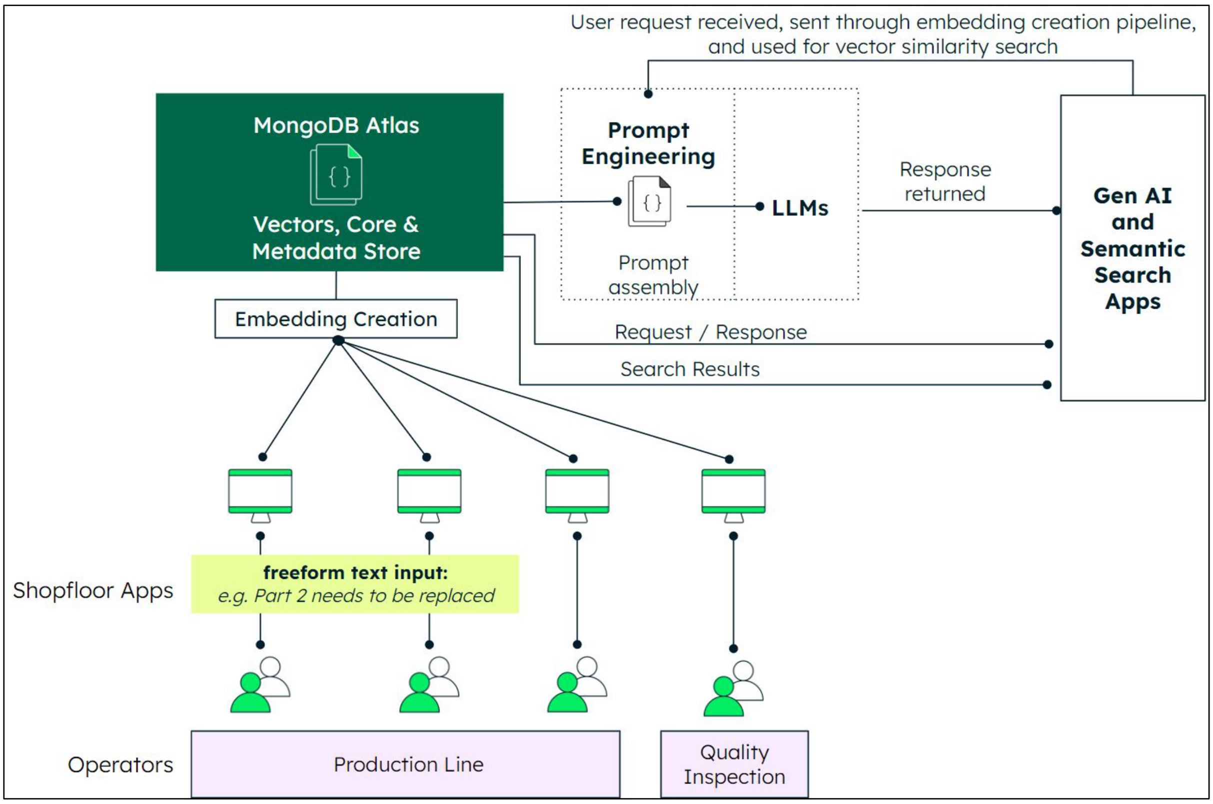
MongoDB Atlas
Cuando apriete o quite herramientas, recuerde ejecutar una prueba corta de la máquina después de asegurar completamente la herramienta para asegurarse de que todas las partes funcionen correctamente.
Al quitar la herramienta, tenga especial cuidado de no dañar la superficie de la herramienta. Use materiales suaves para proteger la herramienta al colocarla.
Durante el mantenimiento y la limpieza de la máquina, especialmente al cambiar herramientas, asegúrese de que no queden pequeños escombros en el sitio de instalación o remoción de la herramienta, como pequeñas piezas de metal o polvo, lo que puede afectar el rendimiento de la máquina.
Service Notes (Spanish)
PDF
PDF To Text (pypdf)
Extracted Text
Translation Model (doctran)
Service Notes
Original Repair Plan
as context
Repair Instruction Generator App (LLM)
Updated plan
Technician/Operator App
Onsite user gets updated repair plan on their app
Atlas Trigger

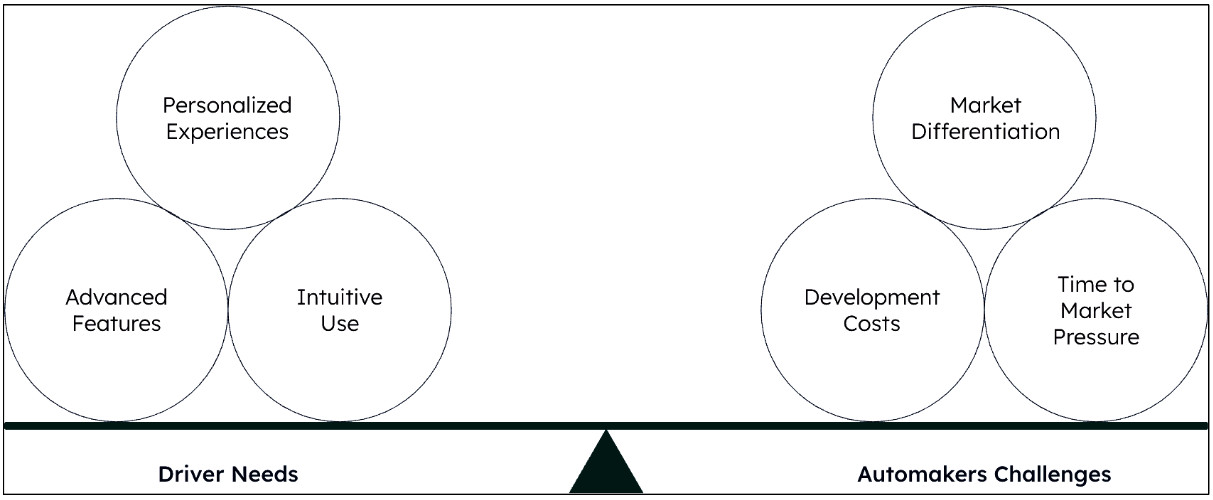



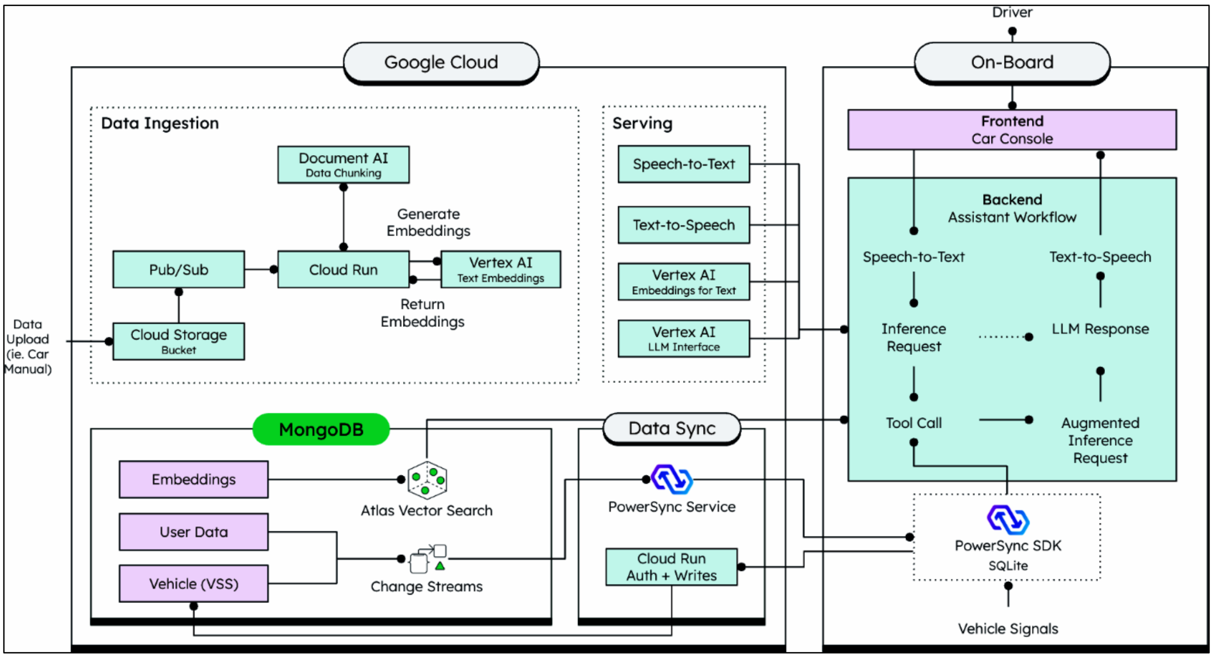
Stage
1 Prioritization of Assets / Criticality Analysis
2 Failure Prediction
3 Maintenance Plan Generation
4 Maintenance Guidance Generation
Input
Structured
Machine info
Production data
Unstructured
Expert interviews
Old work orders
Structured
Sensor Data
Maintenance Logs
Structured
Failure Info
Inventory Data
Unstructured
Machine Manual
Old work orders
Unstructured
Standard Operating Procedures
Service Notes
Developer Data Platform
Atlas Vector Search
Atlas Stream Processing
Atlas Database
Aggregation Framework
LLM and Frameworks
App Framework
Embedding, LLM and Inference models
Output
Machine Priority
Machine Failure
Machine Repair Plan
Repair Instructions

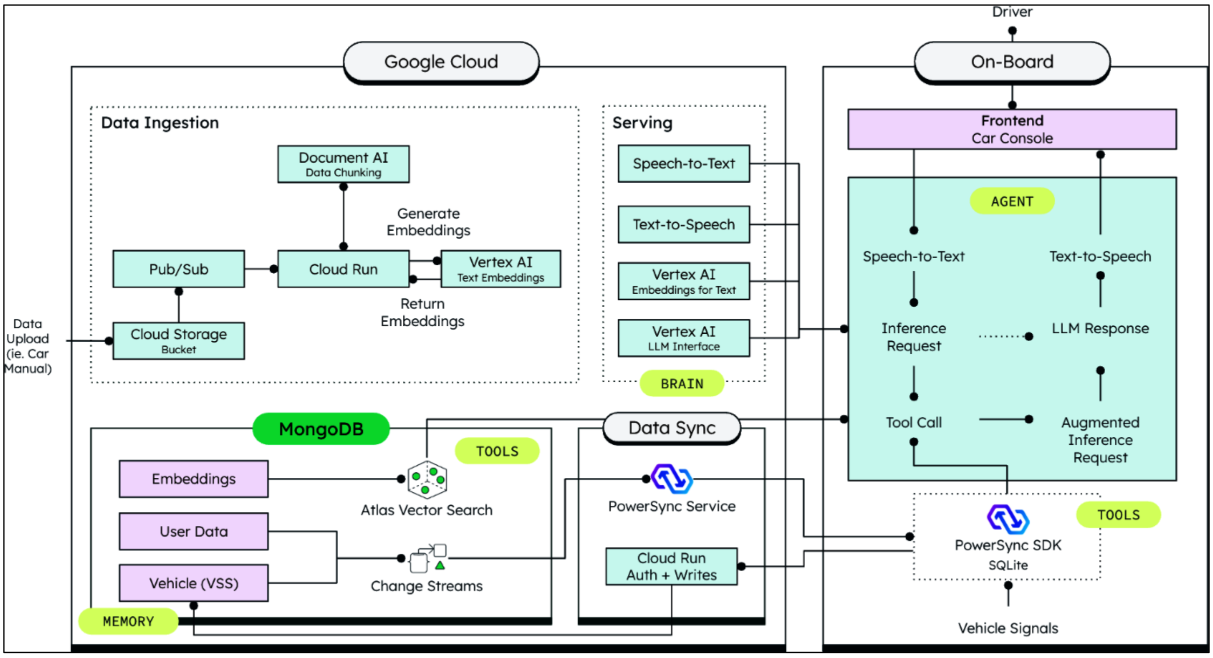

Agentic System
Optimization findings
Optimization findings
Process Optimization Agent
Process Lead
Quality Feedback
Quality feedback
Quality Assurance Agent
Quality Lead
Maintenance findings
Maintenance findings
Quality feedback
Predictive Maintenance Agent
Maintenance Lead
Sensor data
Machine

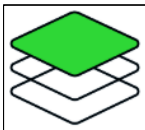

**Documents and Manuals:** Operation manuals, maintenance guides, process documents
**Email Archives:** Archived email communications related to specific projects, issues, or processes
**Project Reports:** Reports from completed projects, including success stories, lessons learned, and best practices
**Standard Operating Procedures (SOPs):** Any existing SOPs that experienced workers have followed and contributed to
**Incident Reports and Case Studies:** Records of past incidents, near-misses, and case studies
**Meeting Notes and Minutes:** Notes from meetings, particularly those involving experienced workers
**Interviews and Personal Conversations:** Transcripts or recordings of interviews with retiring workers, capturing their insights and experiences
**Photos and Videos:** Visual documentation of machinery, equipment, and processes
**External Resources:** Information from external sources, such as industry best practices
**Technical Specifications and Blueprints:** Detailed technical specifications, blueprints, and engineering drawings
**Supplier and Vendor Documentation:** Information provided by suppliers and vendors, including manuals and documentation
**Regulatory and Compliance Documents:** Documents related to industry regulations, safety guidelines, and compliance standards.
**Customer Feedback and Quality Reports:** Feedback from customers, quality assurance reports, and data on product defects or improvements
**Employee Training Materials:** Training materials used for onboarding and skill development
Embedding Creation
MongoDB Atlas
Vectors, Core & Metadata Store
Example LLM Prompts
How to calibrate the de-palletizing robot?
How to fix the shot peening machine?
Give me summary of technical manual of my collaborative robot


User request received, sent through embedding creation pipeline, and used for vector similarity search
MongoDB Atlas
Vectors, Core & Metadata Store
Prompt Engineering
Prompt assembly
LLMs
Response returned
Gen AI and Semantic Search Apps
Embedding Creation
Request / Response
Search Results
Shopfloor Apps
freeform text input:
e.g. Part 2 needs to be replaced
Operators
Production Line
Quality Inspection


Personalized Experiences
Advanced Features
Intuitive Use
Market Differentiation
Development Costs
Time to Market Pressure
Driver Needs
Automakers Challenges

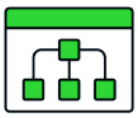


## Fragmentation

Current RAG implementations fragment technical documentation into isolated pieces



## Linkages

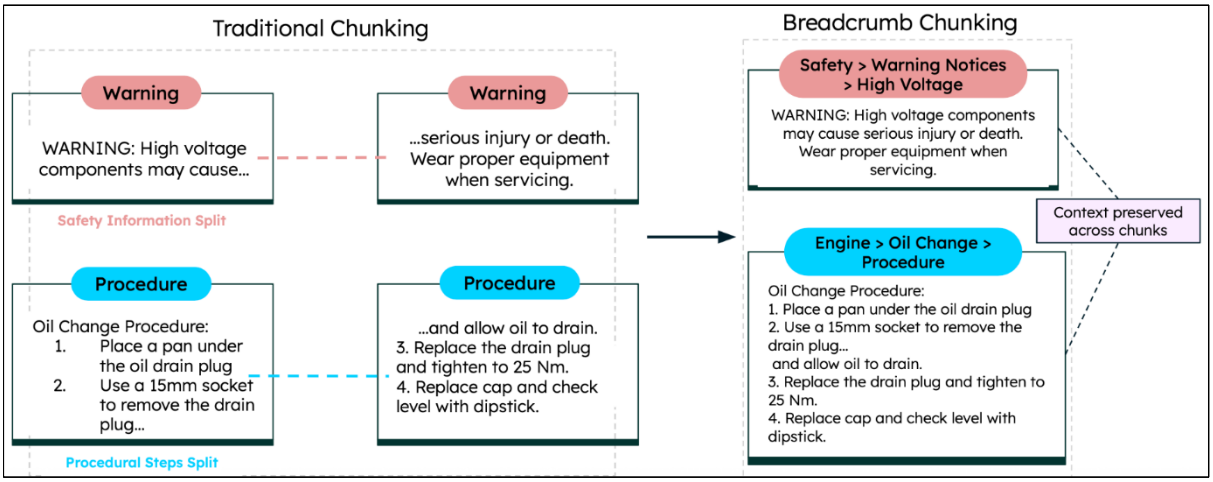
Technical manuals contain deeply interconnected information (procedures, warnings, specifications)

## Risks

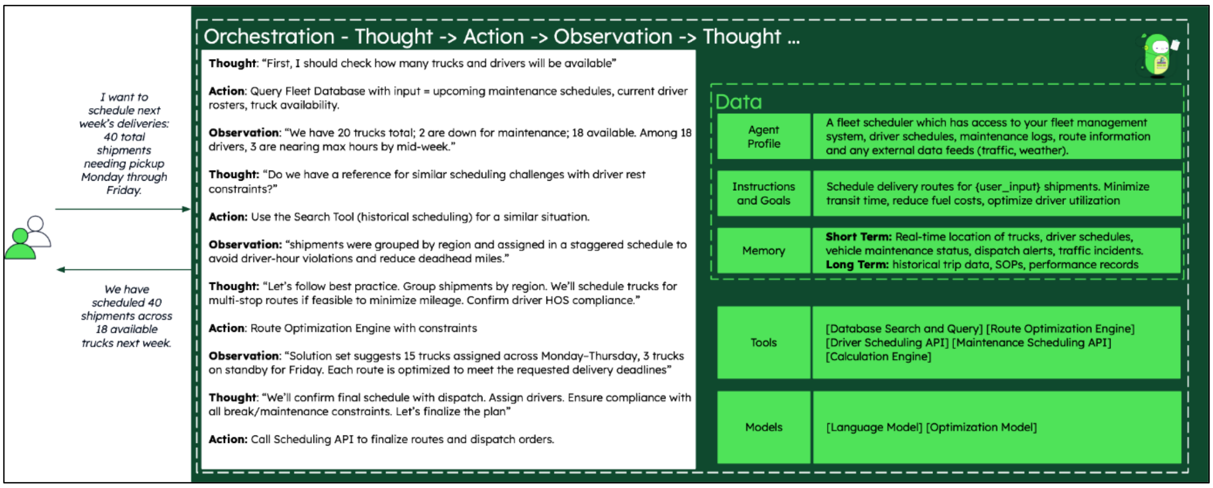
Without context preservation, retrieval produces technically incomplete or unsafe answers

## Chunks!

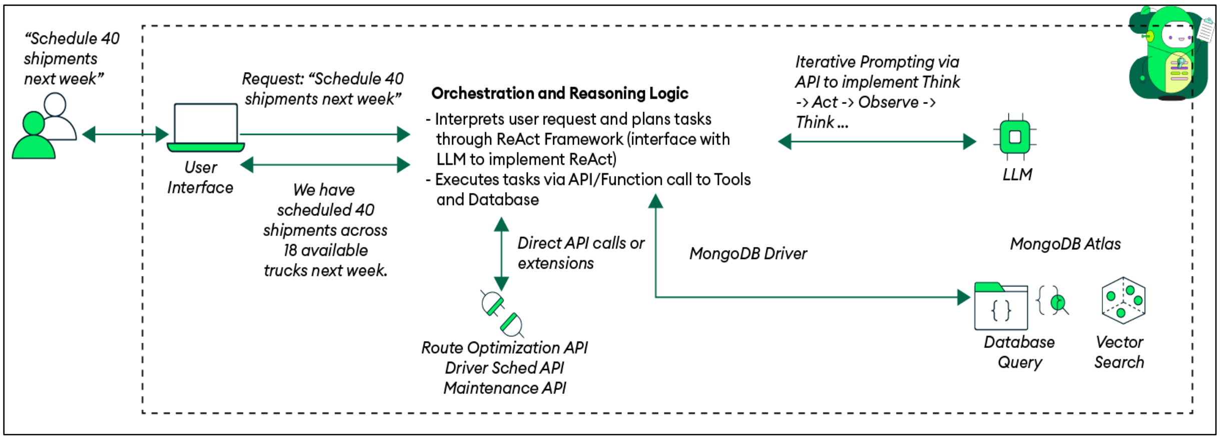
Chunks are the **foundation** of RAG - their quality determines the quality of responses



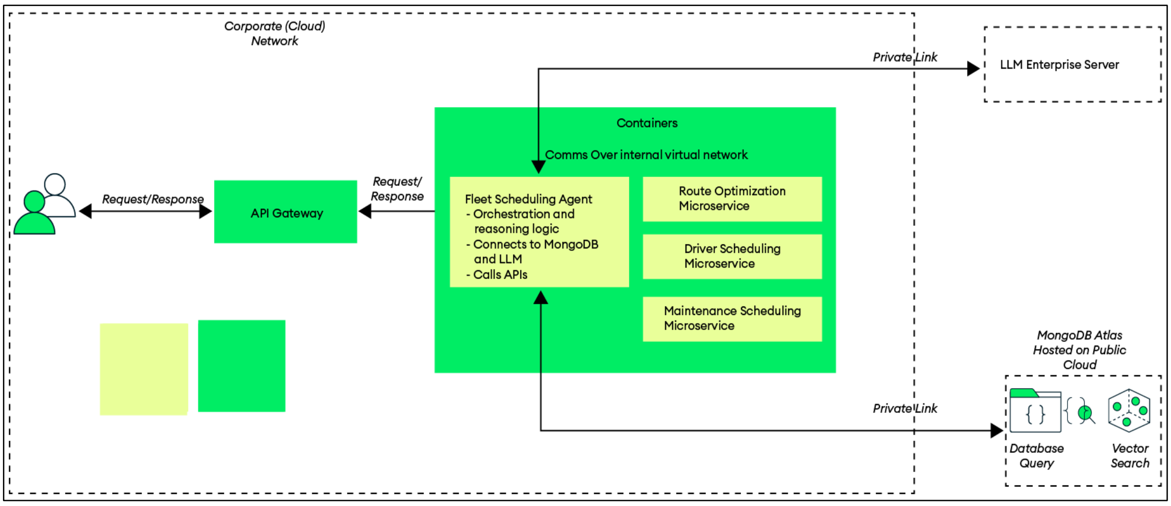
Traditional Chunking
Warning
WARNING: High voltage components may cause...
Warning
...serious injury or death. Wear proper equipment when servicing.
Safety Information Split
Procedure
Oil Change Procedure:
1. Place a pan under the oil drain plug
2. Use a 15mm socket to remove the drain plug...
Procedure
...and allow oil to drain.
3. Replace the drain plug and tighten to 25 Nm.
4. Replace cap and check level with dipstick.
Procedural Steps Split
Breadcrumb Chunking
Safety > Warning Notices > High Voltage
WARNING: High voltage components may cause serious injury or death. Wear proper equipment when servicing.
Context preserved across chunks
Engine > Oil Change > Procedure
Oil Change Procedure:
1. Place a pan under the oil drain plug
2. Use a 15mm socket to remove the drain plug...
and allow oil to drain.
3. Replace the drain plug and tighten to 25 Nm.
4. Replace cap and check level with dipstick.

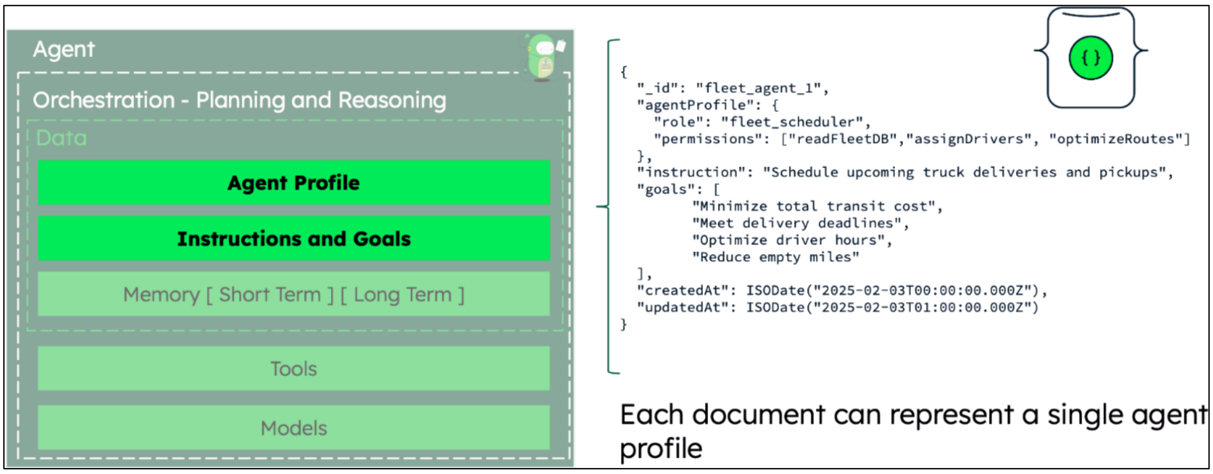



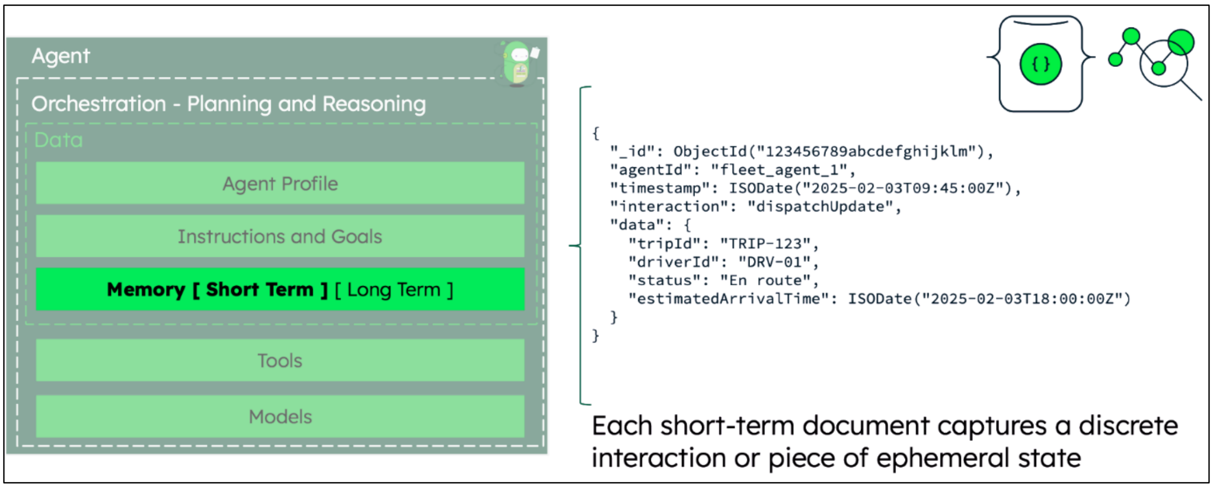
I want to schedule next week's deliveries: 40 total shipments needing pickup Monday through Friday.
We have scheduled 40 shipments across 18 available trucks next week.
Orchestration - Thought -> Action -> Observation -> Thought ...
Thought: "First, I should check how many trucks and drivers will be available"
Action: Query Fleet Database with input = upcoming maintenance schedules, current driver rosters, truck availability.
Observation: "We have 20 trucks total; 2 are down for maintenance; 18 available. Among 18 drivers, 3 are nearing max hours by mid-week."
Thought: "Do we have a reference for similar scheduling challenges with driver rest constraints?"
Action: Use the Search Tool (historical scheduling) for a similar situation.
Observation: "shipments were grouped by region and assigned in a staggered schedule to avoid driver-hour violations and reduce deadhead miles."
Thought: "Let's follow best practice. Group shipments by region. We'll schedule trucks for multi-stop routes if feasible to minimize mileage. Confirm driver HOS compliance."
Action: Route Optimization Engine with constraints
Observation: "Solution set suggests 15 trucks assigned across Monday-Thursday, 3 trucks on standby for Friday. Each route is optimized to meet the requested delivery deadlines"
Thought: "We'll confirm final schedule with dispatch. Assign drivers. Ensure compliance with all break/maintenance constraints. Let's finalize the plan"
Action: Call Scheduling API to finalize routes and dispatch orders.
Data
Agent Profile
A fleet scheduler which has access to your fleet management system, driver schedules, maintenance logs, route information and any external data feeds (traffic, weather).
Instructions and Goals
Schedule delivery routes for {user_input} shipments. Minimize transit time, reduce fuel costs, optimize driver utilization
Memory
Short Term: Real-time location of trucks, driver schedules, vehicle maintenance status, dispatch alerts, traffic incidents.
Long Term: historical trip data, SOPs, performance records
Tools
[Database Search and Query] [Route Optimization Engine] [Driver Scheduling API] [Maintenance Scheduling API] [Calculation Engine]
Models
[Language Model] [Optimization Model]

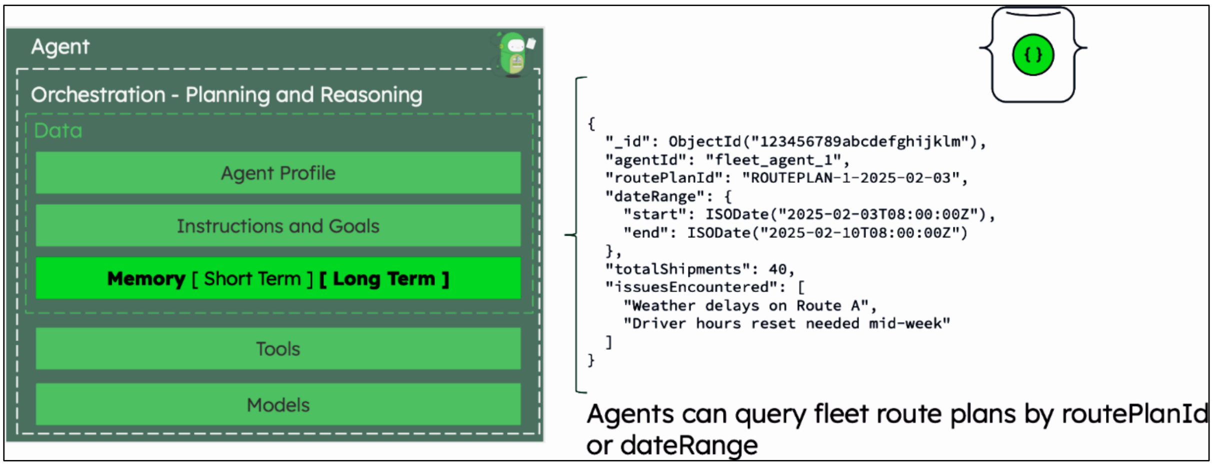



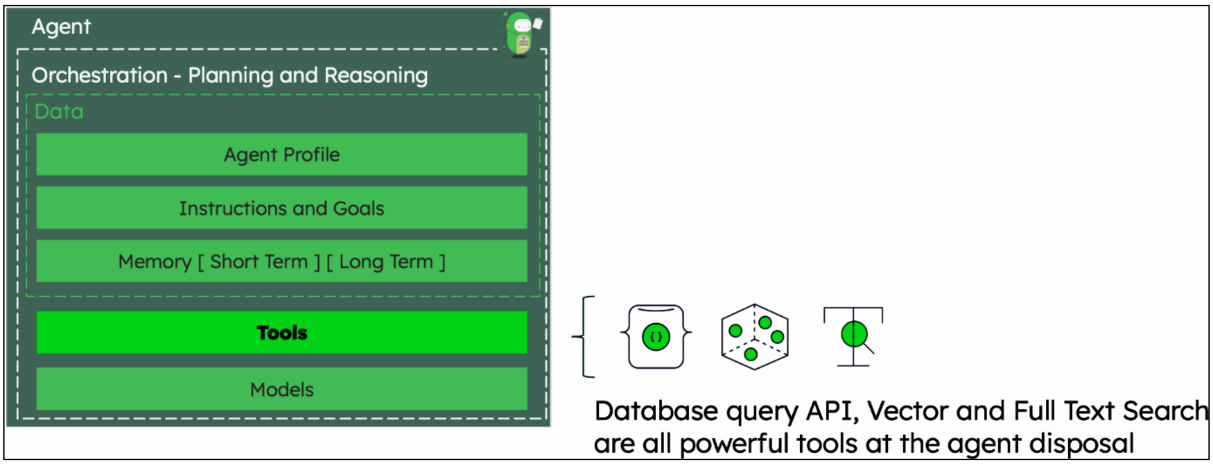
"Schedule 40 shipments next week"
Request: "Schedule 40 shipments next week"
User Interface
We have scheduled 40 shipments across 18 available trucks next week.
Orchestration and Reasoning Logic
- Interprets user request and plans tasks through ReAct Framework (interface with LLM to implement ReAct)
- Executes tasks via API/Function call to Tools and Database
Iterative Prompting via API to implement Think -> Act -> Observe -> Think ...
LLM
Direct API calls or extensions
Route Optimization API
Driver Sched API
Maintenance API
MongoDB Driver
MongoDB Atlas
Database Query
Vector Search

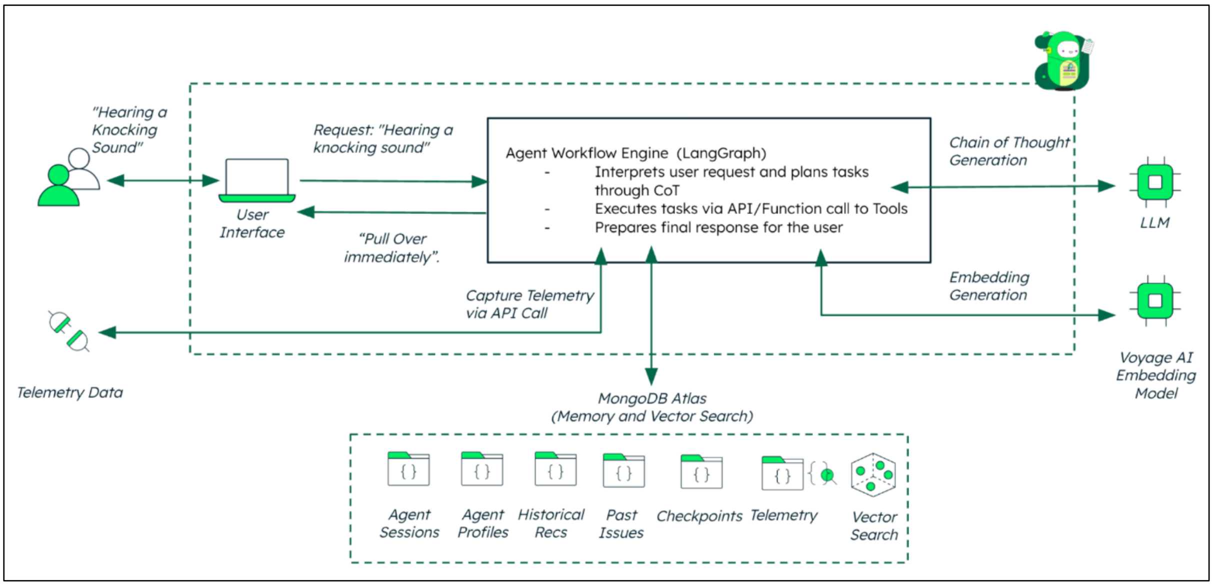

Corporate (Cloud) Network
Private Link
LLM Enterprise Server
Containers
Comms Over internal virtual network
Request/Response
API Gateway
Request/ Response
Fleet Scheduling Agent
- Orchestration and reasoning logic
- Connects to MongoDB and LLM
- Calls APIs
Route Optimization Microservice
Driver Scheduling Microservice
Maintenance Scheduling Microservice
MongoDB Atlas Hosted on Public Cloud
Private Link
Database Query
Vector Search

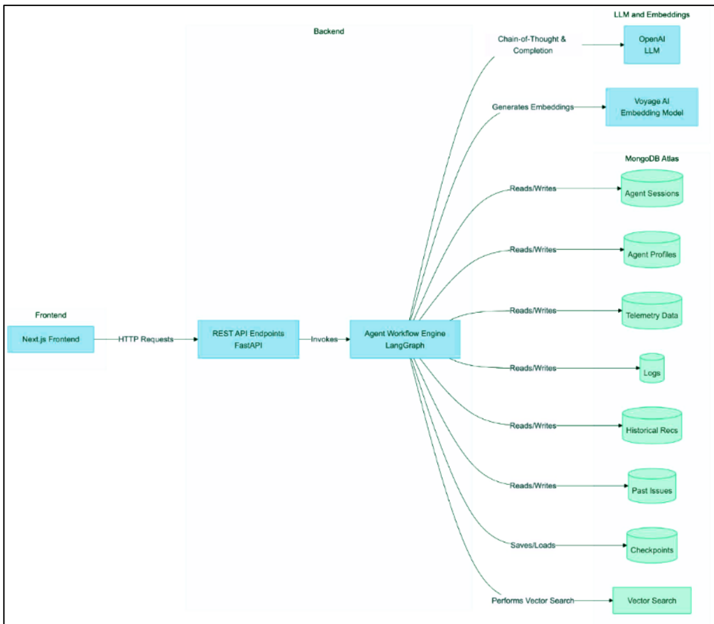

Agent
Orchestration - Planning and Reasoning
Data
Agent Profile
Instructions and Goals
Memory [ Short Term ] [ Long Term ]
Tools
Models
{
"_id": "fleet_agent_1",
"agentProfile": {
"role": "fleet_scheduler",
"permissions": ["readFleetDB","assignDrivers", "optimizeRoutes"]
},
"instruction": "Schedule upcoming truck deliveries and pickups",
"goals": [
"Minimize total transit cost",
"Meet delivery deadlines",
"Optimize driver hours",
"Reduce empty miles"
],
"createdAt": ISODate("2025-02-03T00:00:00.000Z"),
"updatedAt": ISODate("2025-02-03T01:00:00.000Z")
}
Each document can represent a single agent profile


Agent
Orchestration - Planning and Reasoning
Data
Agent Profile
Instructions and Goals
Memory [ Short Term ] [ Long Term ]
Tools
Models
{
"_id": ObjectId("123456789abcdefghijklm"),
"agentId": "fleet_agent_1",
"timestamp": ISODate("2025-02-03T09:45:00Z"),
"interaction": "dispatchUpdate",
"data": {
"tripId": "TRIP-123",
"driverId": "DRV-01",
"status": "En route",
"estimatedArrivalTime": ISODate("2025-02-03T18:00:00Z")
}
}
Each short-term document captures a discrete interaction or piece of ephemeral state


Agent
Orchestration - Planning and Reasoning
Data
Agent Profile
Instructions and Goals
Memory [ Short Term ] [ Long Term ]
Tools
Models
{
"_id": ObjectId("123456789abcdefghijklm"),
"agentId": "fleet_agent_1",
"routePlanId": "ROUTEPLAN-1-2025-02-03",
"dateRange": {
"start": ISODate("2025-02-03T08:00:00Z"),
"end": ISODate("2025-02-10T08:00:00Z")
},
"totalShipments": 40,
"issuesEncountered": [
"Weather delays on Route A",
"Driver hours reset needed mid-week"
]
}
Agents can query fleet route plans by routePlanId or dateRange


Agent
Orchestration - Planning and Reasoning
Data
Agent Profile
Instructions and Goals
Memory [ Short Term ] [ Long Term ]
Tools
Models
Database query API, Vector and Full Text Search are all powerful tools at the agent disposal


"Hearing a Knocking Sound"
User Interface
Request: "Hearing a knocking sound"
"Pull Over immediately".
Agent Workflow Engine (LangGraph)
- Interprets user request and plans tasks through CoT
- Executes tasks via API/Function call to Tools
- Prepares final response for the user
Chain of Thought Generation
LLM
Embedding Generation
Voyage AI Embedding Model
Capture Telemetry via API Call
Telemetry Data
MongoDB Atlas (Memory and Vector Search)
Agent Sessions
Agent Profiles
Historical Recs
Past Issues
Checkpoints
Telemetry
Vector Search

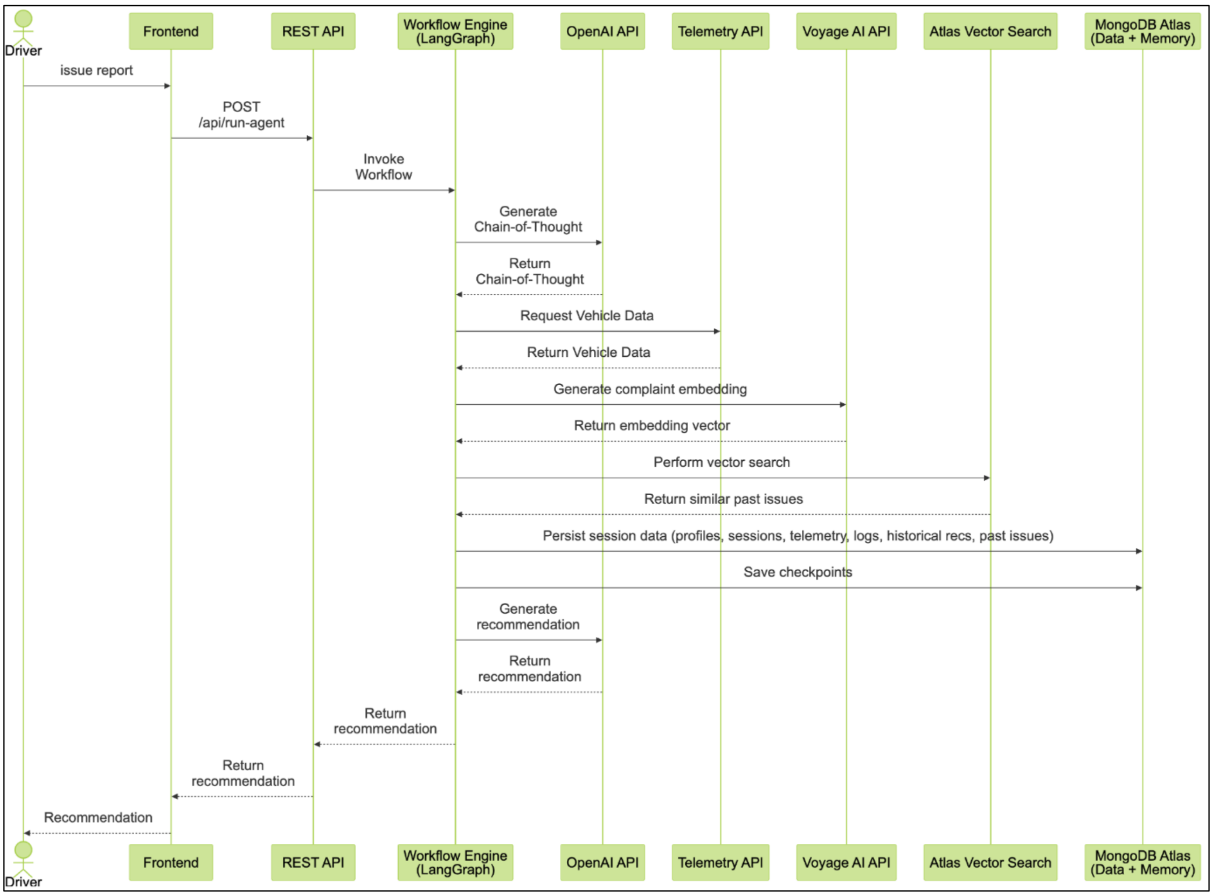



This agent runs a multi-step diagnostic workflow that gets triggered when a driver submits a complaint about the vehicle performance.

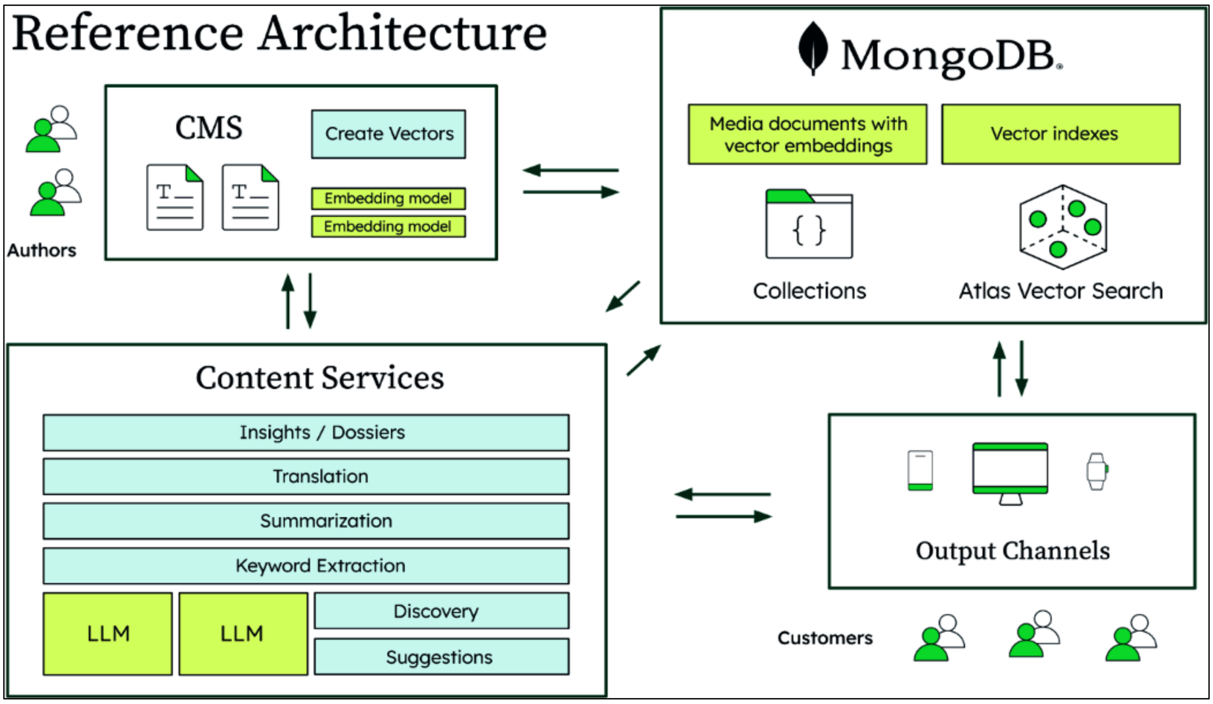
It creates a chain of thought, ingests vehicle data, generates embeddings, and performs a vector search in MongoDB Atlas to find similar past incidents.

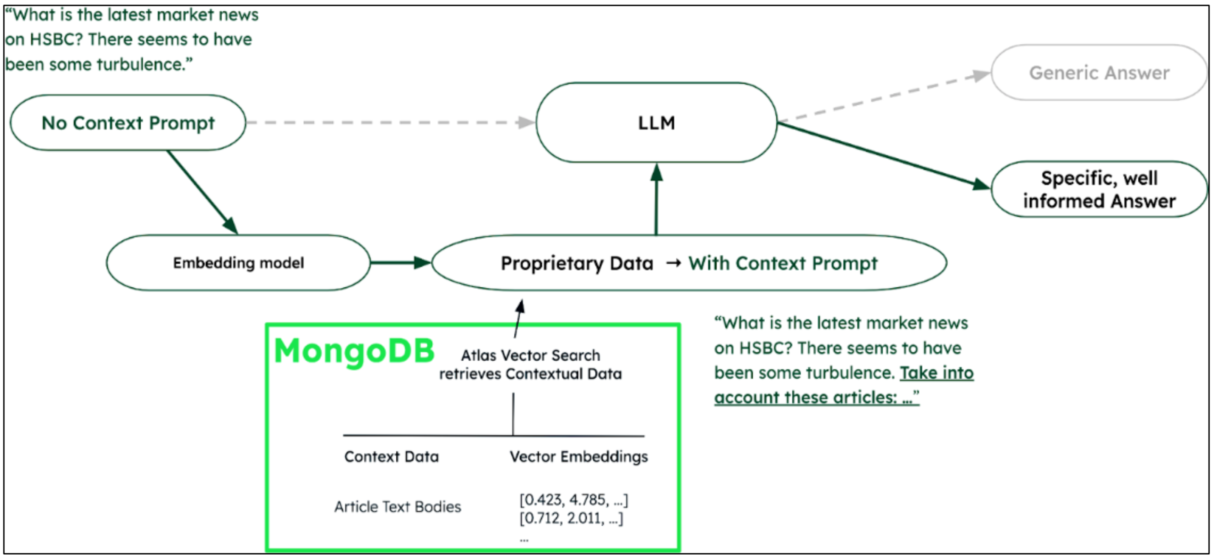
Relevant cases along with telemetry data is passed to OpenAI gpt-4o LLM to generate a final recommendation for the driver (for example, pull off immediately or keep driving and schedule a regular maintenance).

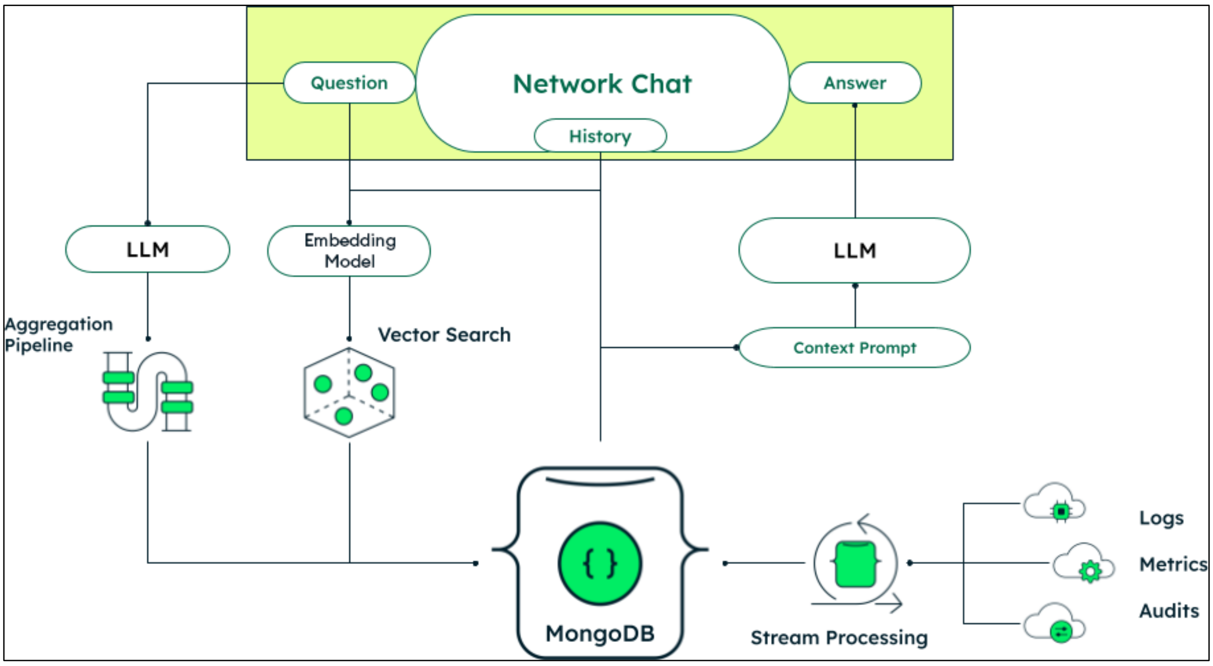
All data, including telemetry, past issues, session logs, agent profiles and recommendations are stored in MongoDB Atlas, ensuring traceability and the ability to refine diagnostics over time.



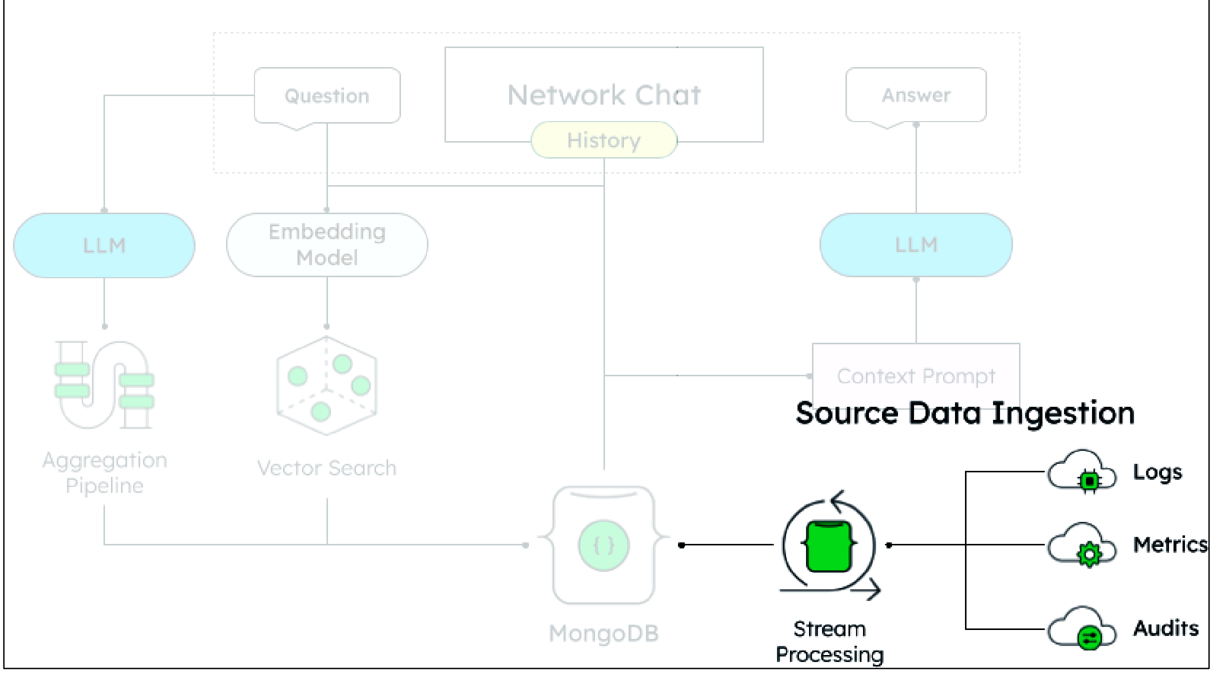
# Chapter 8: AI-Driven Strategies for Media and Telecommunication Industries

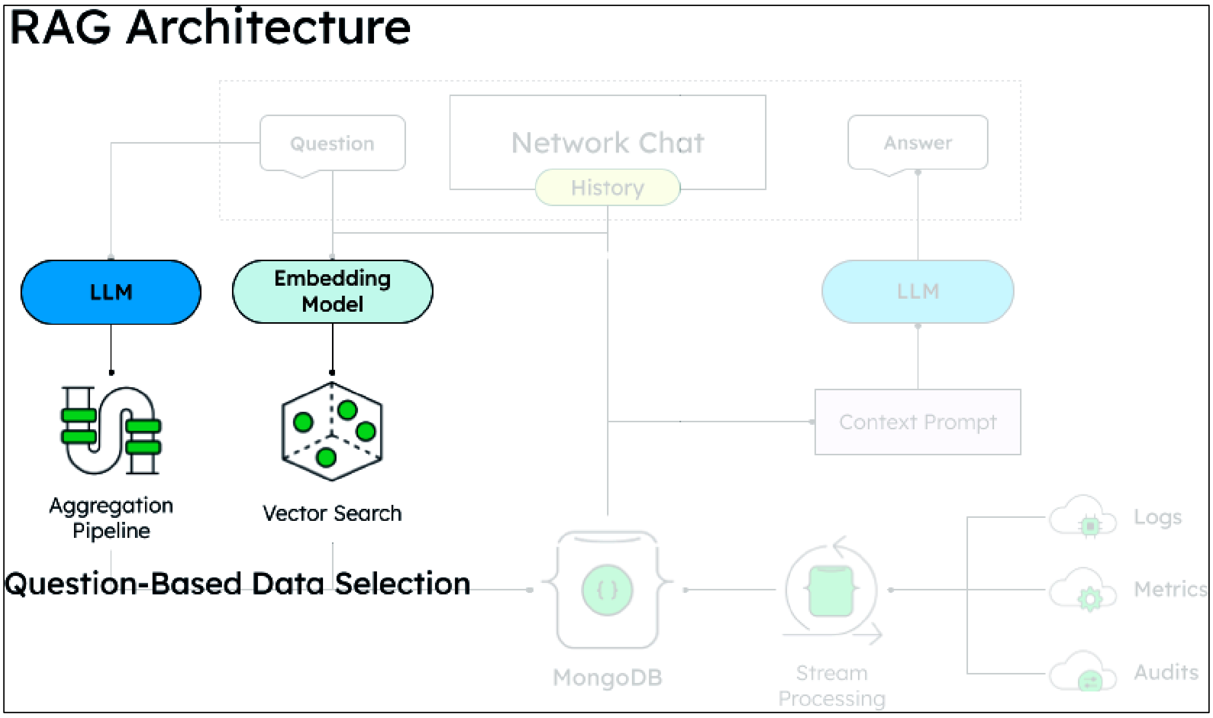


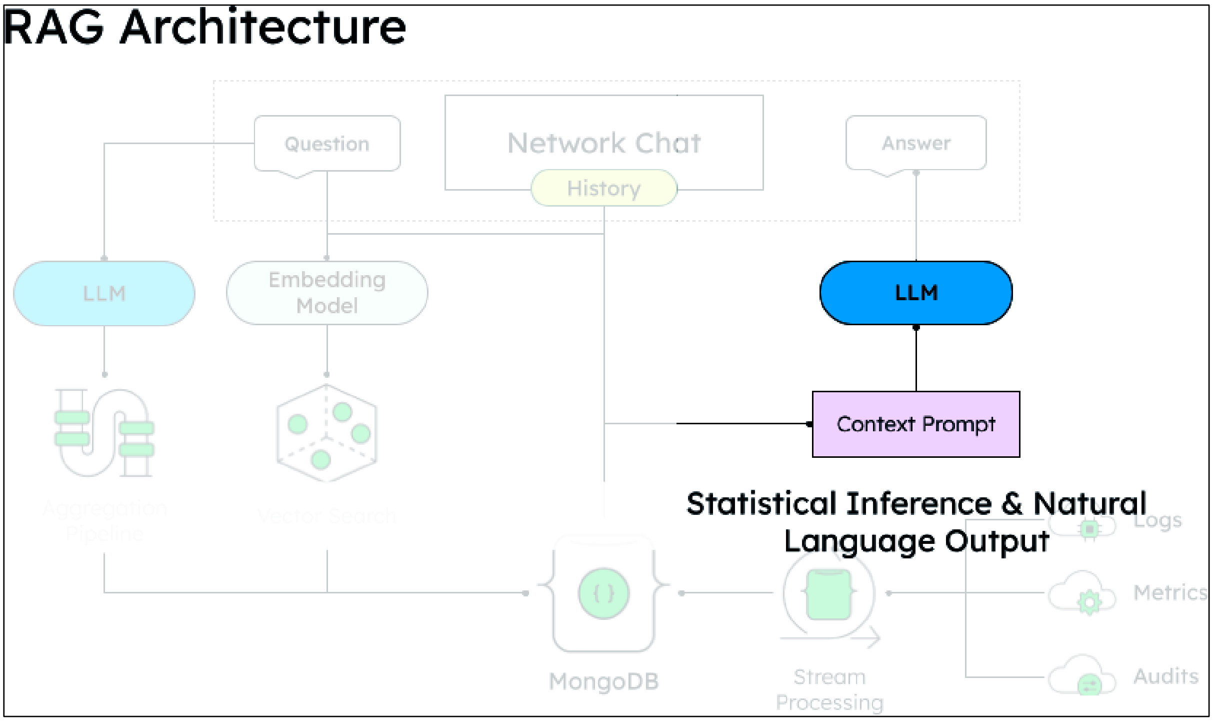



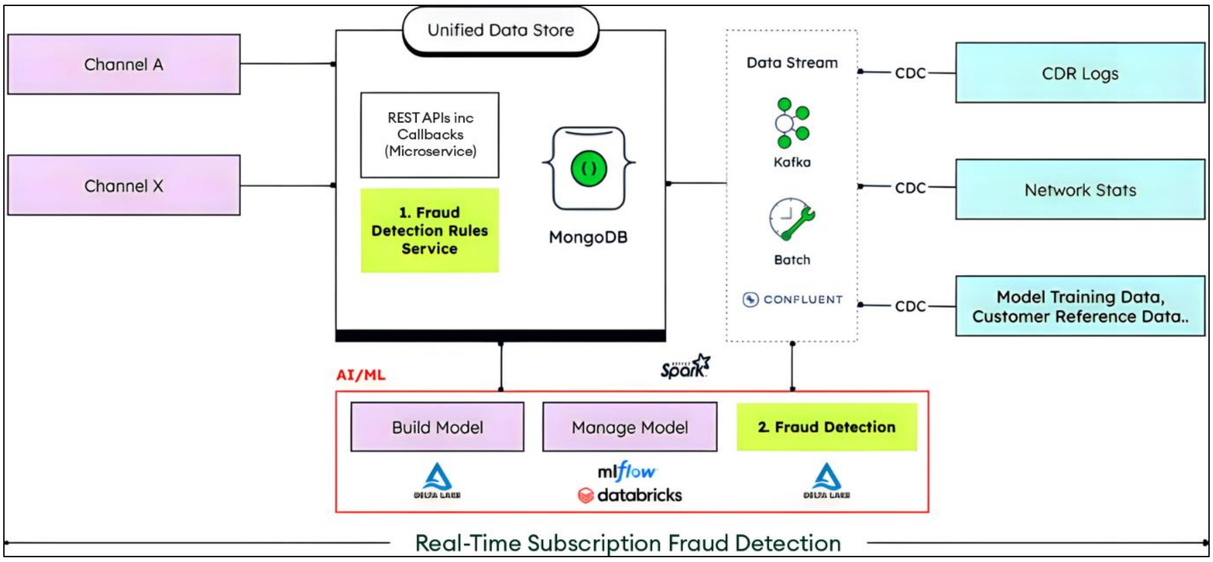



# RAG Architecture



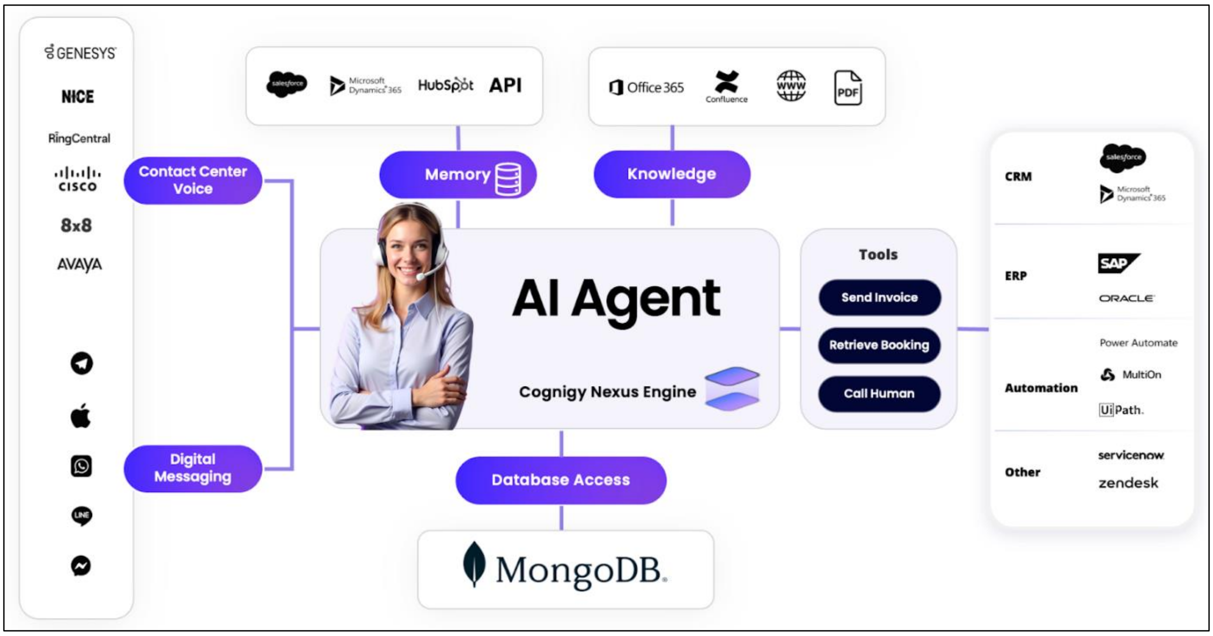

RAG Architecture
Question
Network Chat
History
Answer
LLM
Embedding Model
LLM
Context Prompt
Aggregation Pipeline
Vector Search
Logs
Question-Based Data Selection
Metrics
MongoDB
Stream Processing
Audits

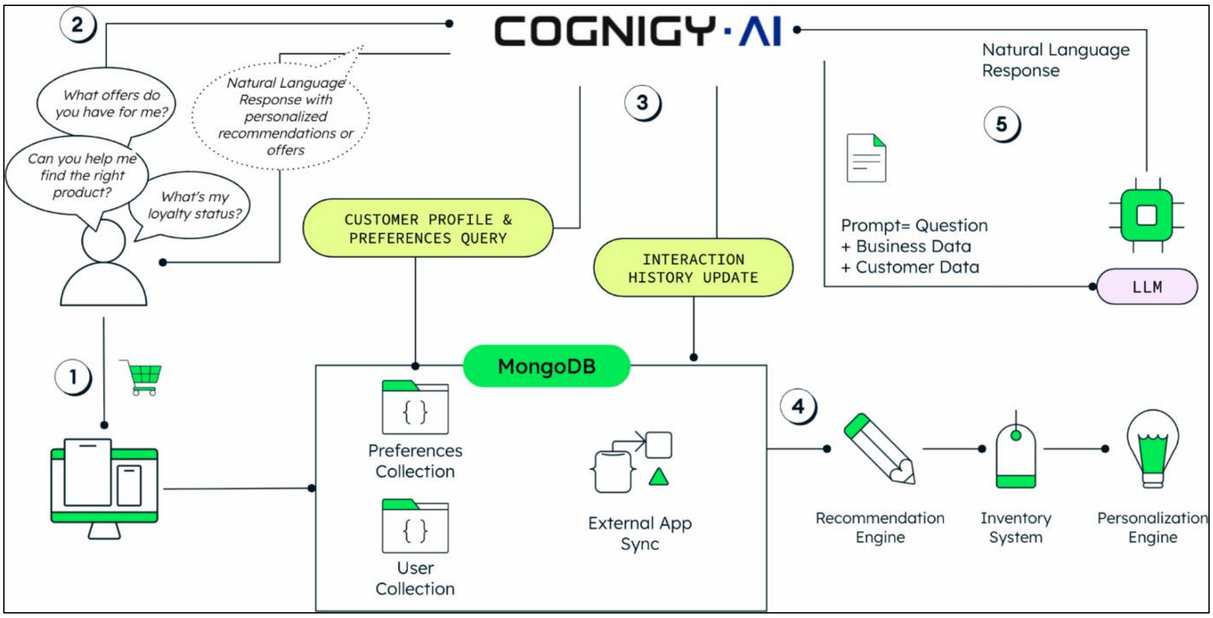

RAG Architecture
Question
Network Chat
History
Answer
LLM
Embedding Model
LLM
Context Prompt
Aggregation Pipeline
Vector Search
Statistical Inference & Natural Language Output
Logs
Metrics
MongoDB
Stream Processing
Audits



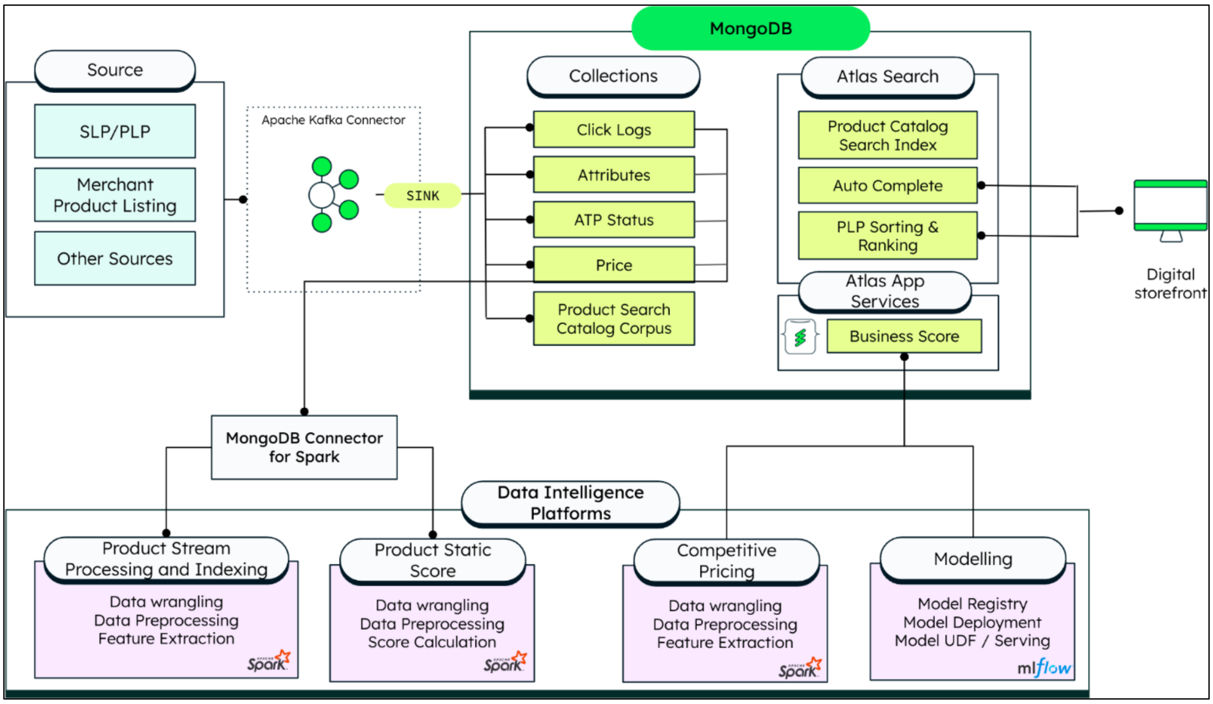
Unified Data Store
Channel A
Channel X
REST APIs inc Callbacks (Microservice)
1. Fraud Detection Rules Service
MongoDB
Data Stream
Kafka
Batch
CONFLUENT
CDC
CDC
CDC
CDR Logs
Network Stats
Model Training Data, Customer Reference Data..
AI/ML
Spark
Build Model
Manage Model
2. Fraud Detection
mlflow
databricks
Real-Time Subscription Fraud Detection

# Chapter 9: Cognigy's Voice and Chatbots in the Time of Agentic AI

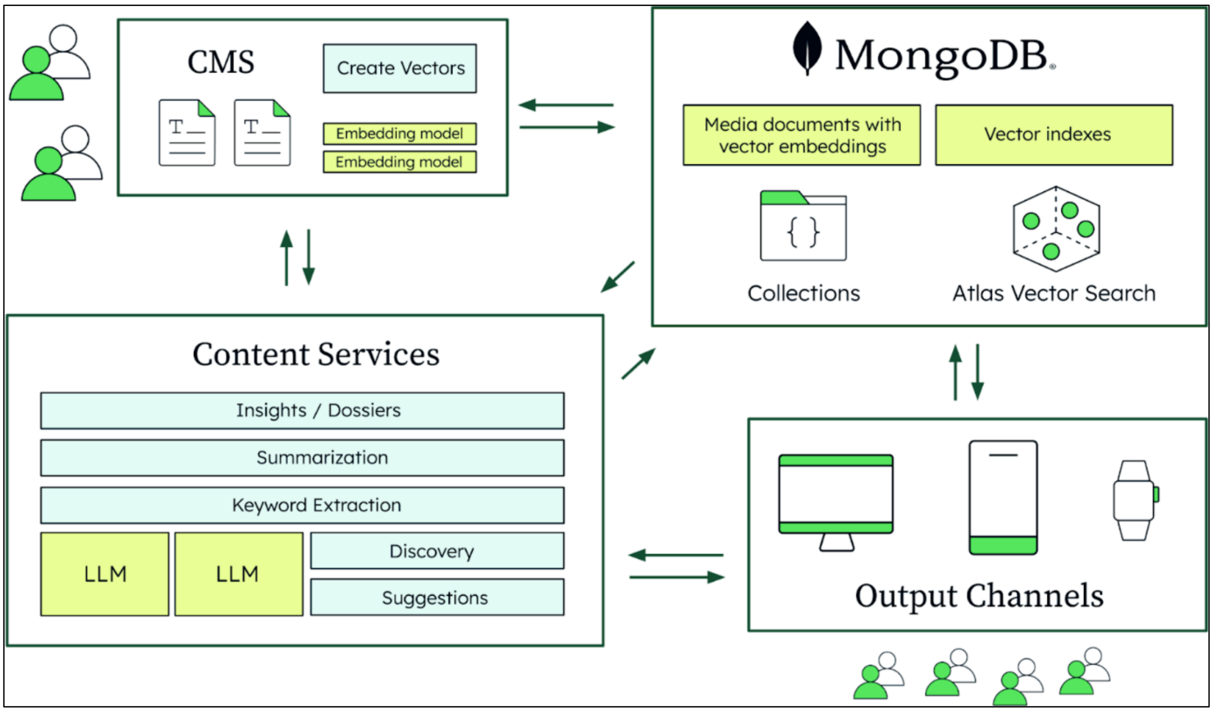


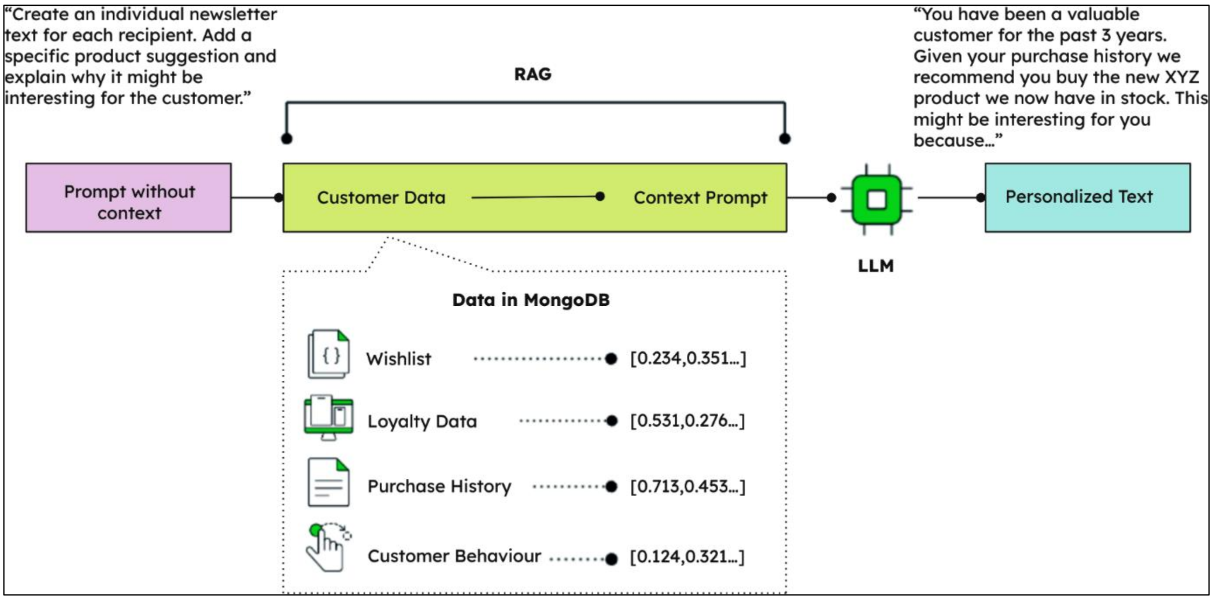
| Use Case | Channel | KPI | Impact |
|---|---|---|---|
| Sportswear and Apparel | Chat + Voice | Deflection rate | 25% |
| Insurance | Chat + Voice | ID&V | 95% |
| Insurance | Chat + Voice | AHT reduction | 27% |
| Industrial Technology and Mobility Solutions | Chat + Voice | Deflection rate | 76% |
| Energy Utilities | Chat + Voice | Deflection rate | 63% |
| Airlines | Chat | Deflection rate | 70% |
| Apparel and Luxury Goods | Chat | Deflection rate | 43% |
| Home Healthcare Services | Voice | AHT reduction | 50-80% |
| Home Healthcare Services | Voice | Deflection rate | 35% |
| Airlines | Chat + Voice | Deflection rate | 80% |
| Online Optics and Eyewear | Chat + Voice | ID&V | 70% |
| Online Optics and Eyewear | Chat + Voice | AHT reduction | 10% |
| Online Optics and Eyewear | Chat + Voice | Deflection rate | 50% |
| Telecommunications | Chat | Response time decrease | 95% |
| Robotic Process Automation (RPA) | Chat | Deflection rate | 26% |
| Employee Wellbeing Solutions | Chat + Voice | Deflection rate | 43% |

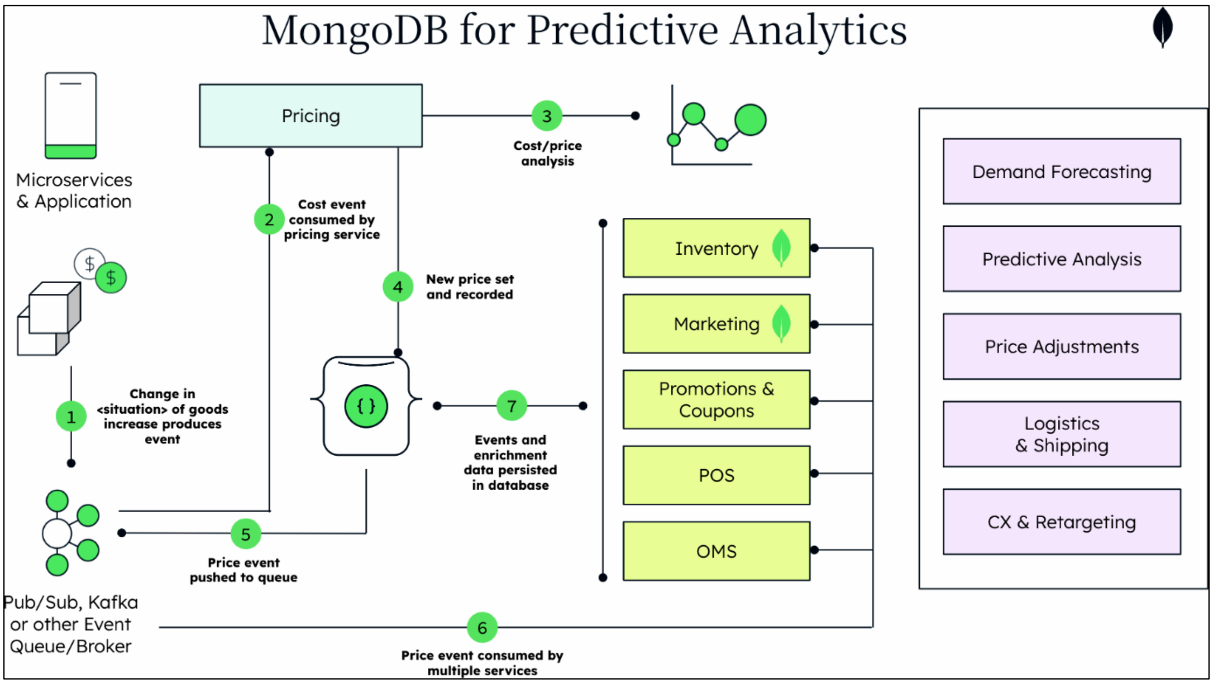


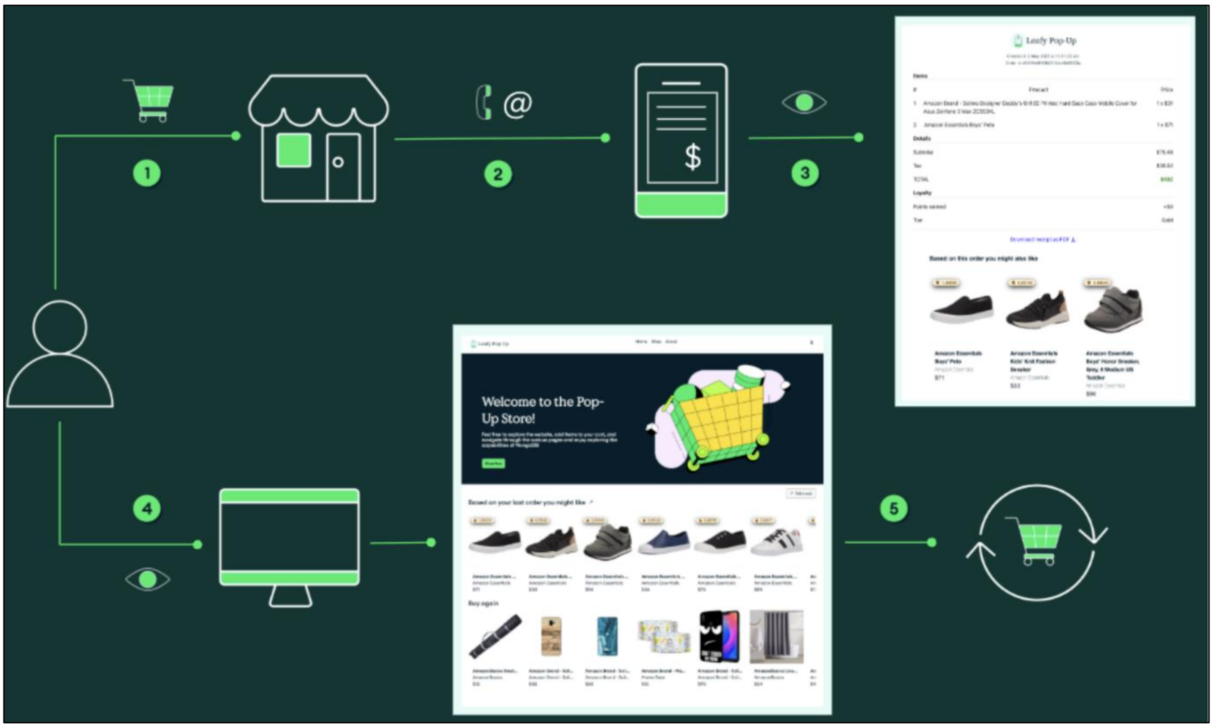
MongoDB
Source
SLP/PLP
Merchant Product Listing
Other Sources
Apache Kafka Connector
SINK
Collections
Click Logs
Attributes
ATP Status
Price
Product Search Catalog Corpus
Atlas Search
Product Catalog Search Index
Auto Complete
PLP Sorting & Ranking
Atlas App Services
Business Score
Digital storefront
MongoDB Connector for Spark
Data Intelligence Platforms
Product Stream Processing and Indexing
Data wrangling
Data Preprocessing
Feature Extraction
Product Static Score
Data wrangling
Data Preprocessing
Score Calculation
Competitive Pricing
Data wrangling
Data Preprocessing
Feature Extraction
Modelling
Model Registry
Model Deployment
Model UDF / Serving
mlflow

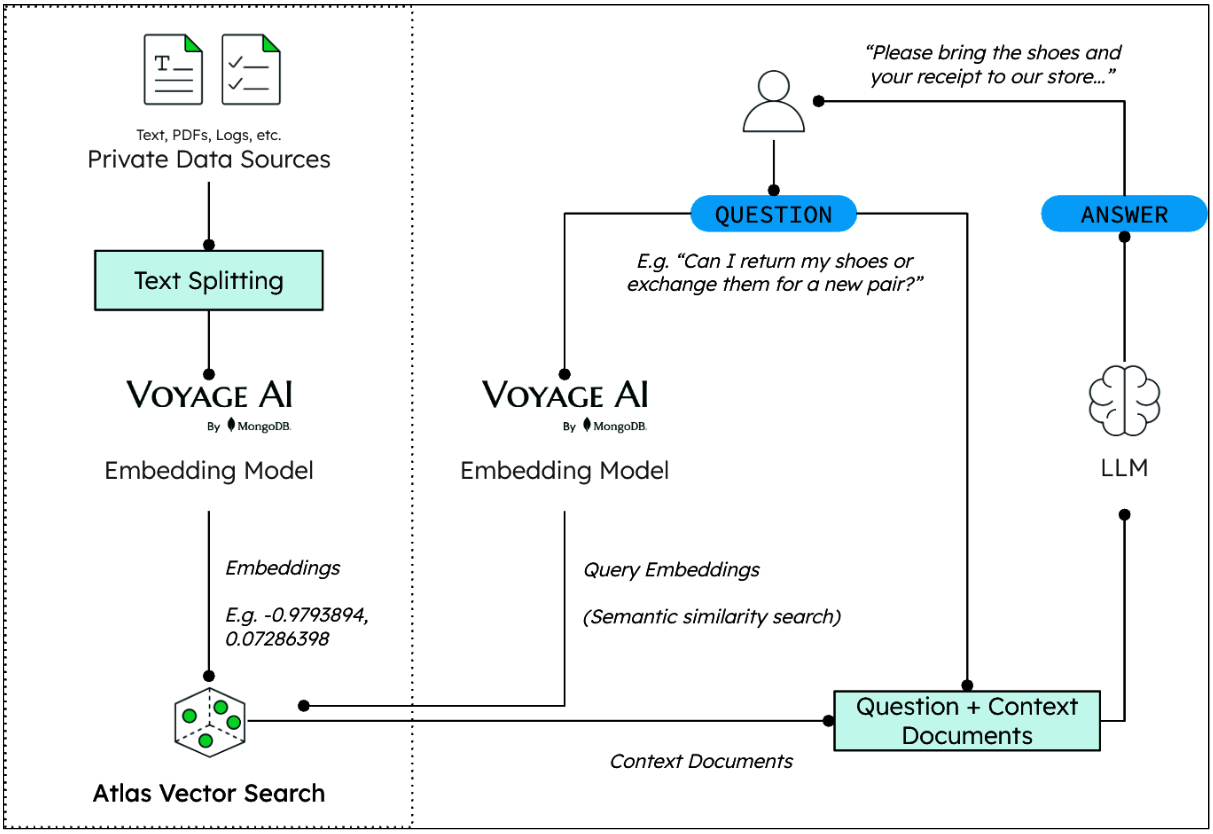



CMS
Create Vectors
Embedding model
Embedding model
MongoDB.
Media documents with vector embeddings
Vector indexes
Collections
Atlas Vector Search
Content Services
Insights / Dossiers
Summarization
Keyword Extraction
LLM
LLM
Discovery
Suggestions
Output Channels

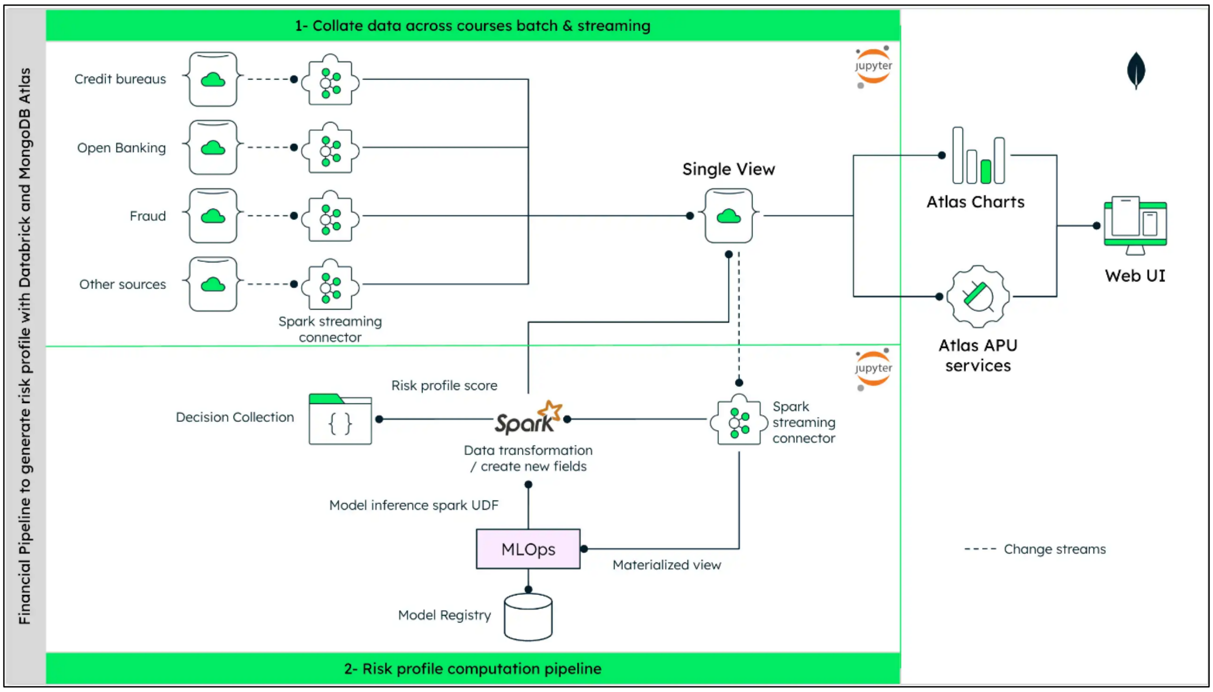

"Create an individual newsletter text for each recipient. Add a specific product suggestion and explain why it might be interesting for the customer."
RAG
Prompt without context
Customer Data
Context Prompt
LLM
Personalized Text
"You have been a valuable customer for the past 3 years. Given your purchase history we recommend you buy the new XYZ product we now have in stock. This might be interesting for you because…"
Data in MongoDB
Wishlist
[0.234,0.351…]
Loyalty Data
[0.531,0.276…]
Purchase History
[0.713,0.453…]
Customer Behaviour
[0.124,0.321…]

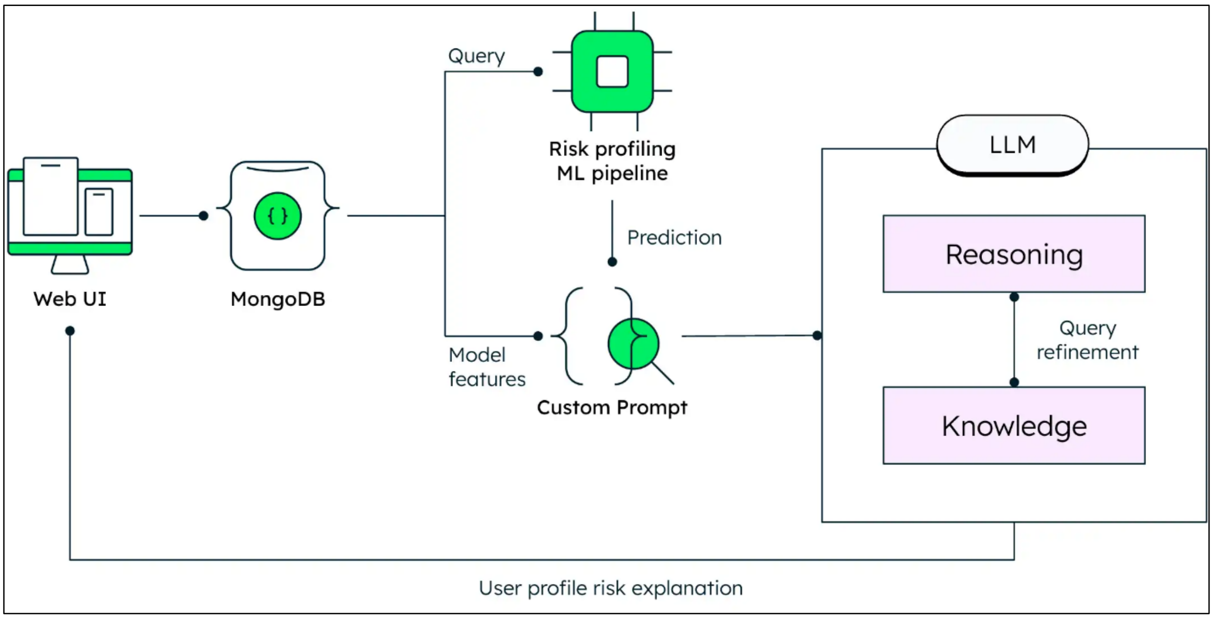

MongoDB for Predictive Analytics
Microservices & Application
Pricing
Cost/price analysis
Cost event consumed by pricing service
New price set and recorded
Change in <situation> of goods increase produces event
Events and enrichment data persisted in database
Price event pushed to queue
Pub/Sub, Kafka or other Event Queue/Broker
Price event consumed by multiple services
Inventory
Marketing
Promotions & Coupons
POS
OMS
Demand Forecasting
Predictive Analysis
Price Adjustments
Logistics & Shipping
CX & Retargeting

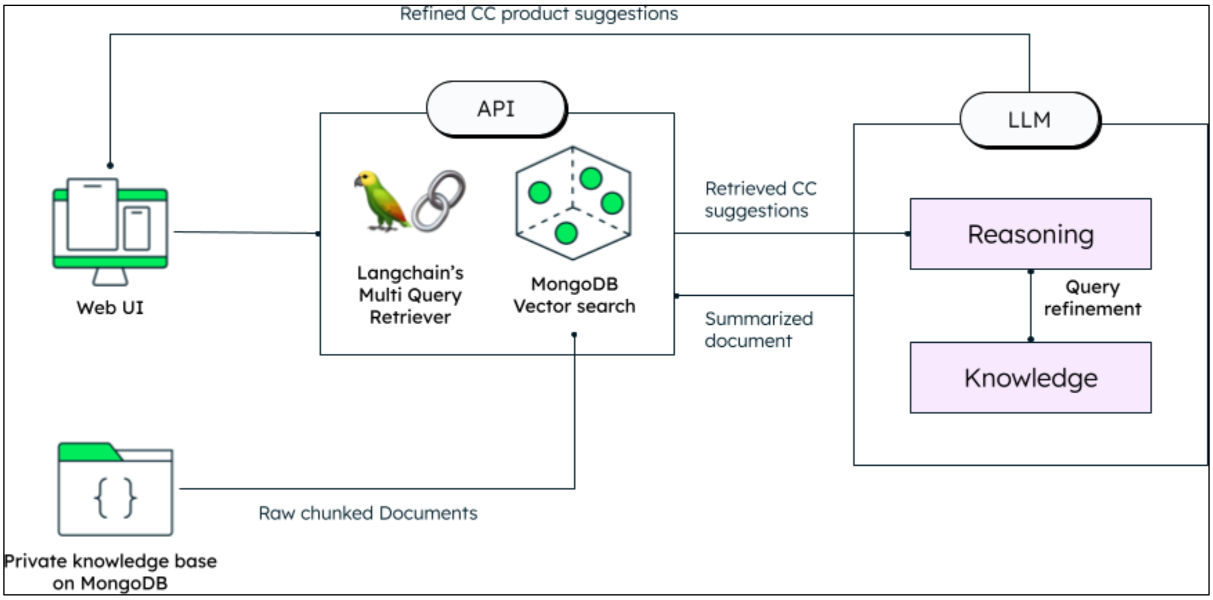

1
2
3
4
5
Welcome to the Pop-Up Store!


Text, PDFs, Logs, etc.
Private Data Sources
Text Splitting
VOYAGE AI
By MongoDB.
Embedding Model
Embeddings
E.g. -0.9793894, 0.07286398
Atlas Vector Search
"Please bring the shoes and your receipt to our store..."
QUESTION
ANSWER
E.g. "Can I return my shoes or exchange them for a new pair?"
VOYAGE AI
By MongoDB.
Embedding Model
LLM
Query Embeddings
(Semantic similarity search)
Question + Context Documents
Context Documents

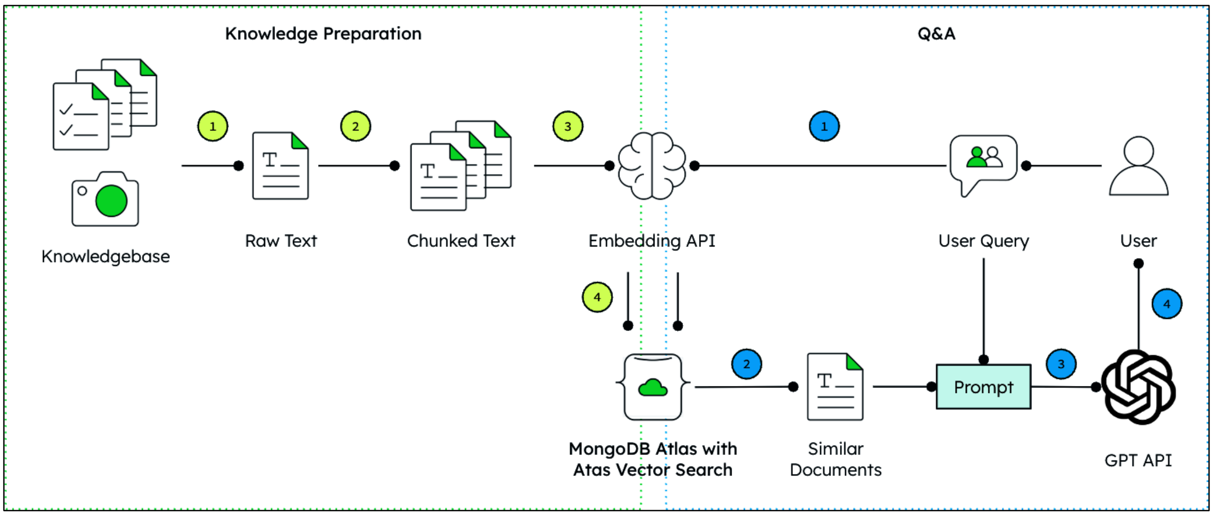
# Chapter 11: Financial Services and the Next Wave of AI



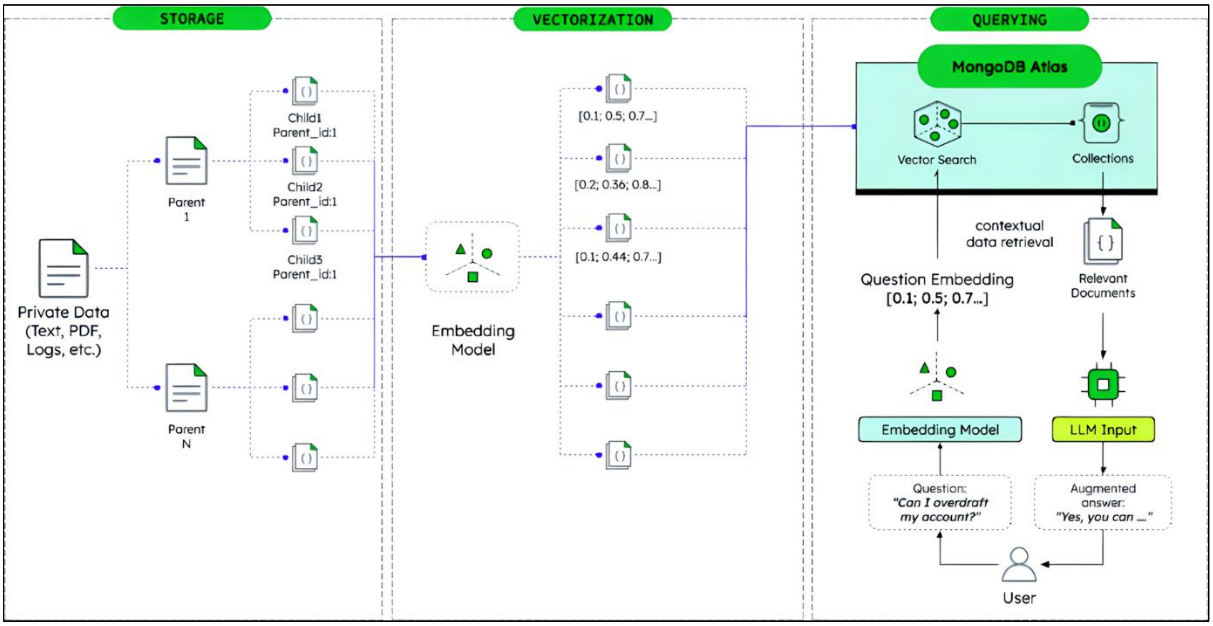





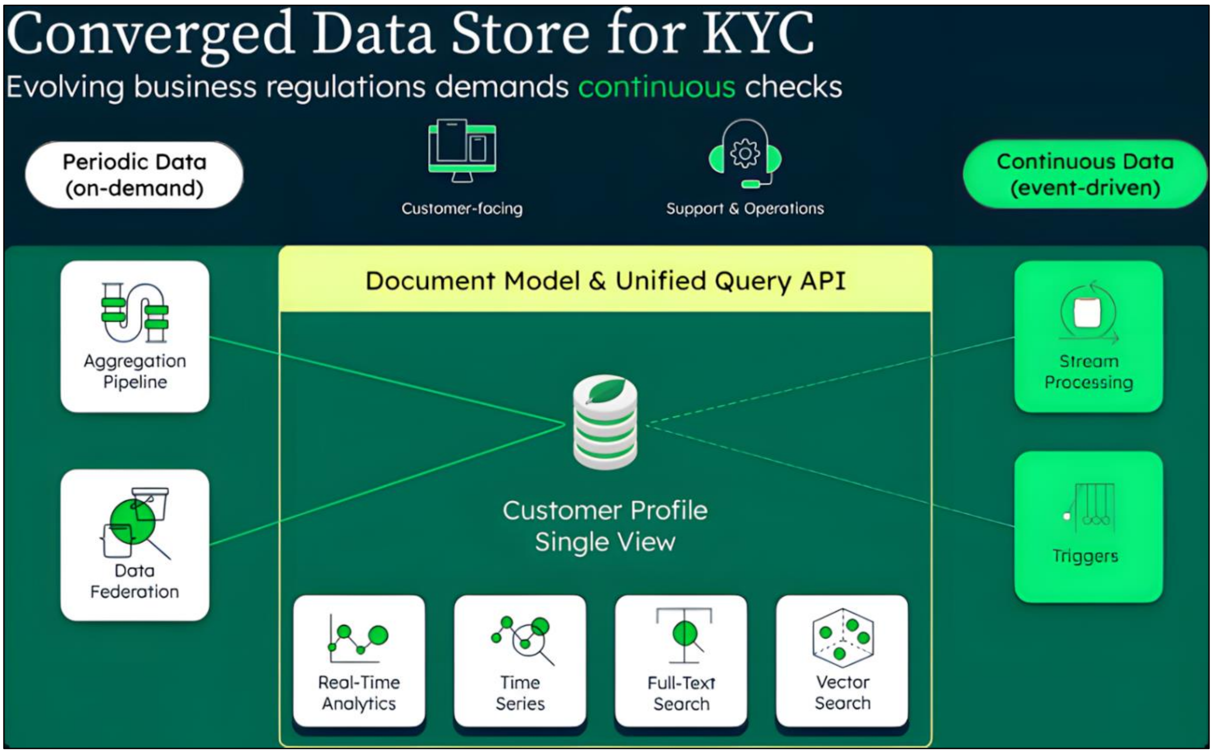
Refined CC product suggestions
API
LLM
Web UI
Langchain's
Multi Query
Retriever
MongoDB
Vector search
Retrieved CC
suggestions
Summarized
document
Reasoning
Query
refinement
Knowledge
Raw chunked Documents
Private knowledge base
on MongoDB

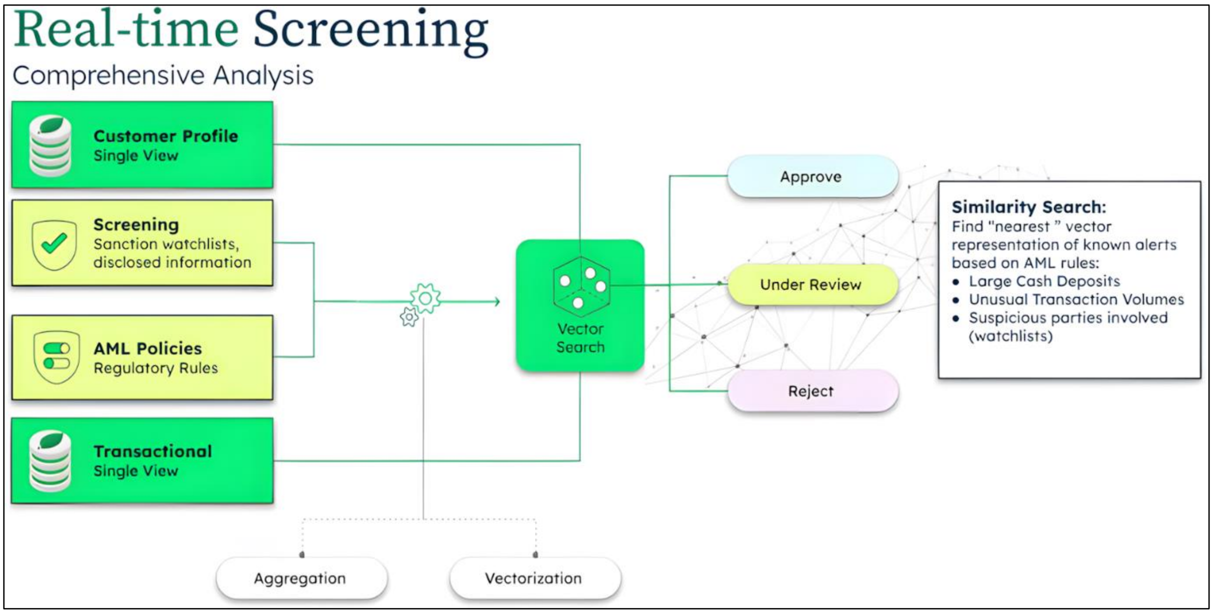



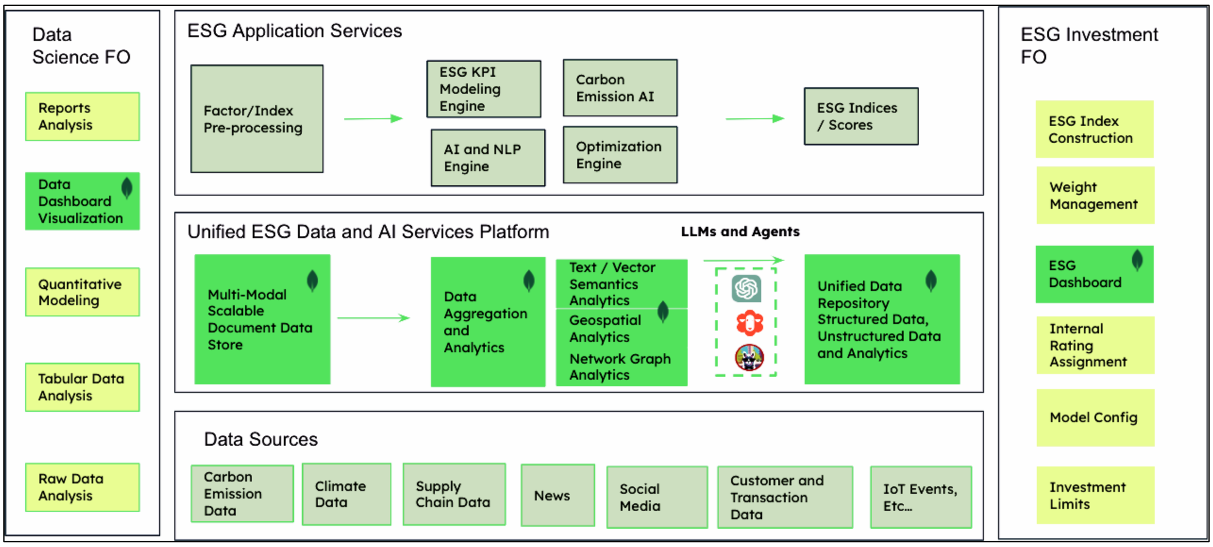
Knowledge Preparation
Q&A
1
2
3
1
Knowledgebase
Raw Text
Chunked Text
Embedding API
User Query
User
4
4
2
3
Prompt
MongoDB Atlas with
Atas Vector Search
Similar
Documents
GPT API

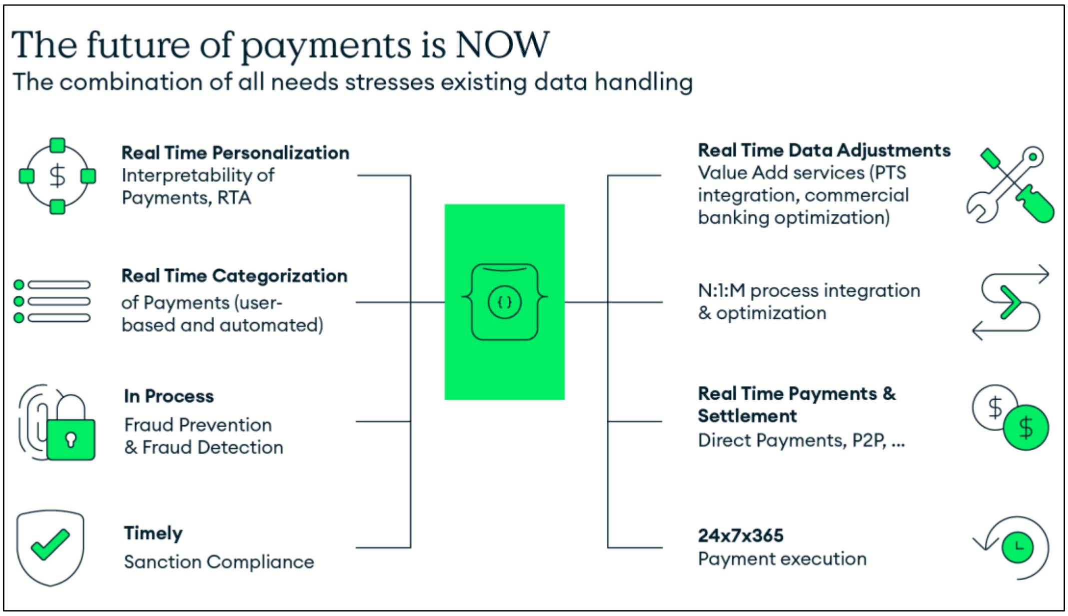



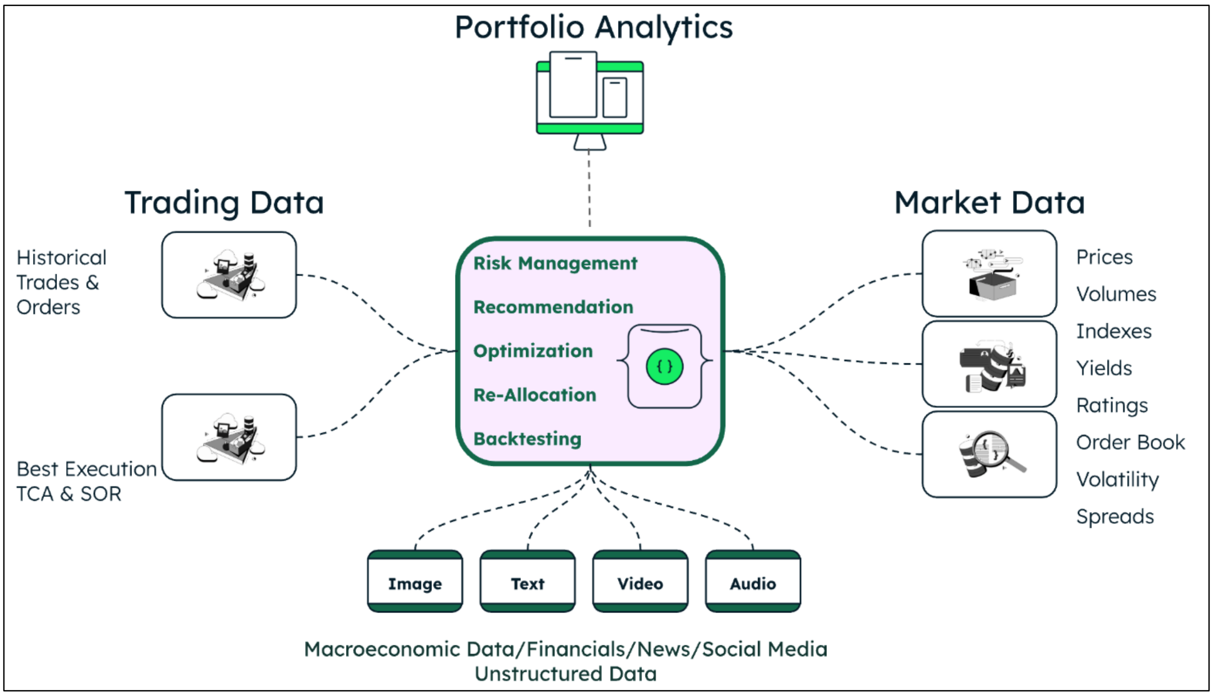
STORAGE
VECTORIZATION
QUERYING
MongoDB Atlas
Vector Search
Collections
Child1
Parent_id:1
Child2
Parent_id:1
Child3
Parent_id:1
Parent
1
Parent
N
Private Data
(Text, PDF,
Logs, etc.)
Embedding
Model
[0.1; 0.5; 0.7...]
[0.2; 0.36; 0.8...]
[0.1; 0.44; 0.7...]
contextual
data retrieval
Question Embedding
[0.1; 0.5; 0.7...]
Relevant
Documents
Embedding Model
LLM Input
Question:
"Can I overdraft
my account?"
Augmented
answer:
"Yes, you can ..."
User


Converged Data Store for KYC
Evolving business regulations demands continuous checks
Periodic Data (on-demand)
Customer-facing
Support & Operations
Continuous Data (event-driven)
Document Model & Unified Query API
Aggregation Pipeline
Data Federation
Customer Profile Single View
Stream Processing
Triggers
Real-Time Analytics
Time Series
Full-Text Search
Vector Search

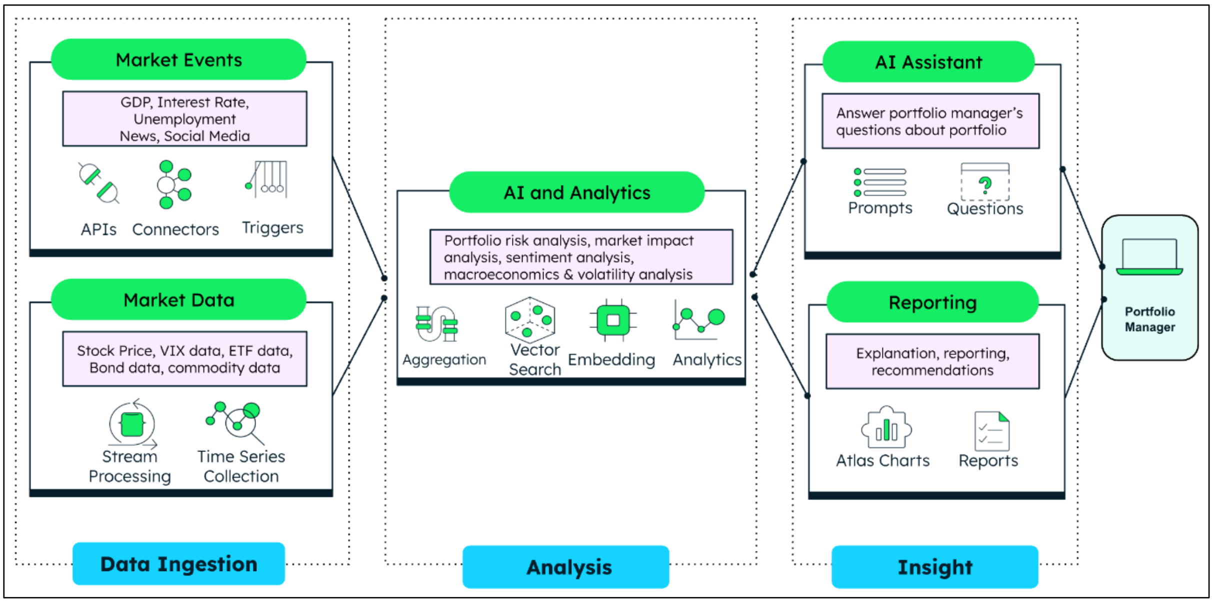

Real-time Screening
Comprehensive Analysis
Customer Profile
Single View
Screening
Sanction watchlists, disclosed information
AML Policies
Regulatory Rules
Transactional
Single View
Vector Search
Approve
Under Review
Reject
Similarity Search:
Find "nearest " vector representation of known alerts based on AML rules:
• Large Cash Deposits
• Unusual Transaction Volumes
• Suspicious parties involved (watchlists)
Aggregation
Vectorization


Data Science FO
Reports Analysis
Data Dashboard Visualization
Quantitative Modeling
Tabular Data Analysis
Raw Data Analysis
ESG Application Services
Factor/Index Pre-processing
ESG KPI Modeling Engine
AI and NLP Engine
Carbon Emission AI
Optimization Engine
ESG Indices / Scores
Unified ESG Data and AI Services Platform
LLMs and Agents
Multi-Modal Scalable Document Data Store
Data Aggregation and Analytics
Text / Vector Semantics Analytics
Geospatial Analytics
Network Graph Analytics
Unified Data Repository Structured Data, Unstructured Data and Analytics
Data Sources
Carbon Emission Data
Climate Data
Supply Chain Data
News
Social Media
Customer and Transaction Data
IoT Events, Etc...
ESG Investment FO
ESG Index Construction
Weight Management
ESG Dashboard
Internal Rating Assignment
Model Config
Investment Limits

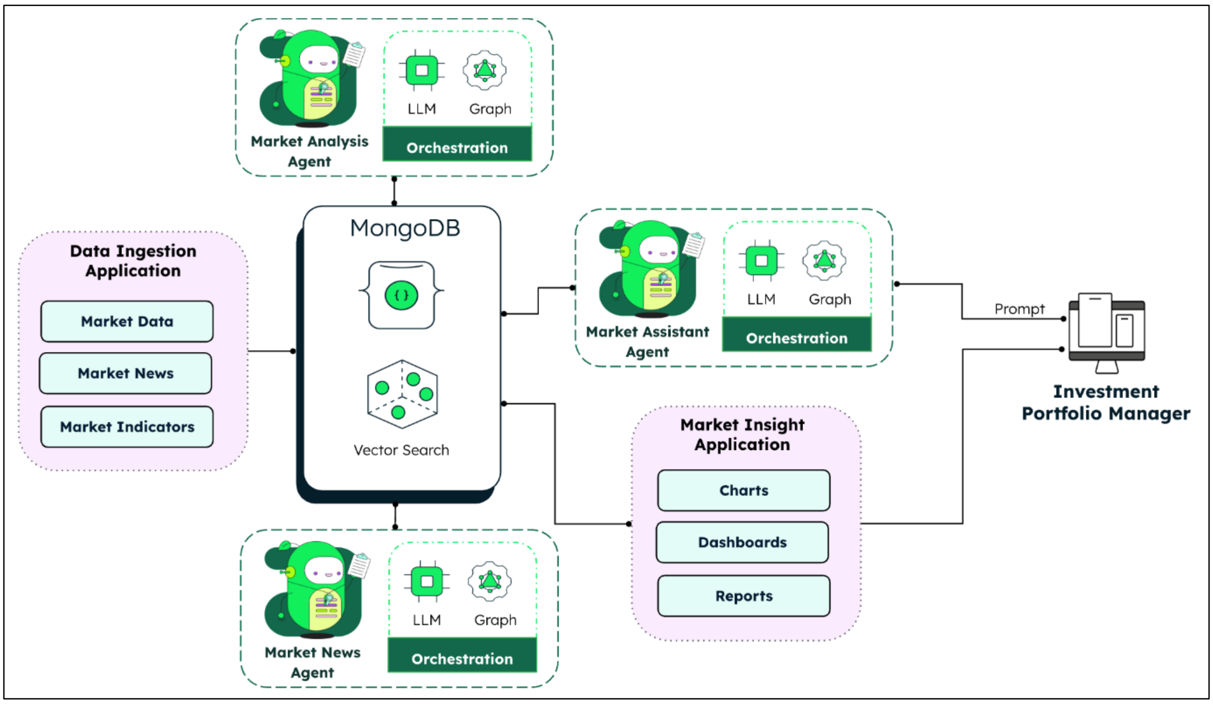

The future of payments is NOW
The combination of all needs stresses existing data handling
Real Time Personalization
Interpretability of Payments, RTA
Real Time Categorization
of Payments (user-based and automated)
In Process
Fraud Prevention & Fraud Detection
Timely
Sanction Compliance
Real Time Data Adjustments
Value Add services (PTS integration, commercial banking optimization)
N:1:M process integration & optimization
Real Time Payments & Settlement
Direct Payments, P2P, ...
24x7x365
Payment execution

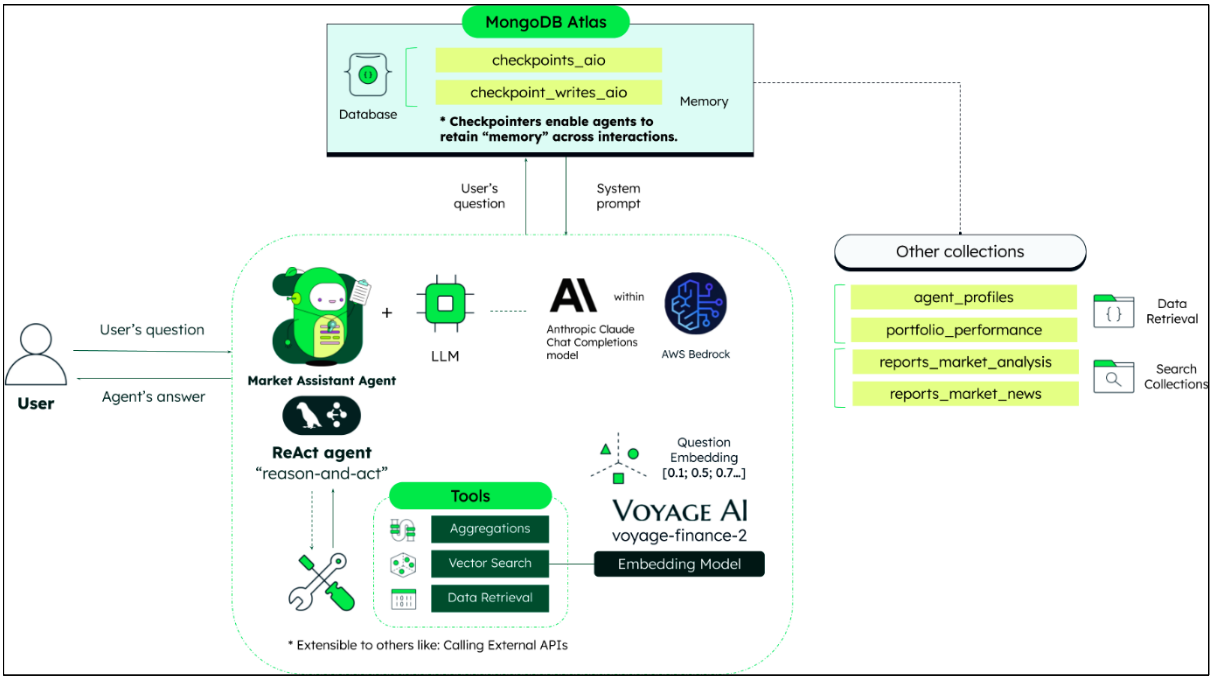


Portfolio Analytics
Trading Data
Historical Trades & Orders
Best Execution TCA & SOR
Risk Management
Recommendation
Optimization
Re-Allocation
Backtesting
Market Data
Prices
Volumes
Indexes
Yields
Ratings
Order Book
Volatility
Spreads
Image
Text
Video
Audio
Macroeconomic Data/Financials/News/Social Media
Unstructured Data

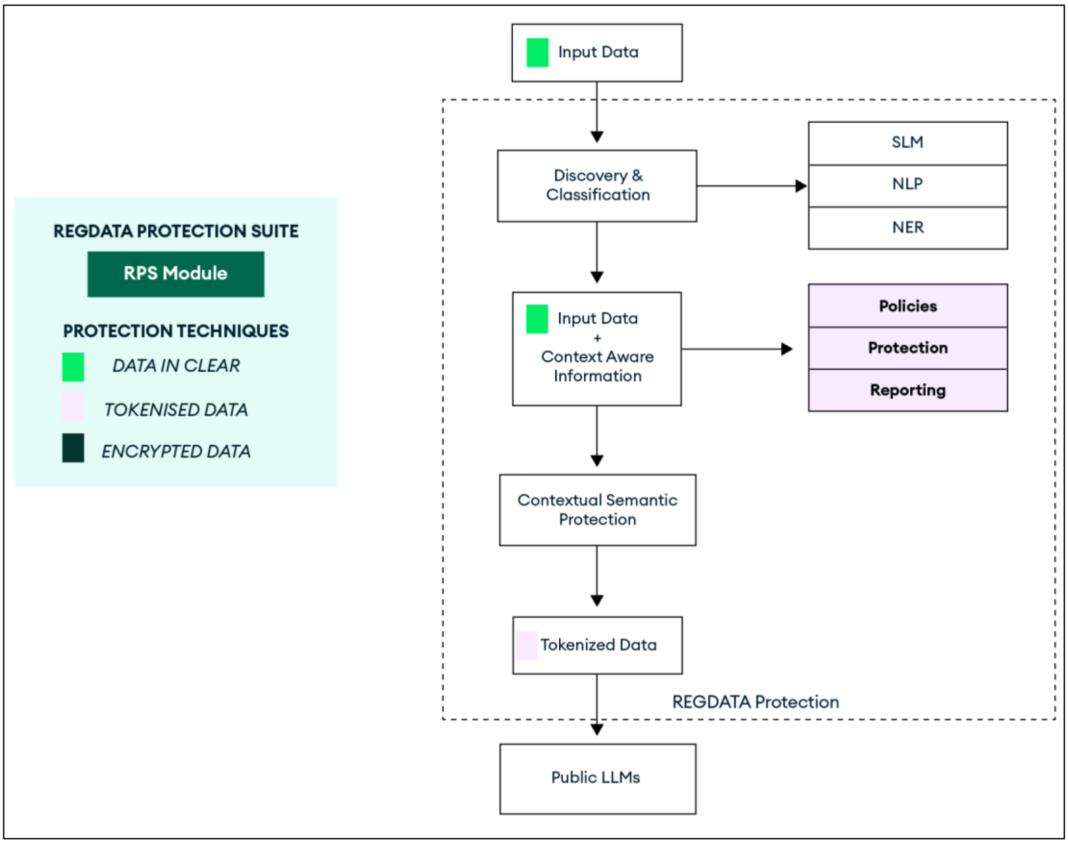



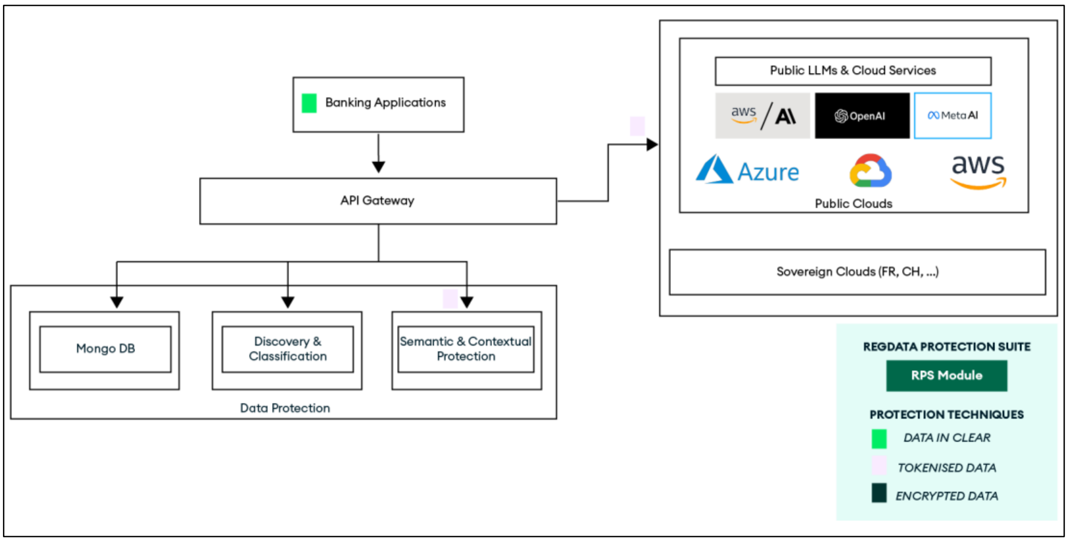
Market Events
GDP, Interest Rate, Unemployment News, Social Media
APIs
Connectors
Triggers
Market Data
Stock Price, VIX data, ETF data, Bond data, commodity data
Stream Processing
Time Series Collection
Data Ingestion
AI and Analytics
Portfolio risk analysis, market impact analysis, sentiment analysis, macroeconomics & volatility analysis
Aggregation
Vector Search
Embedding
Analytics
Analysis
AI Assistant
Answer portfolio manager's questions about portfolio
Prompts
Questions
Reporting
Explanation, reporting, recommendations
Atlas Charts
Reports
Insight
Portfolio Manager

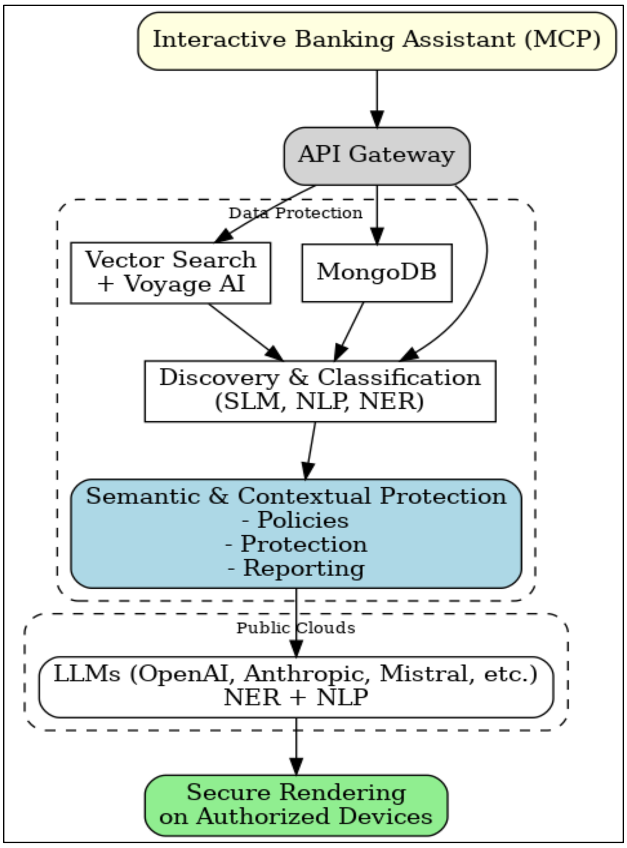

Market Analysis Agent
LLM
Graph
Orchestration
MongoDB
Vector Search
Data Ingestion Application
Market Data
Market News
Market Indicators
Market Assistant Agent
LLM
Graph
Orchestration
Prompt
Investment Portfolio Manager
Market Insight Application
Charts
Dashboards
Reports
Market News Agent
LLM
Graph
Orchestration


MongoDB Atlas
Database
checkpoints_aio
checkpoint_writes_aio
Memory
* Checkpointers enable agents to retain "memory" across interactions.
User's question
System prompt
Other collections
agent_profiles
portfolio_performance
reports_market_analysis
reports_market_news
Data Retrieval
Search Collections
User
User's question
Agent's answer
Market Assistant Agent
LLM
within
Anthropic Claude Chat Completions model
AWS Bedrock
ReAct agent
"reason-and-act"
Question Embedding
[0.1; 0.5; 0.7...]
VOYAGE AI
voyage-finance-2
Embedding Model
Tools
Aggregations
Vector Search
Data Retrieval
* Extensible to others like: Calling External APIs

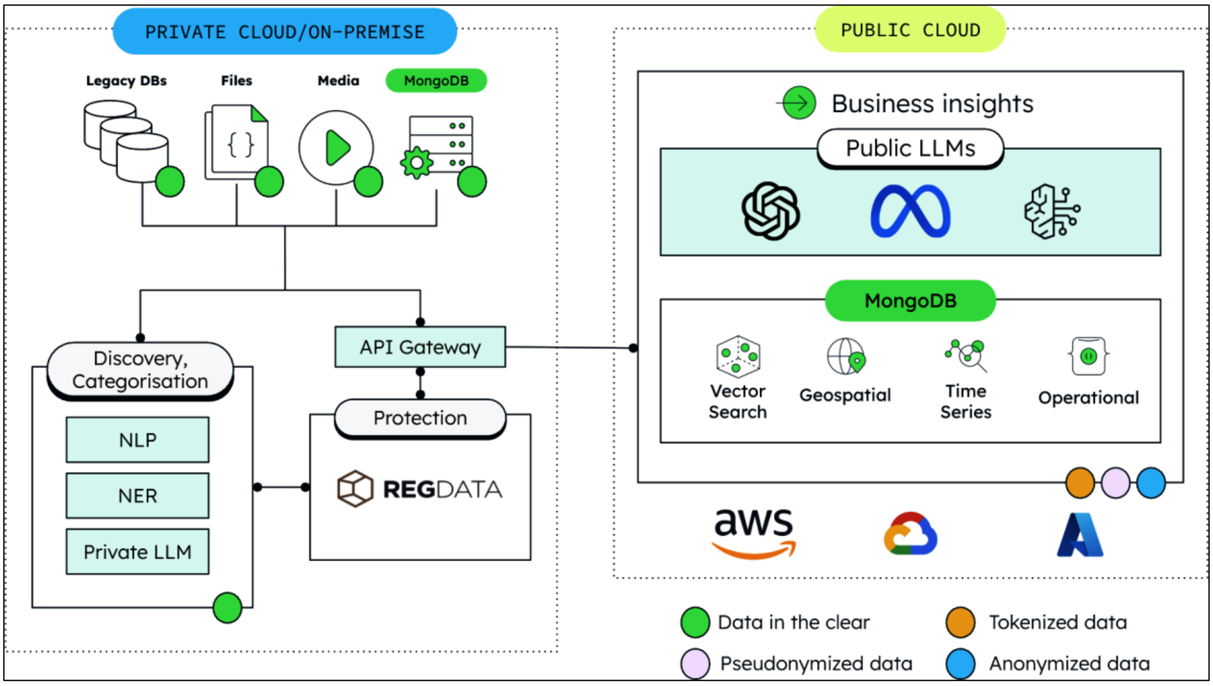
# Chapter 12: RegData, MongoDB, and Voyage AI: Semantic Data Protection in FSI



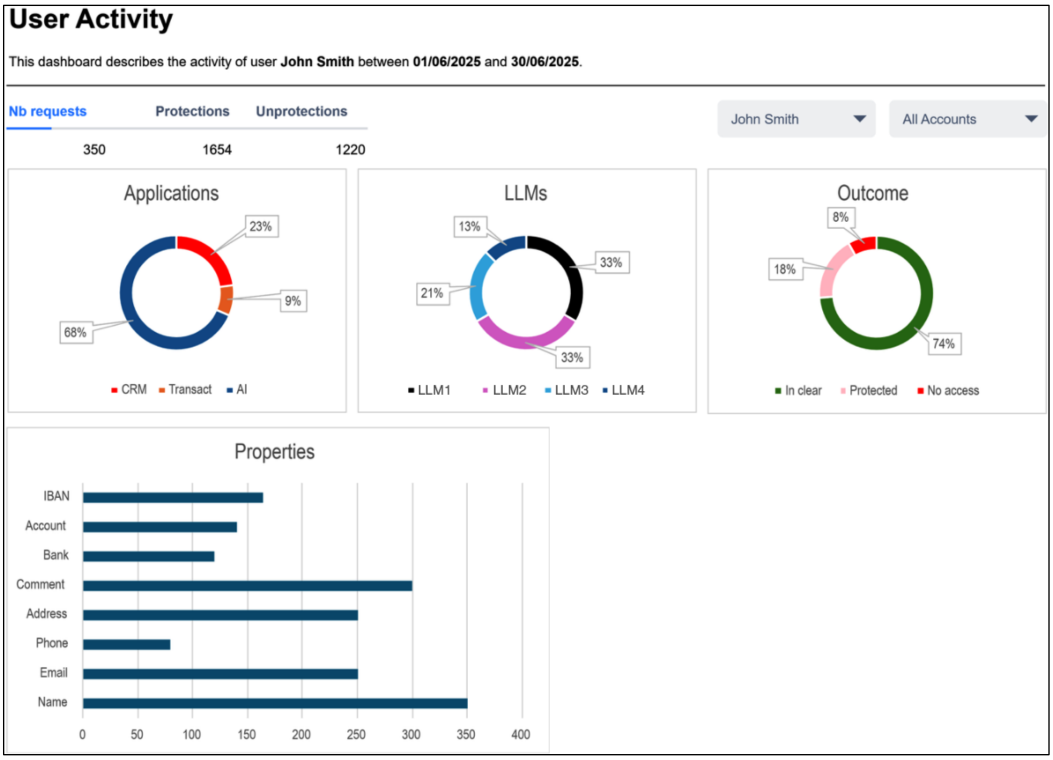





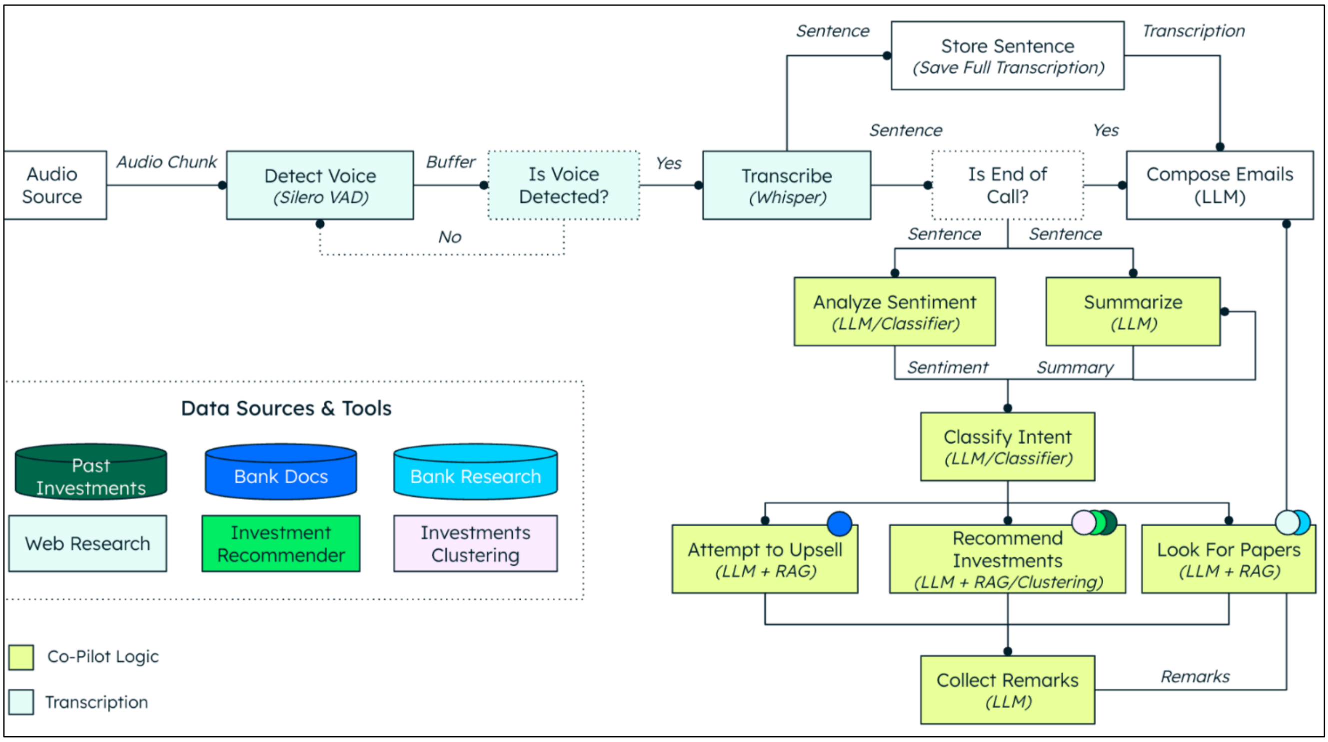
Interactive Banking Assistant (MCP)
API Gateway
Data Protection
Vector Search
+ Voyage AI
MongoDB
Discovery & Classification
(SLM, NLP, NER)
Semantic & Contextual Protection
- Policies
- Protection
- Reporting
Public Clouds
LLMs (OpenAI, Anthropic, Mistral, etc.)
NER + NLP
Secure Rendering
on Authorized Devices

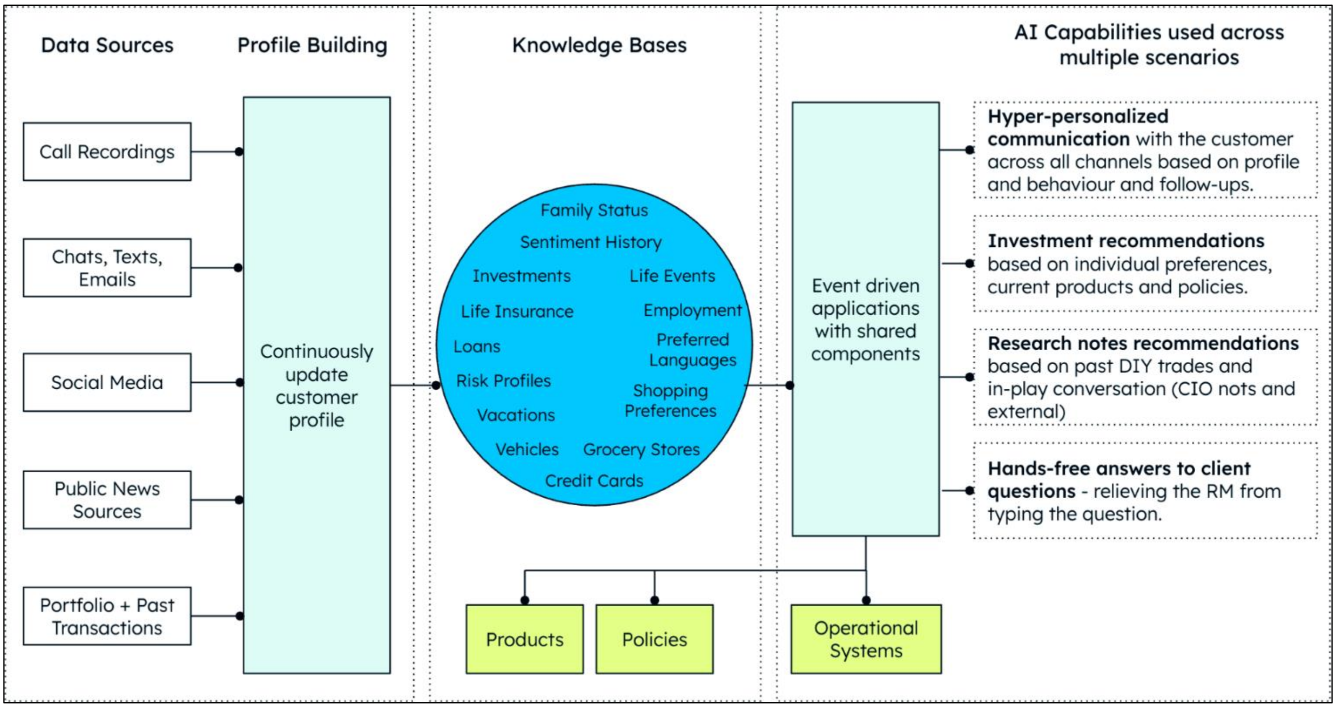



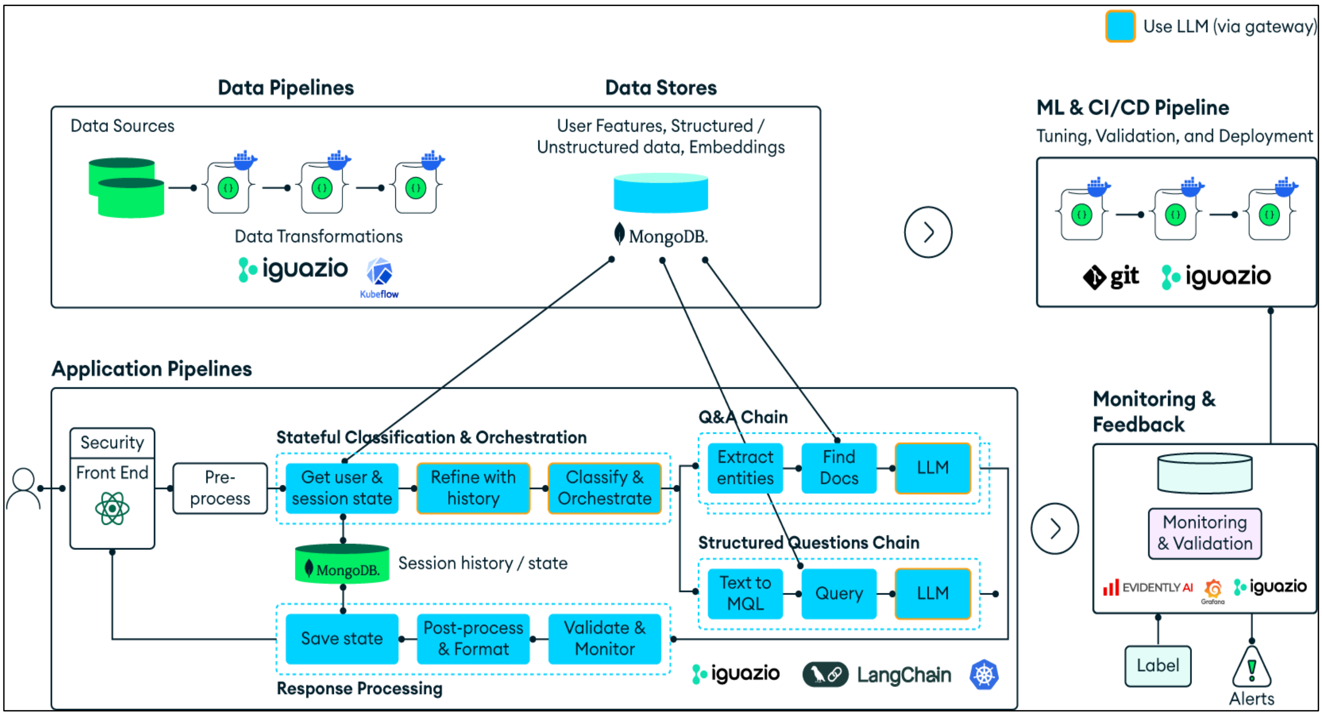
PRIVATE CLOUD/ON-PREMISE
Legacy DBs
Files
Media
MongoDB
Discovery,
Categorisation
NLP
NER
Private LLM
API Gateway
Protection
REGDATA
PUBLIC CLOUD
Business insights
Public LLMs
MongoDB
Vector
Search
Geospatial
Time
Series
Operational
aws
Data in the clear
Tokenized data
Pseudonymized data
Anonymized data


User Activity
This dashboard describes the activity of user John Smith between 01/06/2025 and 30/06/2025.
Nb requests
Protections
Unprotections
John Smith
All Accounts
350
1654
1220
Applications
23%
9%
68%
CRM
Transact
AI
LLMs
13%
33%
21%
33%
LLM1
LLM2
LLM3
LLM4
Outcome
8%
18%
74%
In clear
Protected
No access
Properties
IBAN
Account
Bank
Comment
Address
Phone
Email
Name
0
50
100
150
200
250
300
350
400

# Chapter 13: Driving Client Success in Banking with GenAI Copilots

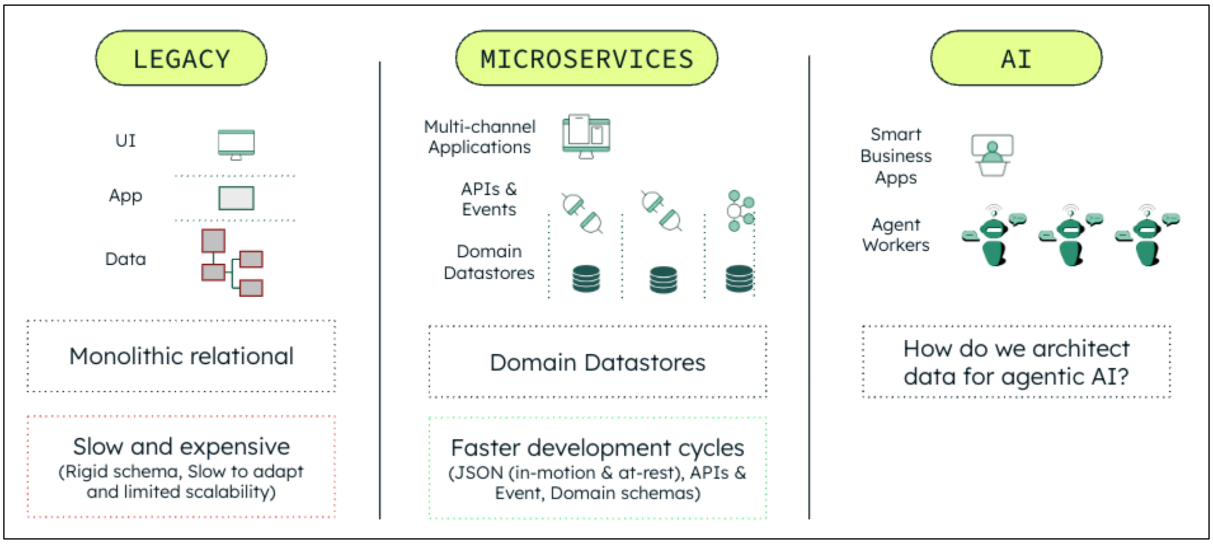


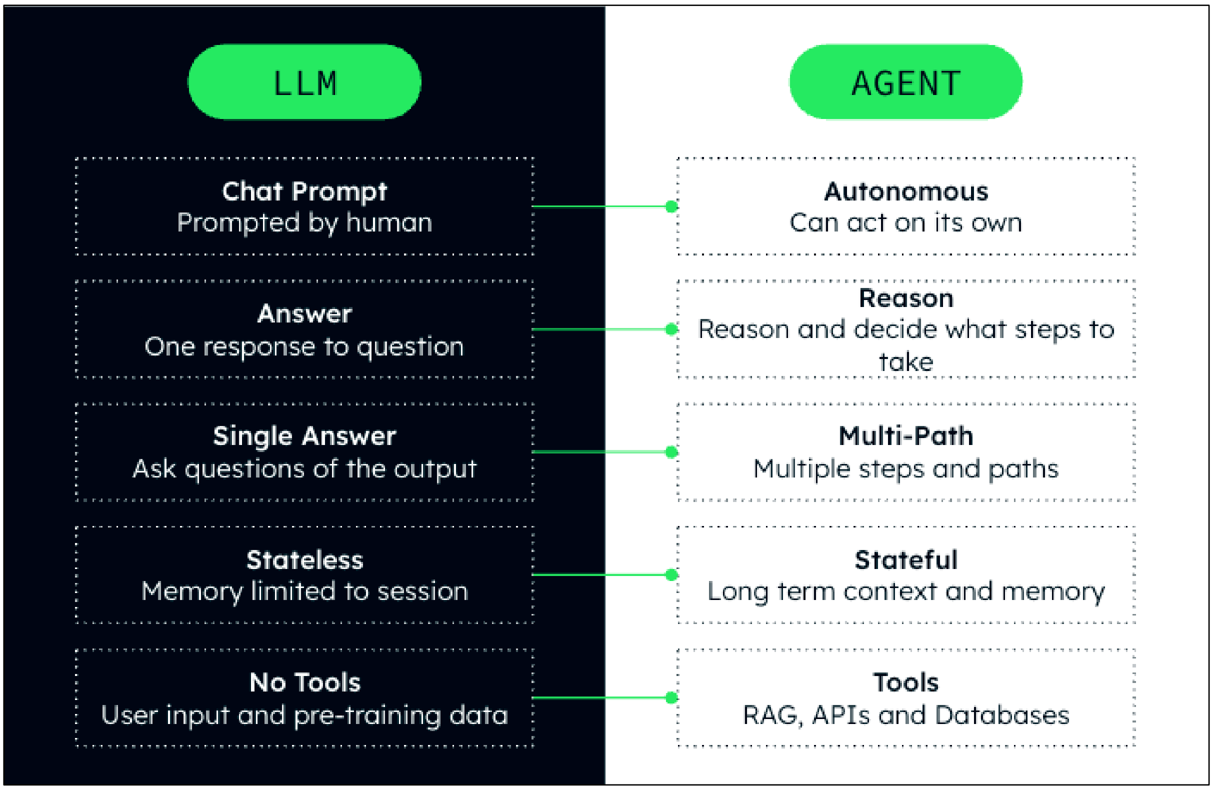



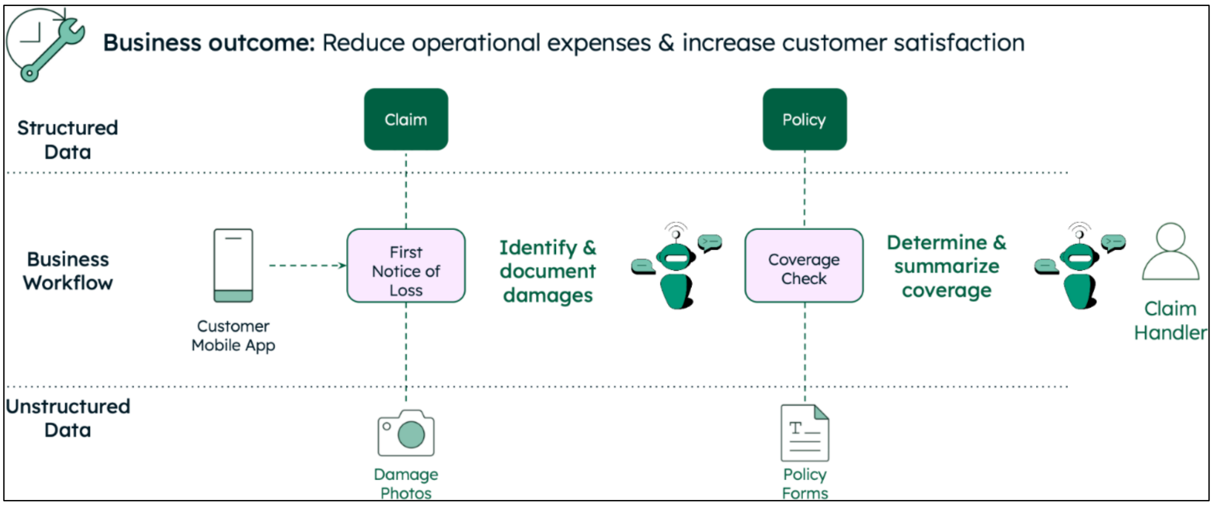


Use LLM (via gateway)
Data Pipelines
Data Stores
Data Sources
User Features, Structured / Unstructured data, Embeddings
MongoDB.
Data Transformations
iguazio
Kubeflow
ML & CI/CD Pipeline
Tuning, Validation, and Deployment
git
iguazio
Application Pipelines
Security
Front End
Pre-process
Stateful Classification & Orchestration
Get user & session state
Refine with history
Classify & Orchestrate
MongoDB.
Session history / state
Q&A Chain
Extract entities
Find Docs
LLM
Structured Questions Chain
Text to MQL
Query
LLM
Save state
Post-process & Format
Validate & Monitor
Response Processing
iguazio
LangChain
Monitoring & Feedback
Monitoring & Validation
EVIDENTLY AI
iguazio
Label
Alerts

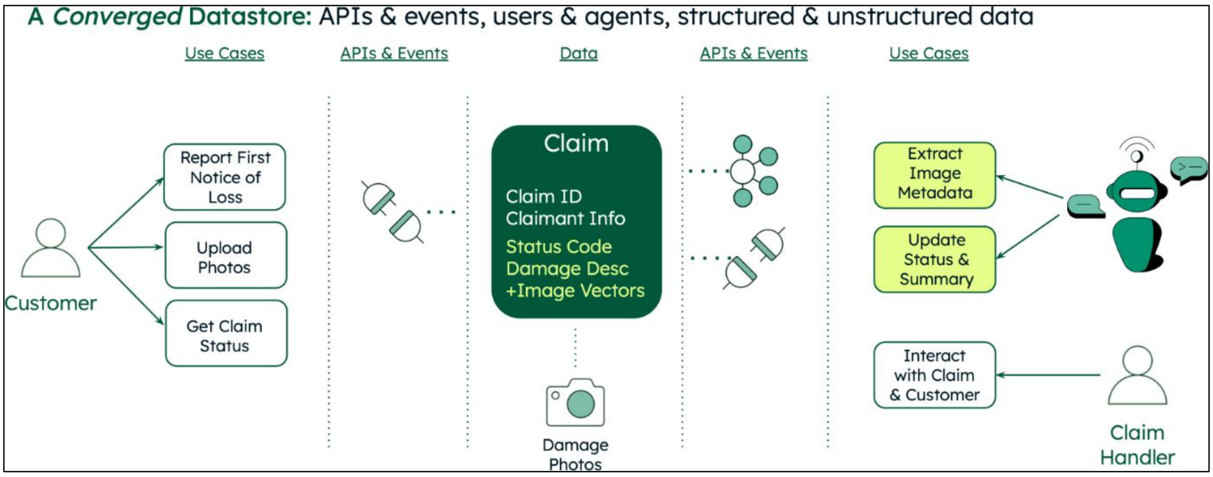
# Chapter 14: Delivering Business Value with AI in Insurance



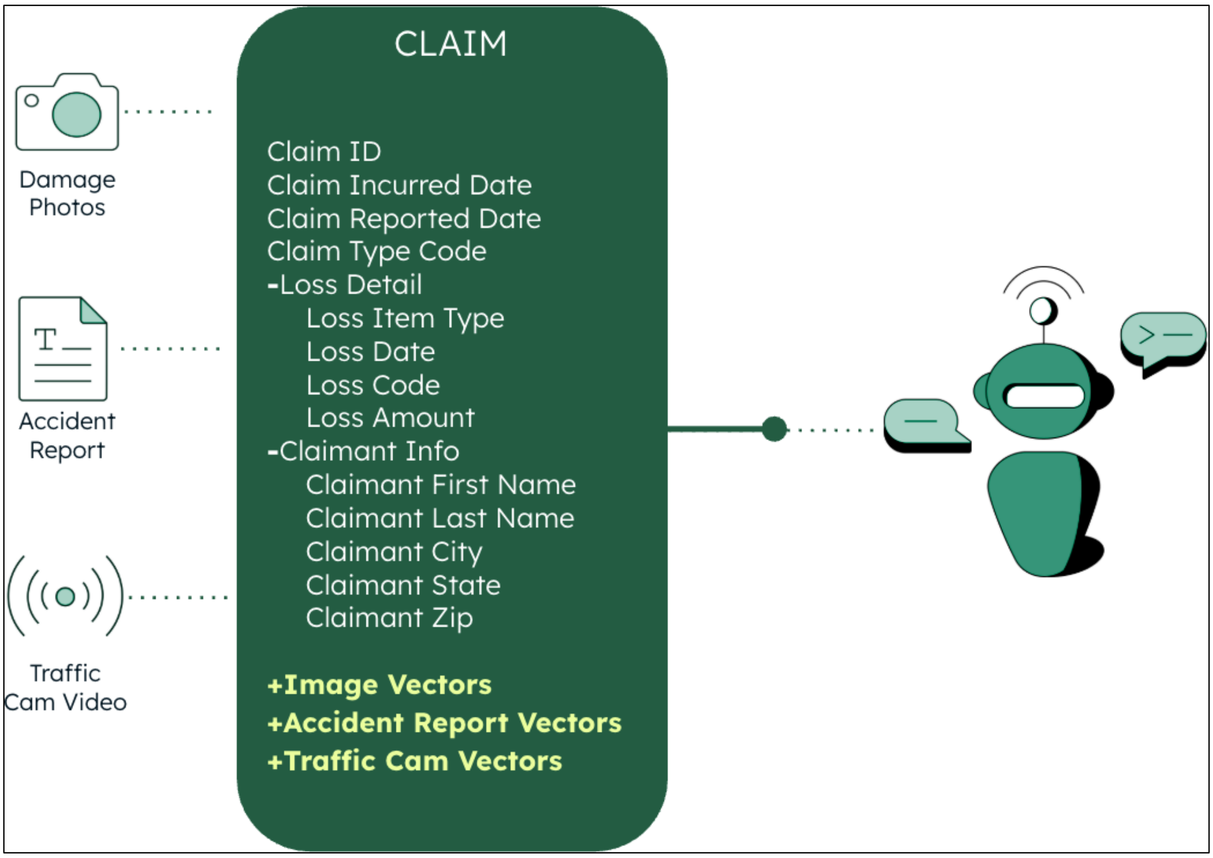



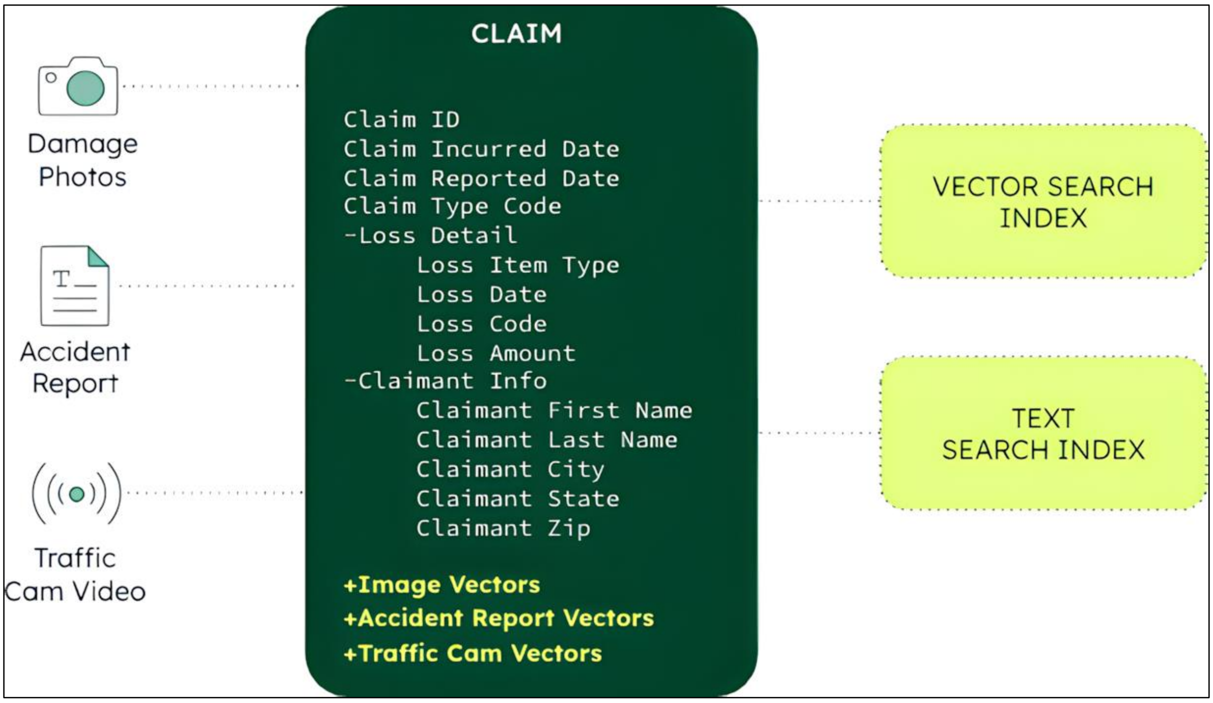


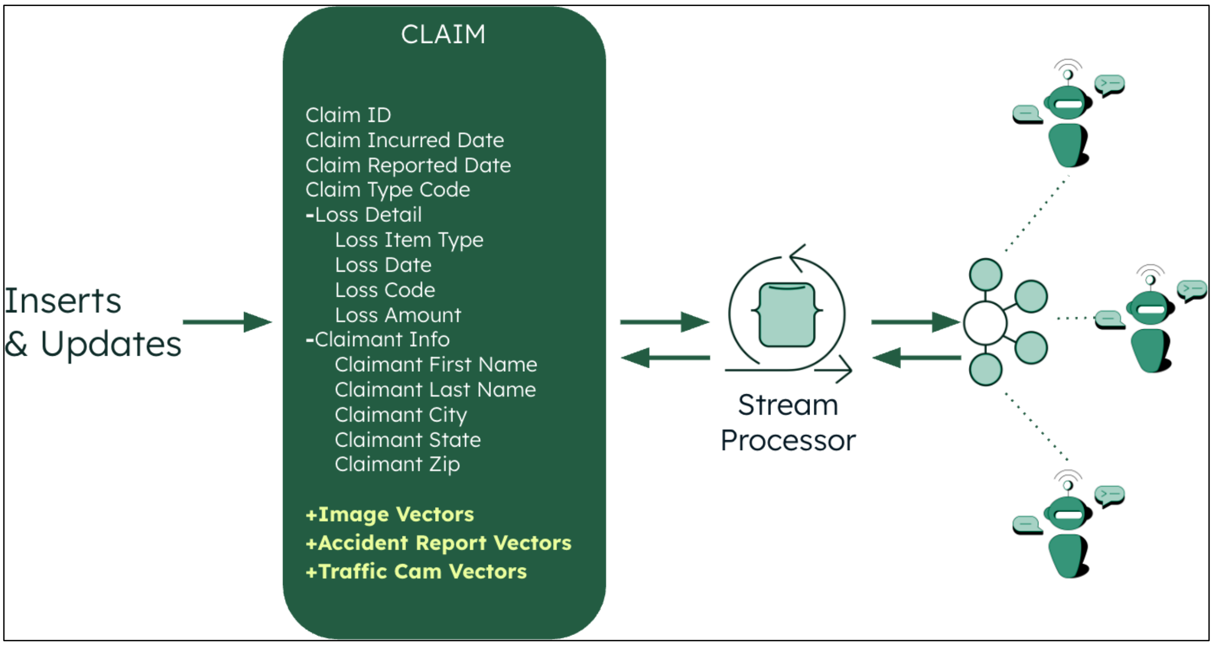
Business outcome: Reduce operational expenses & increase customer satisfaction
Structured Data
Claim
Policy
Business Workflow
Customer Mobile App
First Notice of Loss
Identify & document damages
Coverage Check
Determine & summarize coverage
Claim Handler
Unstructured Data
Damage Photos
Policy Forms

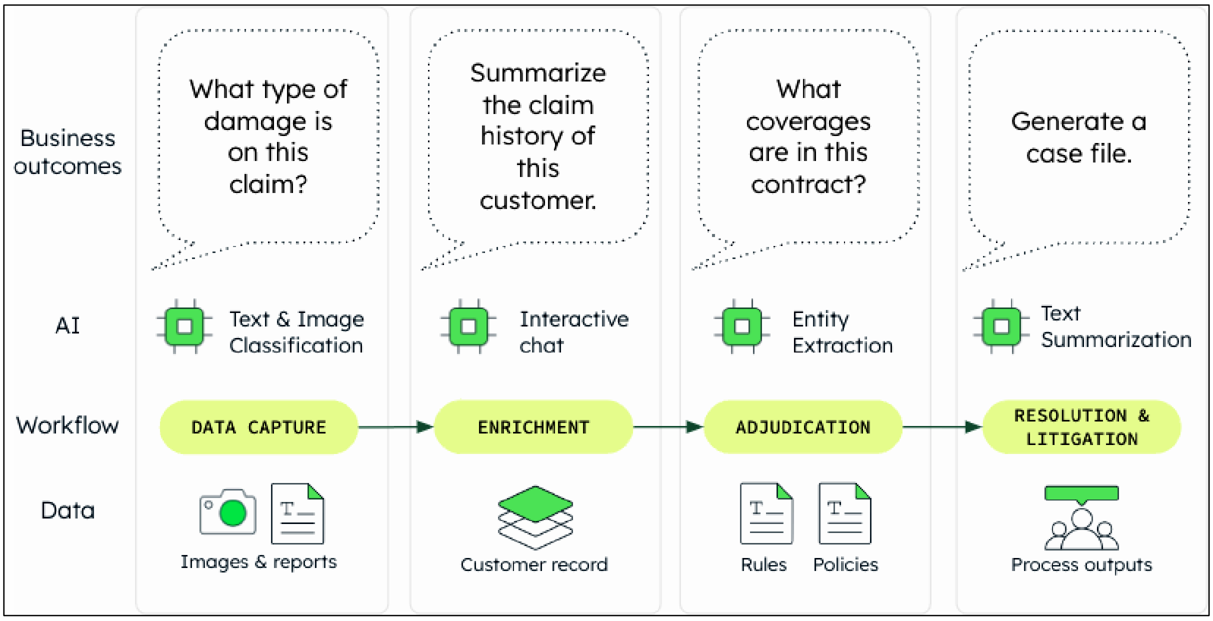



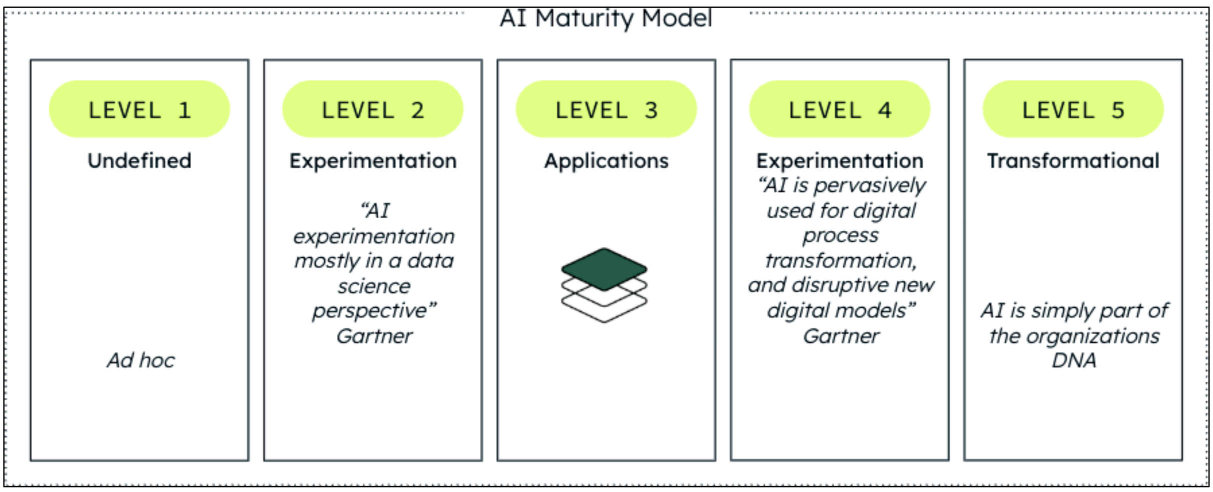
A Converged Datastore: APIs & events, users & agents, structured & unstructured data
Use Cases
APIs & Events
Data
APIs & Events
Use Cases
Customer
Report First Notice of Loss
Upload Photos
Get Claim Status
Claim
Claim ID
Claimant Info
Status Code
Damage Desc
+Image Vectors
Damage Photos
Extract Image Metadata
Update Status & Summary
Interact with Claim & Customer
Claim Handler

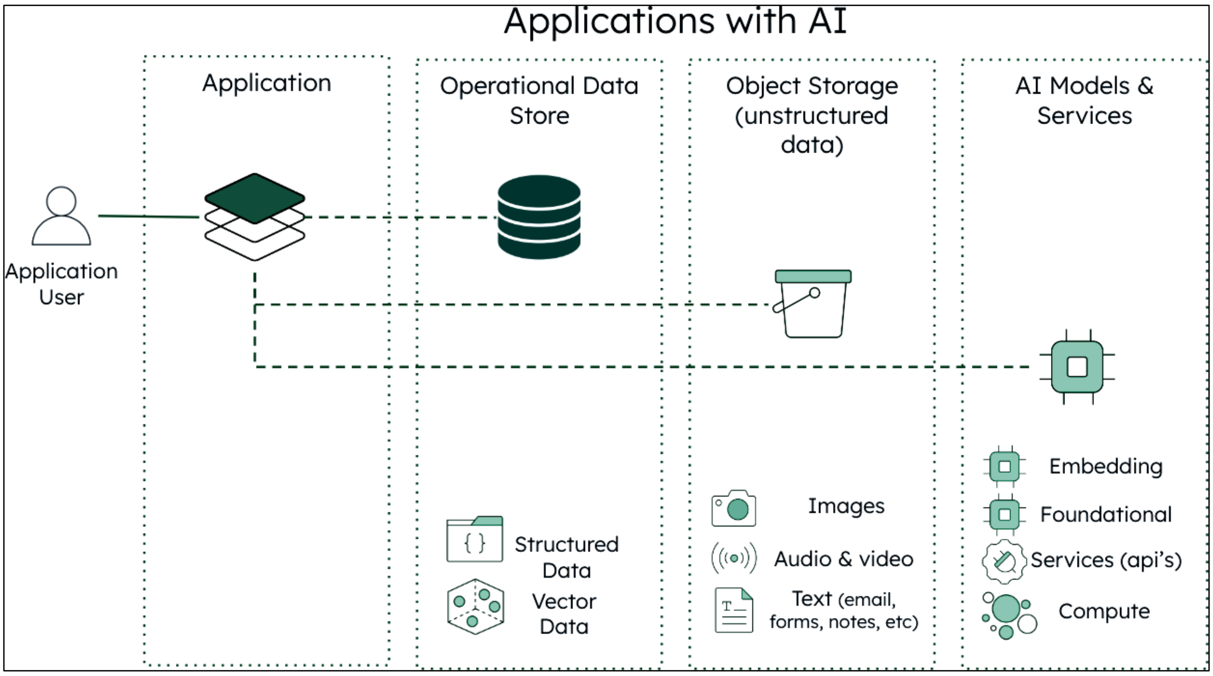

Damage Photos
Accident Report
Traffic Cam Video
CLAIM
Claim ID
Claim Incurred Date
Claim Reported Date
Claim Type Code
-Loss Detail
Loss Item Type
Loss Date
Loss Code
Loss Amount
-Claimant Info
Claimant First Name
Claimant Last Name
Claimant City
Claimant State
Claimant Zip
+Image Vectors
+Accident Report Vectors
+Traffic Cam Vectors


Damage Photos
Accident Report
Traffic Cam Video
CLAIM
Claim ID
Claim Incurred Date
Claim Reported Date
Claim Type Code
-Loss Detail
Loss Item Type
Loss Date
Loss Code
Loss Amount
-Claimant Info
Claimant First Name
Claimant Last Name
Claimant City
Claimant State
Claimant Zip
+Image Vectors
+Accident Report Vectors
+Traffic Cam Vectors
VECTOR SEARCH INDEX
TEXT SEARCH INDEX

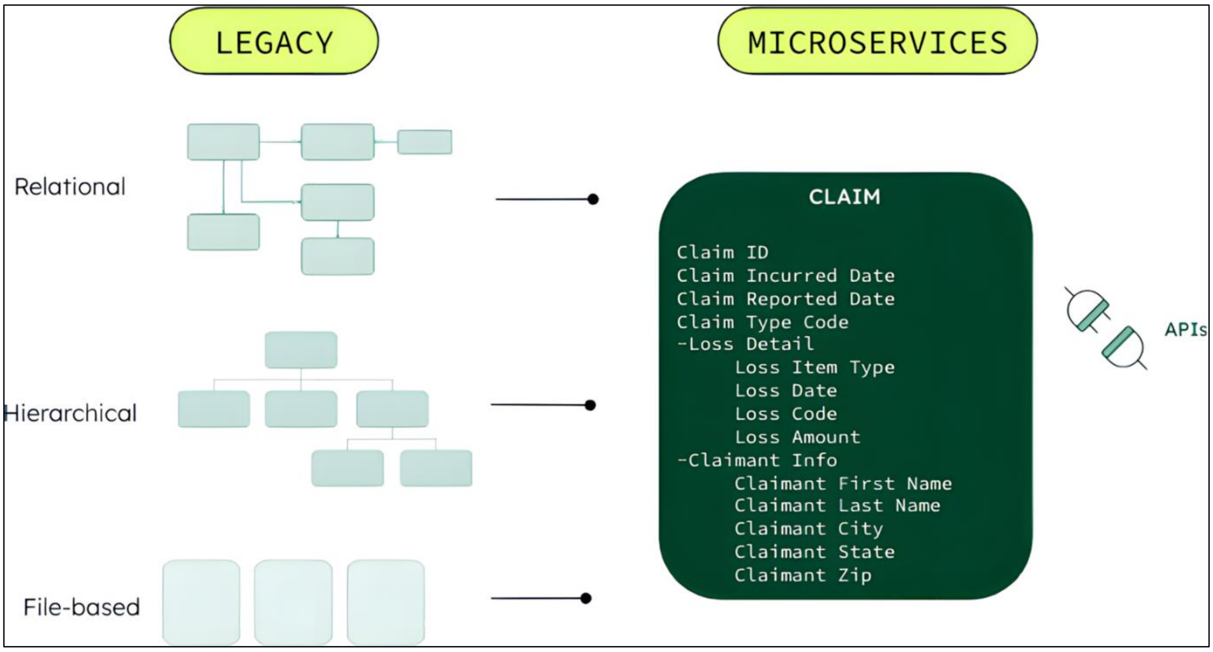

Inserts
& Updates
CLAIM
Claim ID
Claim Incurred Date
Claim Reported Date
Claim Type Code
-Loss Detail
Loss Item Type
Loss Date
Loss Code
Loss Amount
-Claimant Info
Claimant First Name
Claimant Last Name
Claimant City
Claimant State
Claimant Zip
+Image Vectors
+Accident Report Vectors
+Traffic Cam Vectors
Stream
Processor


Business outcomes
What type of damage is on this claim?
Summarize the claim history of this customer.
What coverages are in this contract?
Generate a case file.
AI
Text & Image Classification
Interactive chat
Entity Extraction
Text Summarization
Workflow
DATA CAPTURE
ENRICHMENT
ADJUDICATION
RESOLUTION & LITIGATION
Data
Images & reports
Customer record
Rules
Policies
Process outputs

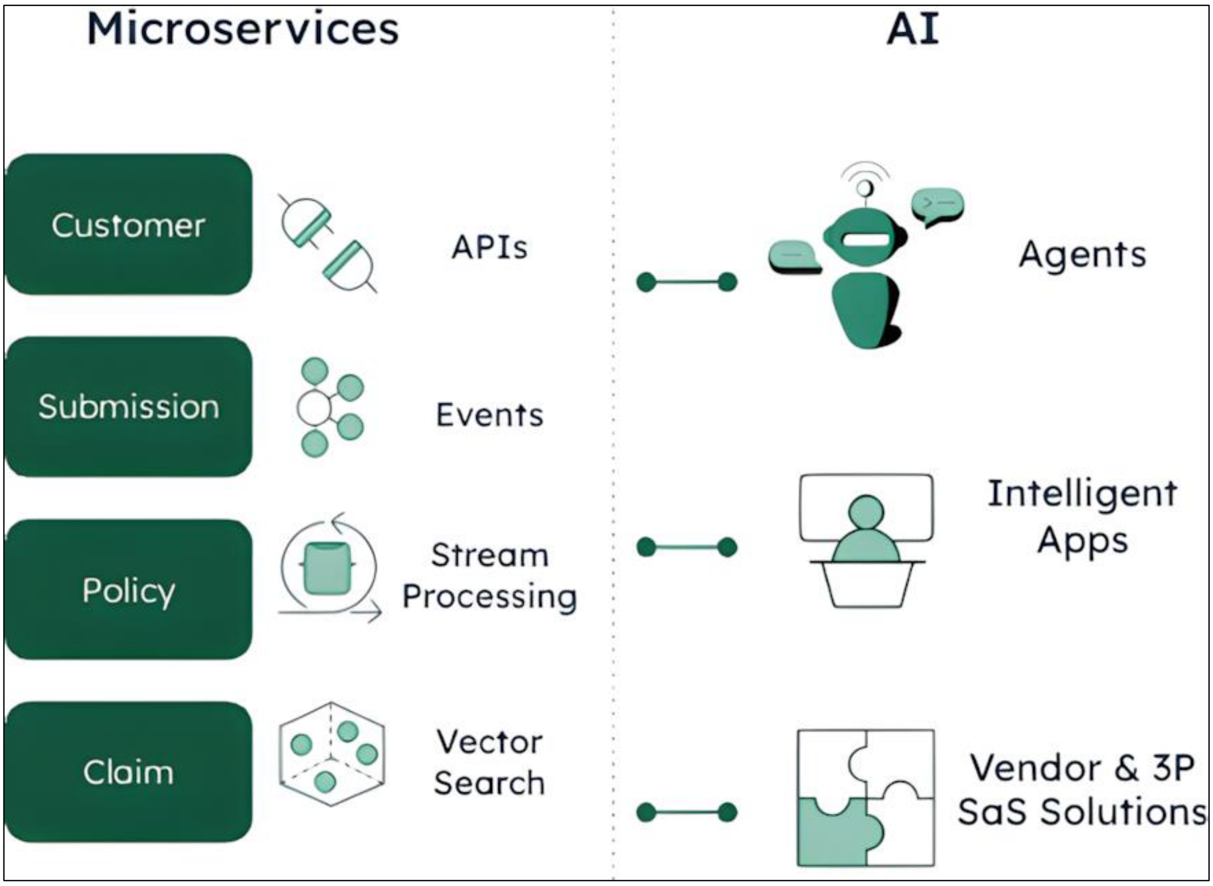

AI Maturity Model
LEVEL 1
Undefined
Ad hoc
LEVEL 2
Experimentation
"AI experimentation mostly in a data science perspective"
Gartner
LEVEL 3
Applications
LEVEL 4
Experimentation
"AI is pervasively used for digital process transformation, and disruptive new digital models"
Gartner
LEVEL 5
Transformational
AI is simply part of the organizations DNA



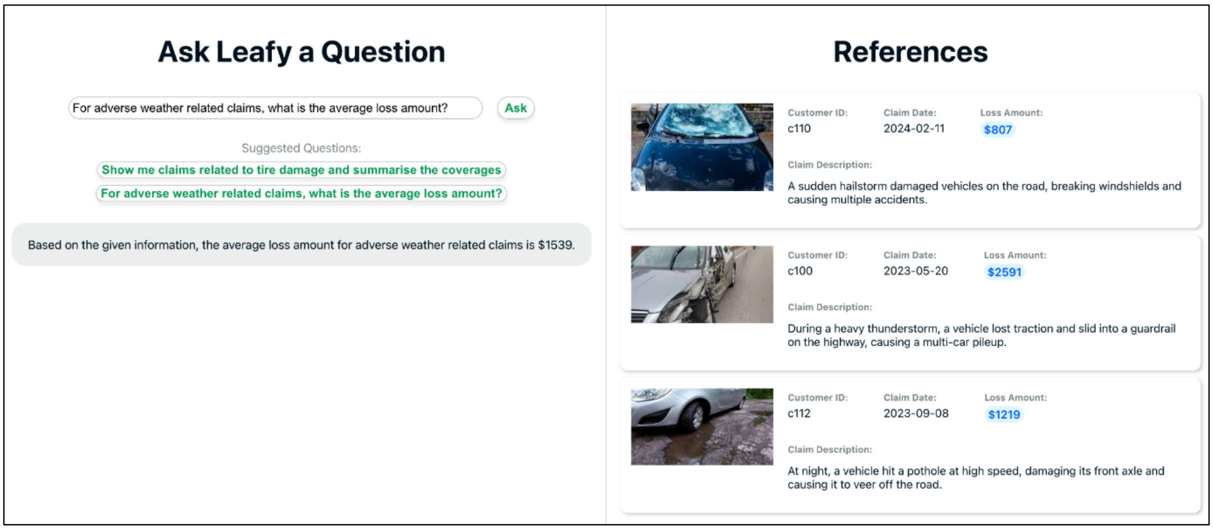
Applications with AI
Application
Operational Data Store
Object Storage (unstructured data)
AI Models & Services
Application User
Structured Data
Vector Data
Images
Audio & video
Text (email, forms, notes, etc)
Embedding
Foundational
Services (api's)
Compute

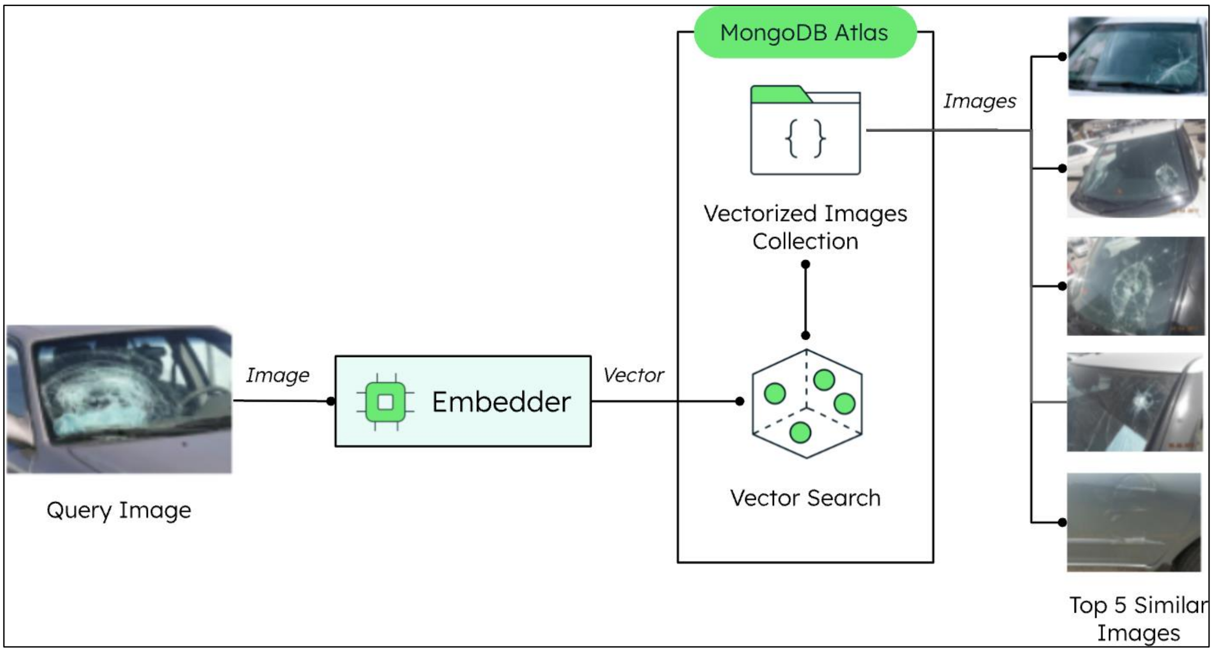
LEGACY
MICROSERVICES
Relational
Hierarchical
File-based
CLAIM
Claim ID
Claim Incurred Date
Claim Reported Date
Claim Type Code
-Loss Detail
Loss Item Type
Loss Date
Loss Code
Loss Amount
-Claimant Info
Claimant First Name
Claimant Last Name
Claimant City
Claimant State
Claimant Zip
APIs

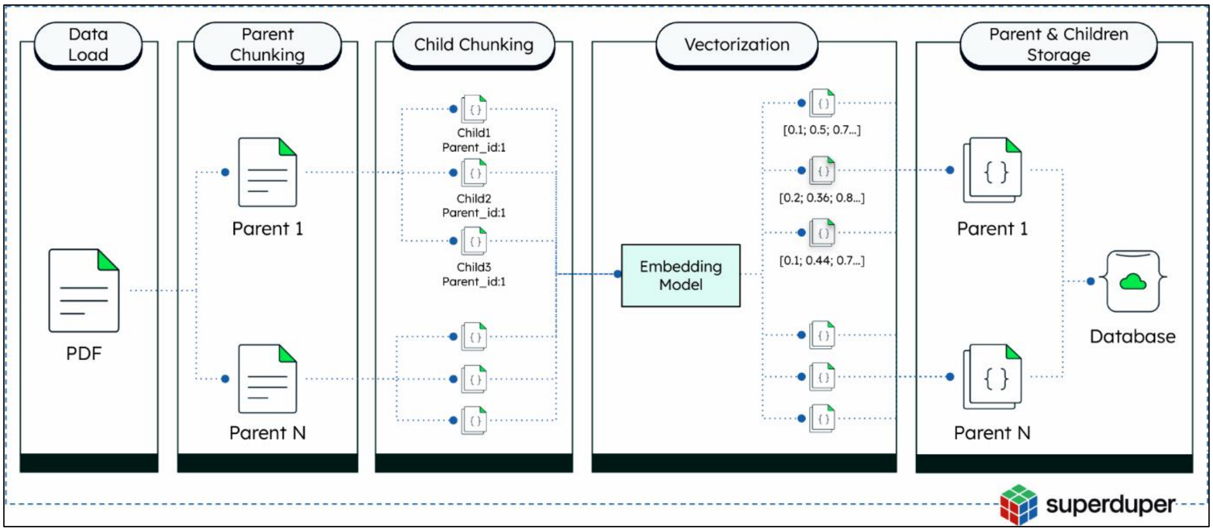


Microservices
AI
Customer
APIs
Agents
Submission
Events
Policy
Stream Processing
Intelligent Apps
Claim
Vector Search
Vendor & 3P SaS Solutions

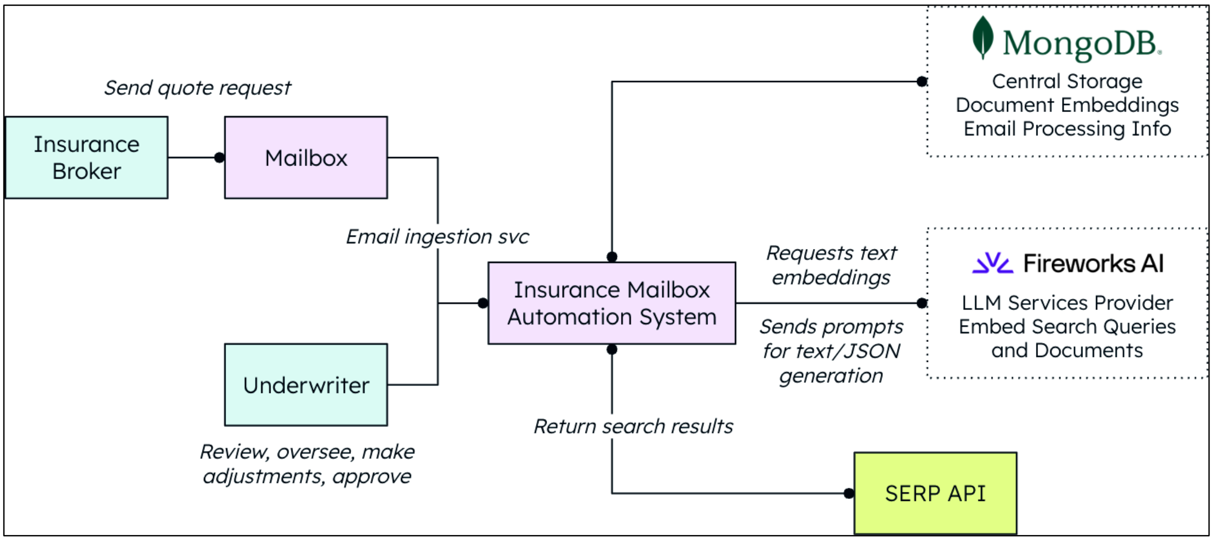



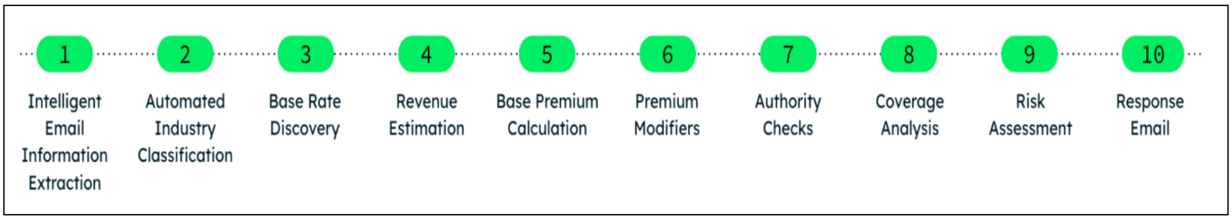
Ask Leafy a Question
For adverse weather related claims, what is the average loss amount?
Ask
Suggested Questions:
Show me claims related to tire damage and summarise the coverages
For adverse weather related claims, what is the average loss amount?
Based on the given information, the average loss amount for adverse weather related claims is $1539.
References
Customer ID: c110
Claim Date: 2024-02-11
Loss Amount: $807
Claim Description: A sudden hailstorm damaged vehicles on the road, breaking windshields and causing multiple accidents.
Customer ID: c100
Claim Date: 2023-05-20
Loss Amount: $2591
Claim Description: During a heavy thunderstorm, a vehicle lost traction and slid into a guardrail on the highway, causing a multi-car pileup.
Customer ID: c112
Claim Date: 2023-09-08
Loss Amount: $1219
Claim Description: At night, a vehicle hit a pothole at high speed, damaging its front axle and causing it to veer off the road.

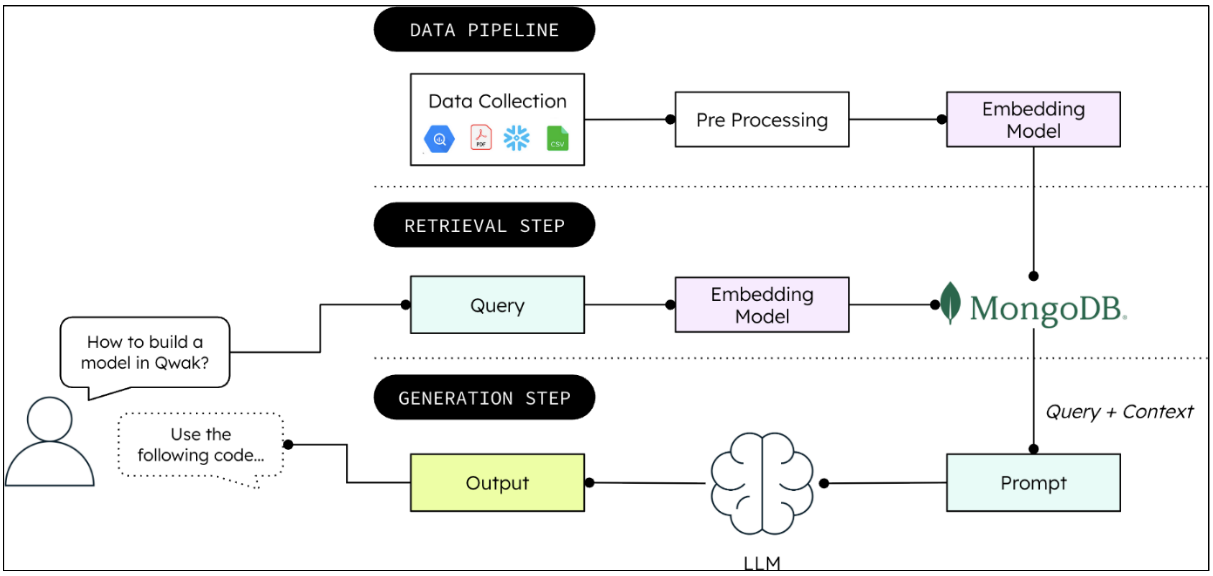

MongoDB Atlas
Images
Vectorized Images Collection
Image
Embedder
Vector
Query Image
Vector Search
Top 5 Similar Images


Data Load
Parent Chunking
Child Chunking
Vectorization
Parent & Children Storage
Child1 Parent_id:1
[0.1; 0.5; 0.7...]
Parent 1
Child2 Parent_id:1
[0.2; 0.36; 0.8...]
Parent 1
Child3 Parent_id:1
[0.1; 0.44; 0.7...]
Embedding Model
PDF
Database
Parent N
Parent N
superduper

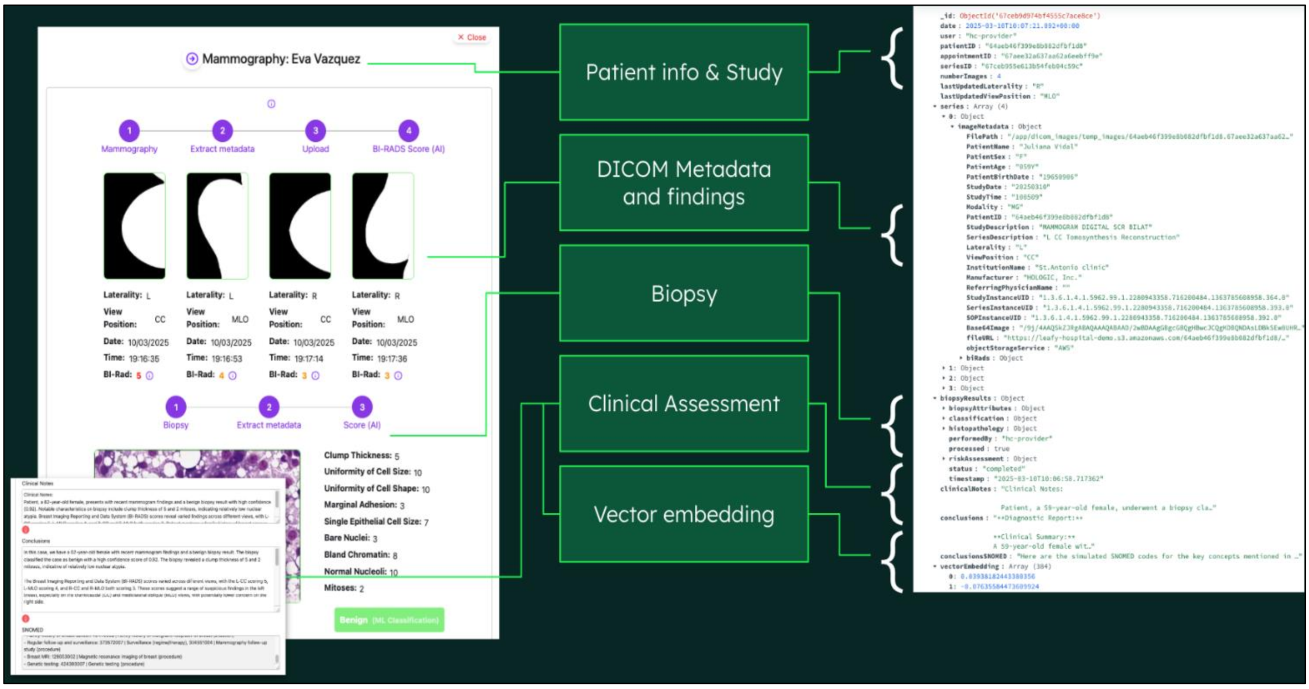
## Chapter 15: Automating Insurance Underwriting with Fireworks AI and MongoDB

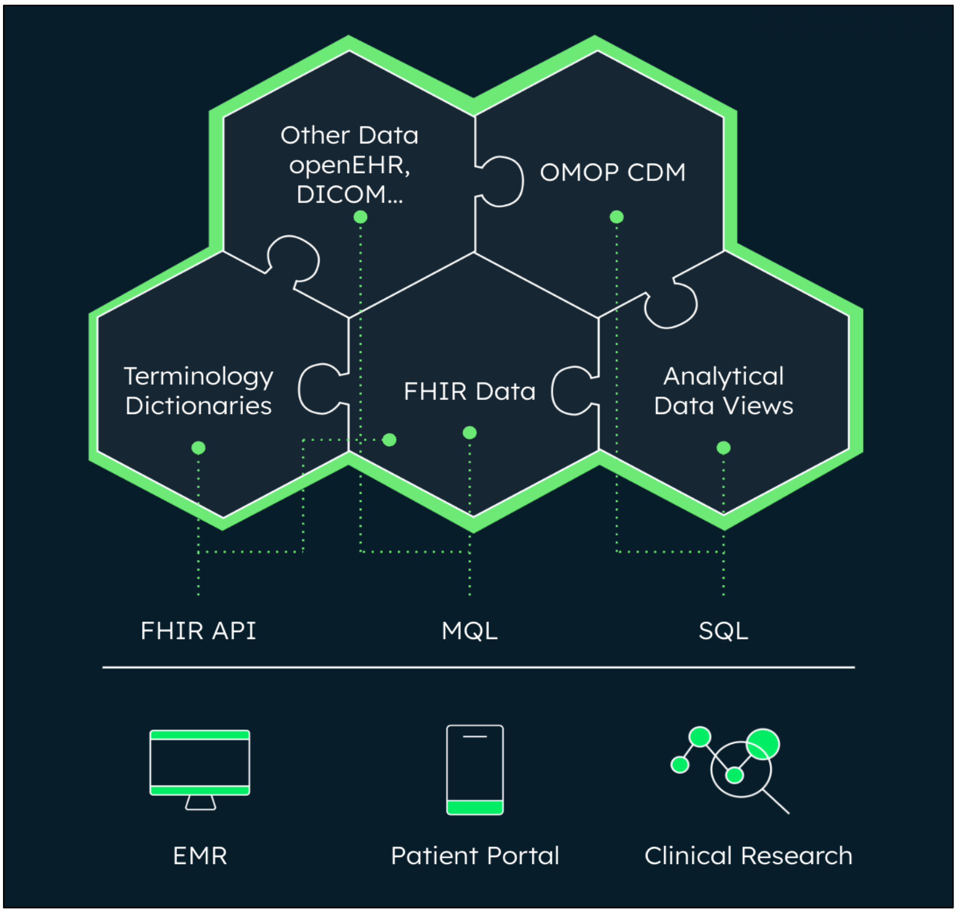


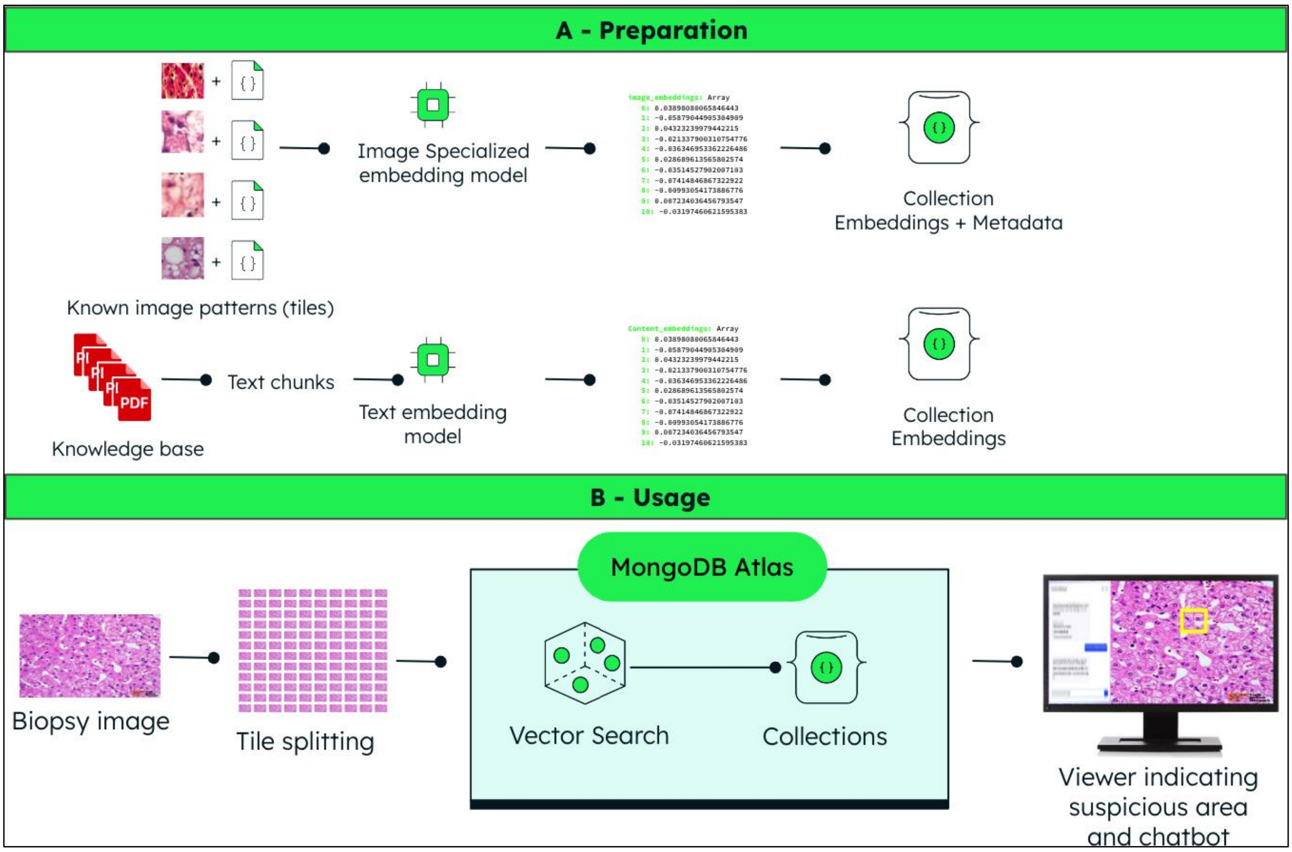



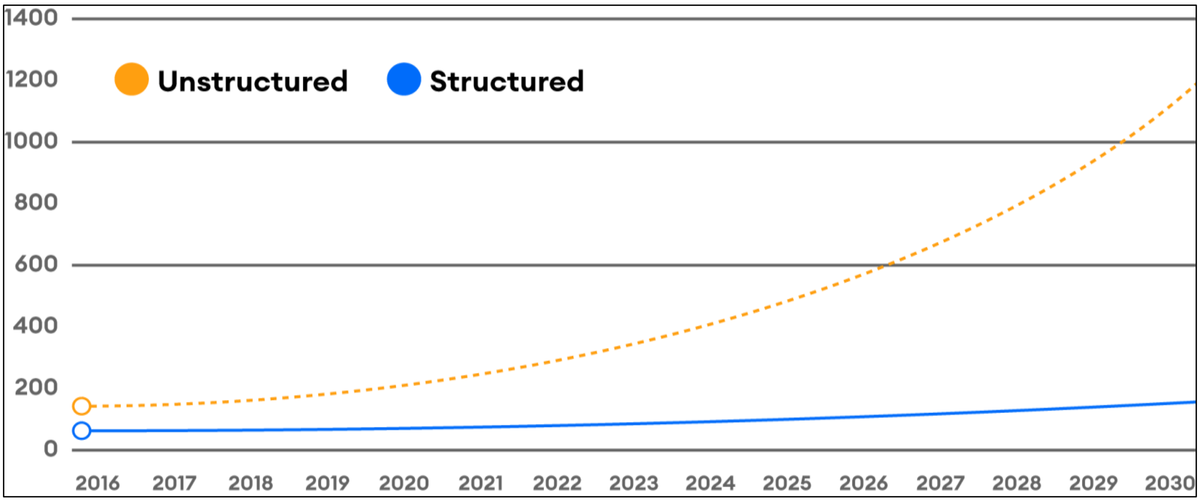
# Chapter 16: AI-Powered Transformation of Healthcare and Life Sciences

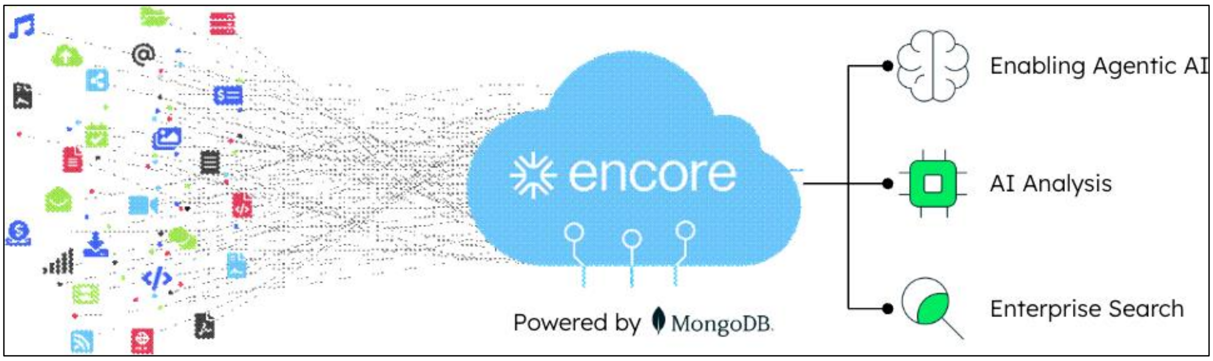


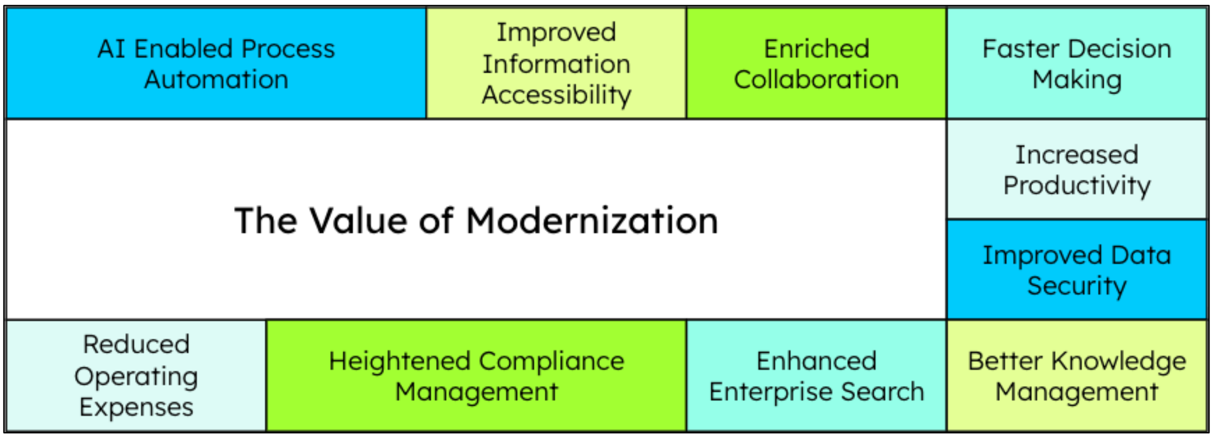




A - Preparation
Image Specialized embedding model
Collection Embeddings + Metadata
Known image patterns (tiles)
Knowledge base
Text chunks
Text embedding model
Collection Embeddings
B - Usage
MongoDB Atlas
Biopsy image
Tile splitting
Vector Search
Collections
Viewer indicating suspicious area and chatbot

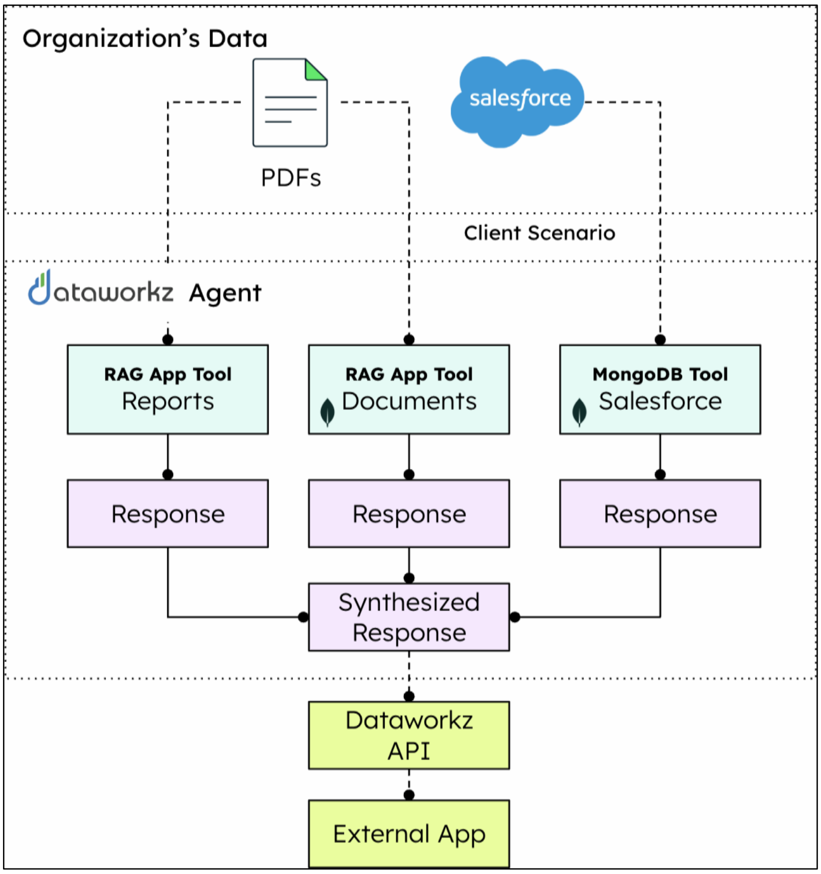
# Chapter 17: Enterprise Document Management with MongoDB and AI

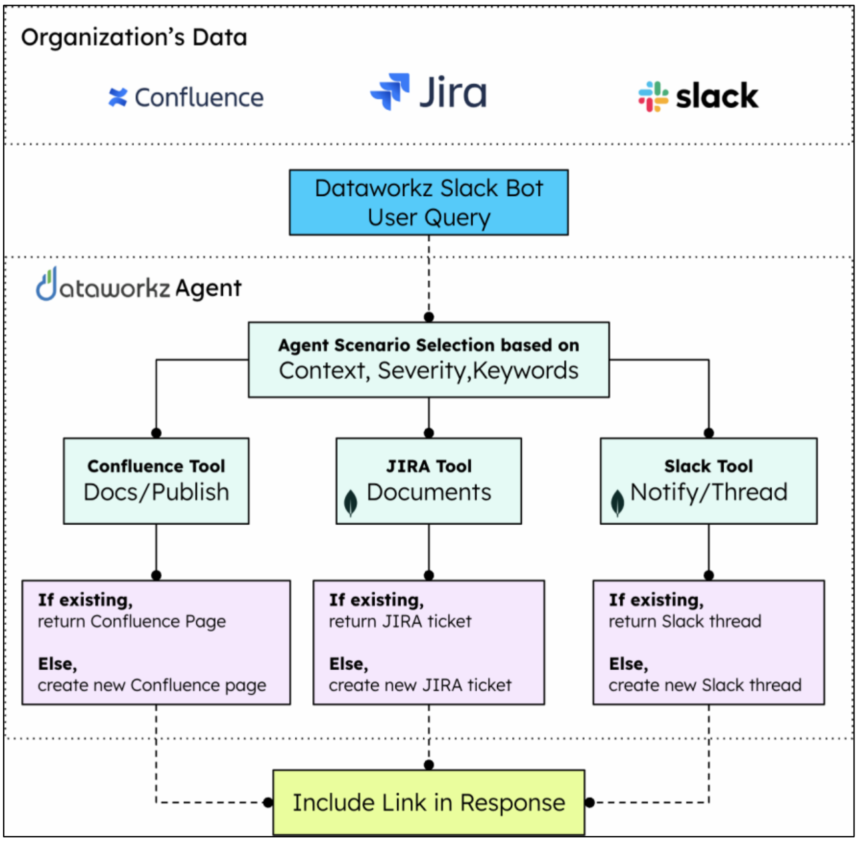


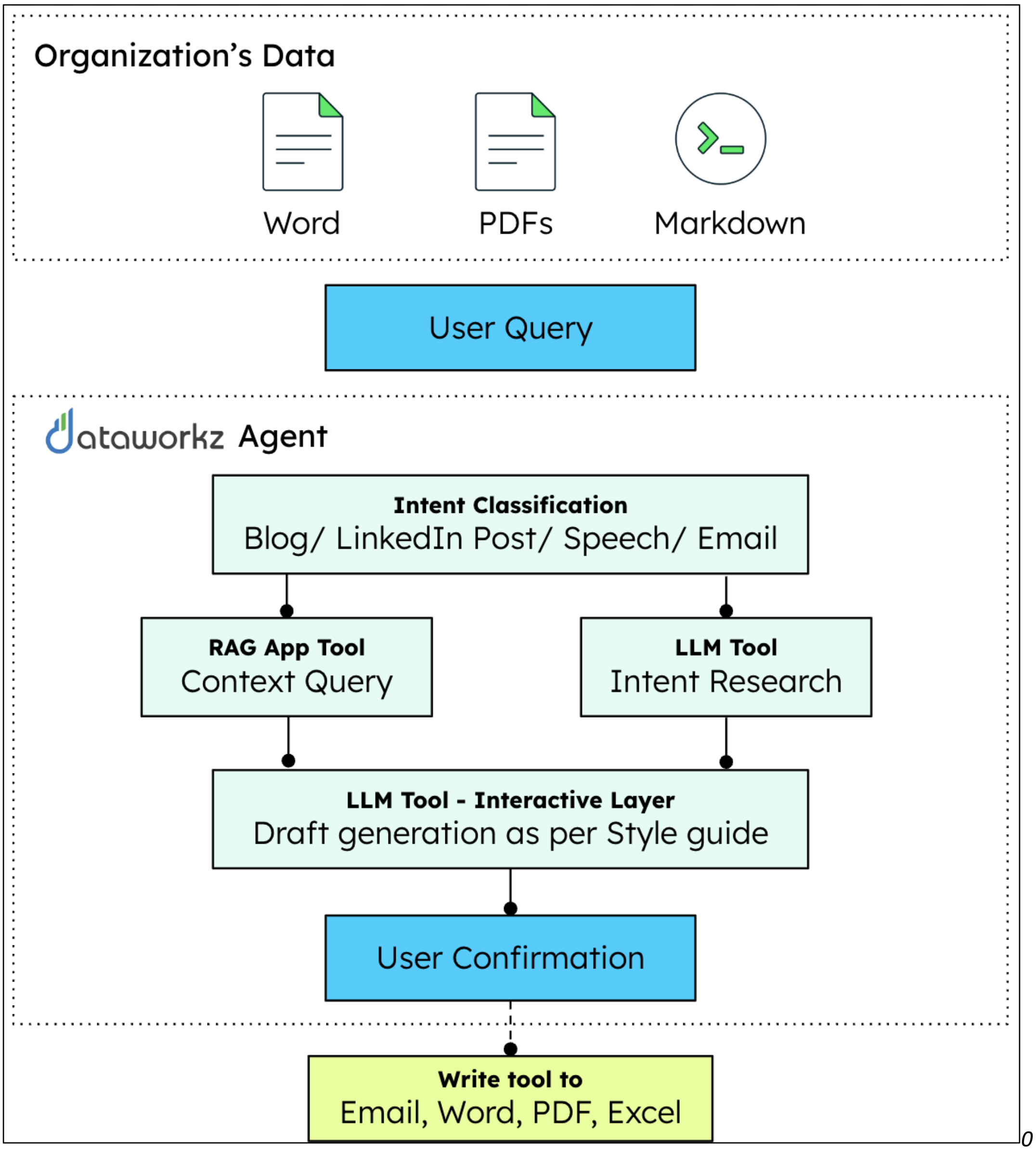



# Chapter 18: Democratizing Agentic AI for Enterprise with Dataworkz and MongoDB




Organization's Data
Word
PDFs
Markdown
User Query
dataworkz Agent
Intent Classification
Blog/ LinkedIn Post/ Speech/ Email
RAG App Tool
Context Query
LLM Tool
Intent Research
LLM Tool - Interactive Layer
Draft generation as per Style guide
User Confirmation
Write tool to
Email, Word, PDF, Excel

## Chapter 19: Outlook: Beyond Today's AI